Abstraction Mechanisms for Virtual Avatars

Master's Thesis

submitted by

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Abstract

Virtual reality applications increasingly rely on avatar representations to facilitate social presence and user embodiment. However, photorealistic avatar rendering can trigger the uncanny valley effect, where near-realistic human representations evoke discomfort and reduced trust rather than engagement. Non-photorealistic rendering (NPR) offers a principled alternative, deliberately abstracting avatar appearance while preserving expressiveness and identity.

This thesis presents the design, implementation and evaluation of a real-time NPR stylisation system for virtual avatars. Eight techniques were developed iteratively: a geometric silhouette outline, toon shading and its X-Toon extension, screen-space normal edge detection, Sobel edge detection, Gaussian prefiltered Sobel, anisotropic Kuwahara painterly filtering and hierarchical multi-cue edge detection combining depth, normal and colour discontinuities. The techniques were implemented and compared across three avatar platforms: a Mixamo humanoid model, the Meta Avatars SDK and Avaturn. This revealed their visual behaviour and robustness under different mesh topologies and rendering pipelines. Avaturn was selected as the primary platform for the user study due to its facial rigging capabilities and higher photorealistic fidelity.

The user study investigates how visual abstraction affects the perceived trustworthiness of an avatar delivering advice. Participants watched short video scenarios in which a speaker recommended one of two options in a practical decision situation. The speaker was presented either as a real person, a photorealistic Avaturn avatar or an NPR-stylised avatar. After each scenario, participants rated the speaker on eeriness, likeability and three dimensions of trust. Participants registered the stylised avatar as visually distinct from its photorealistic counterpart. It was nonetheless rated no less eerie and no more trustworthy than that counterpart and both avatars fell well below the real speaker on every trust measure. Eeriness did predict lower trust within participants across all three dimensions. Abstraction and uncanny affect therefore behaved as separable properties for the composite evaluated here. This qualifies the expectation that stylisation alone relocates an avatar out of the uncanny valley.

Keywords: non-photorealistic rendering, virtual reality, avatar abstraction, trust perception, edge detection, Avaturn, uncanny valley, Kuwahara filter.

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Abbreviations

1D	One-Dimensional
2D	Two-Dimensional
3D	Three-Dimensional
AI	Artificial Intelligence
ANOVA	Analysis of Variance
API	Application Programming Interface
AR	Augmented Reality
CG	Computer-Generated
CGI	Computer-Generated Imagery
DoG	Difference of Gaussians
ESF	Extensible Shader Framework
FBX	Filmbox (Autodesk interchange format)
FPS	Frames Per Second
GI	Global Illumination
GLB	Binary glTF (GL Transmission Format)
GPU	Graphics Processing Unit
HCI	Human-Computer Interaction
HLSL	High-Level Shading Language
HMD	Head-Mounted Display
HSV	Hue, Saturation, Value (colour space)
LOD	Level of Detail
NPR	Non-Photorealistic Rendering
PBR	Physically-Based Rendering
RGB	Red, Green, Blue
SDK	Software Development Kit
SSS	Subsurface Scattering
TBN	Tangent, Bitangent, Normal (matrix)
TOST	Two One-Sided Tests
UDP	User Datagram Protocol
UI	User Interface
URP	Universal Render Pipeline
UV	Texture Coordinate Axes (U and V)
VR	Virtual Reality
XDoG	Extended Difference of Gaussians

1. Introduction

1.1. Motivation

Computer-generated (CG) humanoid characters have become an ordinary part of contemporary digital experiences, especially as generative artificial intelligence (AI) video content is widely encountered on a daily basis. As virtual assistants, avatars and other agents are increasingly integrated into everyday domains, they interact with users instead of remaining passive tools that act only on command [1]. Engaging with a CG human makes its design consequential, since the character is no longer a background presence but an active participant in the exchange. That design governs aesthetic appeal as well as how users perceive and respond to the character. These effects raise several concerns that carry consequences across film, games and other domains where virtual humans appear. One of the difficult ones is how realistically or abstractly such a character should be rendered [2]. That tension between realism and abstraction in avatars is the starting point for the work presented in this thesis.

This design choice can be represented along a fidelity axis. Higher fidelity corresponds to greater human likeness on that axis, meaning the degree to which an avatar visually resembles a real human. Anthropomorphism describes the viewer's side of that axis. Furthermore, an agent's perceived anthropomorphism and its behaviour have been demonstrated to increase users' trust in it [3]. However, this increase in trust does not scale linearly with likeness across the fidelity axis. Mori termed this effect the uncanny valley. Comfort rises with human likeness only up to a point, then drops sharply for entities that appear almost but not quite human, before recovering near-perfect likeness [4]. Consequently, an avatar that approaches full realism may reduce users' confidence, not raise it. The same dip appears for CG characters generally [2]. Avatars fall within this category and their stylisation can affect users' behavioural choice [5]. Together, these results suggest that rendering style influences not only users' subjective confidence but also the decisions they make. Alternative visual styles could protect confidence and choice alike by avoiding the perceptual pitfalls that highly realistic avatars carry.

Non-Photorealistic Rendering (NPR) provides an alternative approach by deliberately abstracting an avatar's visual appearance while preserving its recognisability and expressive capabilities. Moreover, NPR allows designers to position an avatar at different locations along the fidelity axis through controlled stylisation rather than maximising realism. Accordingly, the central question is not whether stylised avatars are visually appealing. It is whether deliberate abstraction can maintain or even improve users' trust while avoiding the perceptual costs of near-photorealistic rendering. That question is becoming increasingly relevant as virtual avatars move beyond research settings into widely used consumer applications such as gaming and marketing [6]. Thus, determining the balance between realism and abstraction is the defining challenge in the design of trustworthy virtual avatars. How this tension is resolved will shape whether virtual avatars can transition from novelty to genuinely trusted participants in social and professional contexts.

1.2. Problem Statement

Although NPR offers a means of controlling avatar appearance, applying NPR stylisation to modern photorealistic avatar systems is not straightforward. Current avatar platforms expose substantially different rendering architectures and each imposes its own constraints on the implementation of stylisation techniques. In particular, some platforms provide only a single shader integration point while keeping the remainder of the rendering pipeline inaccessible. This prevents additional render passes or access to intermediate shading information. Others expose broader pipeline access but introduce a separate set of challenges arising from photogrammetric textures and asset representations. Therefore, the applicability of a given NPR technique depends on both the desired visual effect and the rendering capabilities of the underlying platform.

Thus, the same visual effect must be re-engineered for every pipeline it meets. What survives that re-engineering is not evident before it is attempted. One of the challenges is that NPR offers many operators, and each abstracts a different property of the image at its own cost. Which of them produce a convincing result varies with the

platform, since an operator that works on one avatar's texture may fail on another's. As a result, the choice must be settled by comparison instead of being assumed in advance.

Because the ultimate goal of the abstraction is to convey trust rather than merely to render successfully, resolving these technical challenges leaves two questions open. The first is whether abstraction lessens the drop in comfort that the uncanny valley predicts. The second is whether it can render a stylised avatar as trustworthy as an actual human speaker in a digital environment. This concerns the perceptual and behavioural effect of stylisation, as distinct from its technical feasibility or design. The same concern applies to any domain in which avatar credibility is at stake: social virtual reality (VR), remote consultation, training simulation, advertising and education.

1.3. Research Questions

The following three research questions guide this work:

RQ1: How can NPR stylisation techniques be adapted to avatar platforms with different rendering architectures and what capabilities and limitations does each platform impose?

RQ2: Which NPR technique or combination of techniques best balances visual abstraction against facial and expressive legibility under production rendering constraints?

RQ3: How does NPR stylisation affect perceived eeriness and trustworthiness and can an NPR avatar reach the trustworthiness of a photorealistic avatar and of a real human speaker?

1.4. Contributions

This thesis makes the following contributions:

- **NPR stylisation across three production avatar platforms.** Eight progressively refined techniques were produced, each addressing limitations observed in its predecessor. They range from a geometry-based outline through edge detection to a painterly post-processing filter and all are selectable at runtime without recompilation. Building the same set on Mixamo, the Meta Avatars SDK and Avaturn showed that classical NPR operators behave differently under production rendering constraints. Some required recalibration depending on the texture of the avatar. Others required rebuilding around a substituted input signal on the most restrictive platform, since it offers more limited pipeline access. These differences indicate which operators transfer across production rendering pipelines and which require adaptation tailored to the platform.
- **Perceptual calibration of NPR parameters.** A parameter adjustment study established the final NPR parameter values through preference testing rather than visual inspection alone. Participants ranked rendered avatar variants by perceived trustworthiness, and these rankings also decided which techniques were retained in the composite.
- **A measurement design that isolates rendering style from confounding factors.** The study questionnaire combined validated instruments from earlier trust, likeability and eeriness research with a behavioural advisor ranking measure. The combination addresses a methodological gap identified in prior reviews, where reported trust ratings and behavioural choice have rarely been measured together for stylised avatars. Three realistic decision scenarios carried these measures, each targeting a different dimension of interpersonal trust: ability, benevolence or integrity. Confounds were controlled by alternating which option each scenario recommended and by varying the order in which the options were discussed. The same script structure was applied uniformly across all three. Any difference in trust ratings therefore reflects rendering style rather than variation in scenario content or argumentation pattern.
- **Evidence on NPR stylisation and perceived avatar trust.** A within-subjects user study was conducted using prerecorded video scenarios in which the same avatar and speaker appeared across three rendering conditions. Comparing the three conditions yielded empirical evidence on how NPR stylisation affects

the perceived trustworthiness of an avatar's advice relative to both a photorealistic counterpart and a real human speaker.

1.5. Thesis Outline

The remainder of this thesis is structured as follows:

Chapter 2 introduces the technical and theoretical foundations that the later chapters assume, spanning rendering concepts, the image processing operators behind the stylisation techniques and the constructs of trust and eeriness.

Chapter 3 surveys prior work relevant to avatar realism, trust and NPR rendering, then identifies the specific gap that this thesis addresses.

Chapter 4 details the research design and system architecture, elaborating on the questions above as well as on how the user study was constructed.

Chapter 5 describes the full implementation of the NPR shader system across all three avatar platforms.

Chapter 6 reports the technical evaluation of each stylisation technique and presents the user study results. It then discusses what those outcomes indicate as well as the limitations that qualify them.

Chapter 7 summarises the contributions of this thesis and identifies directions for future work.

2. Background

This chapter introduces the technical and theoretical foundations that the remainder of the thesis assumes. It opens with the notation used throughout, then presents the rendering concepts on which the stylisation techniques depend. Virtual avatars follow, together with the constructs of trust and eeriness that the user study measures. The final two sections cover the image processing operators from which the stylisation techniques derive and the edge-preserving filter behind painterly abstraction.

2.1. Notation and Conventions

Scalars are denoted by italic letters x. Vectors appear in bold, as with the surface normal $\mathbf{N}$. An image is treated as a function $I(x, y)$ over a discrete pixel grid, where x increases towards the right of the screen and y increases upwards, matching the coordinate system that the GPU exposes to a shader. Convolution kernels take an italic uppercase letter and a directional subscript, as in G_x. The convolution of an image I with a kernel K is written $K * I$. Colour values are treated as linear RGB triples in the range $[0, 1]$ unless stated otherwise.

2.2. Rendering Concepts

A render pass is one complete traversal of scene geometry. Its result is written into a render target, an off-screen image buffer. A fragment shader that operates within a single pass sees only the surface point it is shading. Any operator needing a spatial neighbourhood must gather that neighbourhood itself. Additional passes remove this restriction. A post-processing pass renders one full-screen quad and samples the render target left by the preceding pass, which gives every fragment access to the whole rendered image. Unity's Universal Render Pipeline (URP) schedules such injected work through a renderer feature [7]. However, every extra pass costs another traversal of the geometry or of the full screen. Thus, the number of passes a platform permits is an engineering constraint, not a free parameter.

Despite this restriction on passes, the material data itself remains unaffected. It is held in separate texture maps under physically based rendering (PBR). The albedo map stores the base colour of the surface with no lighting baked into it. The normal map encodes fine surface detail that the geometry does not carry, and roughness and metallic maps complete the set. The techniques described in this thesis read the albedo and the normal. A shader confined to a single pass can still measure how either signal changes across the screen. It does so through the screen-space derivative, which supplies the rate of change of any value between neighbouring fragments. Fragments are rasterised in 2×2 quads, and this lets the hardware return that difference at no additional cost.

Yet the surface carrying these maps does not stay stable at runtime, since an animated character never holds a fixed pose. That motion is produced by the rig, the control structure bound to the mesh before any animation is played. A rig displaces vertex positions by two mechanisms, skeletal deformation and blendshapes. Blendshapes, a linear combination of stored expression shapes, handle the finer motion that a skeleton cannot reach. They hold alternative vertex positions that are interpolated against the base mesh [8]. A platform whose rig omits blendshapes therefore cannot carry facial expression at all, whatever its rendering quality.

2.3. Avatars, Trust and Eeriness

An avatar is a digital body that represents a person within a virtual space. The term spans forms ranging from abstract geometric markers to fully articulated humanoid figures. Its visual appearance becomes a central cue from which viewers form an impression of the person that the avatar represents [9]. Consequently, the central social judgement an observer makes about an avatar is one of trust. Mayer et al. defined trust as a willingness to be vulnerable to another party's actions, held on the expectation that the party will act as needed and without any means of monitoring or controlling them [10].

This judgement is not a single quantity. Perceived trustworthiness decomposes into three components: ability, benevolence and integrity. *Ability* denotes the skills and characteristics that let a trustee act effectively within a relevant domain, *benevolence* the extent to which the trustee appears to care about the observer's wellbeing and *integrity* adherence to acceptable principles. The three are separable rather than unrelated, and each can vary while the others hold [10]. McKnight and Chervany added predictability to that set as a fourth trusting belief [11]. Together, these four components describe what an observer believes about the trustee and they leave the perceptual response to appearance unaddressed.

A complementary family of instruments targets that perceptual dimension directly. The Godspeed Questionnaire Series measures five perceptual qualities of an artificial agent: anthropomorphism, animacy, likeability, perceived intelligence and perceived safety. Anthropomorphism is the degree of human likeness an observer perceives in the agent, and its anchors run from machinelike to humanlike and from artificial to lifelike. Animacy denotes whether the agent appears alive rather than inert, and its items range from dead to alive and from apathetic to responsive. Likeability is the immediate affective impression the agent makes, anchored between dislike and like and between unfriendly and friendly. Accordingly, these five subscales hold 24 semantic differential items in total, each rated on a five-point scale [12]. Ho and MacDorman found every subscale except perceived safety too strongly intercorrelated to measure distinct constructs, and none of them corresponded specifically to eeriness. Their own instrument addresses that gap with eight items for eeriness alongside six for perceived humanness. Eeriness, as they frame it, is the unsettled perceptual response that arises when a stimulus approaches but does not reach the peak of human likeness. Yet the two indices proved orthogonal in validation, with $r \approx .02$ [13]. This indicates that high human likeness and strong discomfort can co-occur in the same stimulus.

2.4. Edge Detection

A digital image consists of a grid of pixels, each storing an intensity value. An edge is conventionally defined as a location at which this intensity changes rapidly [14]. Detecting such locations involves estimating the image gradient and applying a threshold to identify pixels whose gradient magnitude is large.

2.4.1. Gradient-Based Detection

Given an image $I(x, y)$, the gradient is the vector $\nabla I = (\partial I/\partial x,\ \partial I/\partial y)$, whose two components measure the rate of intensity change along the horizontal and vertical axes. Its magnitude

$$M = \|\nabla I\| = \sqrt{\left(\frac{\partial I}{\partial x}\right)^2 + \left(\frac{\partial I}{\partial y}\right)^2} \tag{2.1}$$

provides a scalar measure of edge strength at each pixel. This magnitude stays small on uniform surfaces and grows large at boundaries. Moreover, it merges both directional derivatives into a single value, meaning that it detects edges irrespective of their orientation. The partial derivatives are approximated in practice by discrete convolution with finite difference kernels, small filters that estimate local variation from the difference between neighbouring pixel values.

The Sobel operator is a widely used kernel of this kind. It estimates the two directional derivatives with a pair of 3×3 kernels [14]:

$$G_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}, \qquad G_y = \begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}. \tag{2.2}$$

Specifically, G_x responds to horizontal intensity transitions and thereby detects vertically oriented edges, while G_y measures vertical transitions and identifies the horizontal ones. Each kernel is convolved with the image separately and the two responses combine pointwise into the gradient magnitude

$$M = \sqrt{(G_x * I)^2 + (G_y * I)^2}. \tag{2.3}$$

Each Sobel kernel carries weights of ±2 on the two coefficients adjacent to its centre [14]. Notably, this doubling averages the samples nearest the centre before the difference is taken. The Sobel operator consequently smooths slightly as it differentiates and its robustness to noise improves.

A smaller alternative is the Roberts Cross operator, which differentiates along the two diagonals of a 2×2 neighbourhood [15]:

$$R_1 = \begin{bmatrix} +1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad R_2 = \begin{bmatrix} 0 & +1 \\ -1 & 0 \end{bmatrix}. \tag{2.4}$$

Its edge magnitude is commonly approximated as $|R_1 * I| + |R_2 * I|$, which avoids the square root of Equation (2.3) [14]. Furthermore, the operator has no centre sample and reads four pixels instead of nine. Its response is therefore more local and cheaper than that of Sobel, at the cost of greater sensitivity to noise.

Both operators ultimately act on the image intensity function $I(x, y)$ rather than on the three colour channels of a rendered frame. Applying them per channel would allow a difference in hue carrying no change in brightness to register as an edge. Gradient computation is accordingly performed on a single luminance channel formed as a weighted sum of the red, green and blue components:

$$L = 0.2126\,R + 0.7152\,G + 0.0722\,B, \tag{2.5}$$

following the coefficients defined in ITU-R Recommendation BT.709 for high-definition video [16]. The weighting reflects the differing sensitivity of the human visual system to each primary. Differentiating this channel aligns edge strength with perceived brightness contrast as opposed to an arbitrary numerical difference between colour channels.

2.4.2. Screen-Space Normal Derivatives

The kernels above require several texture samples per pixel. The screen-space derivative offers a cheaper alternative for any value already interpolated across the surface. It is exposed by the `ddx` and `ddy` intrinsics, which return the partial derivative of a value with respect to the screen-space x and y coordinates and are available only inside a pixel shader [17]. This derivative costs no texture sample.

Applying these intrinsics to the world-space surface normal $\mathbf{N}$ yields

$$M_N = \sqrt{\|\mathtt{ddx}(\mathbf{N})\|^2 + \|\mathtt{ddy}(\mathbf{N})\|^2}, \tag{2.6}$$

which responds to abrupt changes in surface orientation such as creases, mesh boundaries and regions of high curvature. Equally important, the operator measures change per pixel rather than per unit of surface area. A crease of fixed world size spans fewer pixels as the camera retreats. And the orientation change recorded across one pixel step grows with viewing distance. The extent of the detected edge therefore varies with the viewpoint instead of remaining tied to the surface.

2.4.3. Gaussian Prefiltering

The operators introduced so far are sensitive to small fluctuations in their input. Differentiation amplifies these high-frequency components. Therefore, a noisy input yields an unstable gradient. Marr and Hildreth established that smoothing must precede differentiation and they adopted the Gaussian as the unique filter that optimally balances localisation in space against localisation in frequency [18]. This smoothing need not run as a separate step, since both convolution and differentiation are linear operations. The smoothed gradient may be written as

$$\nabla(G_\sigma * I) = (\nabla G_\sigma) * I, \tag{2.7}$$

where G_σ is a Gaussian kernel with standard deviation σ. Canny built an influential detector on this principle. The detector smooths the image with a Gaussian first. It then computes the gradient of the smoothed result. Non-maximum suppression then thins the gradient response. Hysteresis thresholding follows and keeps connected contours [19]. The Gaussian also supports a second edge operator, built on subtraction rather than differentiation. Subtracting a strongly blurred copy of an image from a lightly blurred one gives the difference

of Gaussians (DoG). Marr and Hildreth established that this difference behaves as a second-derivative operator, and they used it to localise edges [18]. Winnemöller et al. weighted the coarser blur before the subtraction, and this weight sets how much centre-surround difference an edge must show [20]. A later compendium named that weighted form the extended difference of Gaussians (XDoG) and credits the modification with a much wider range of stylistic effects [21].

Moreover, the Gaussian is separable. A 2D convolution decomposes into one horizontal pass and one vertical pass and this is what makes the operator inexpensive. The saving depends on storing the horizontal result so that neighbouring pixels can share it. However, a shader that must produce its output in one execution has nowhere to store it. Every fragment consequently repeats the full 2D neighbourhood itself and the redundant work grows with the size of the kernel.

2.5. Edge-Preserving Filtering

Gradient operators mark boundaries by drawing lines over an otherwise unchanged surface. A second family abstracts the surface itself and the Kuwahara filter is the member of that family used in this thesis. It removes detail even inside high-contrast regions. Shape boundaries are still protected where contrast is low. The level of abstraction thus stays roughly uniform across the image [22]. The filter divides the neighbourhood around each pixel into N overlapping subregions and computes the mean μ_i and variance σ_i^2 of each. It then emits the mean of whichever subregion is most homogeneous:

$$F_{\text{Kuwahara}}(x, y) = \mu_j, \qquad j = \arg\min_i \sigma_i^2. \tag{2.8}$$

Any subregion straddling a boundary carries a higher variance and loses this comparison. Interior regions therefore flatten towards uniform fills while boundaries stay sharp and the result resembles painterly brushwork. The classical formulation sets $N = 4$, and its isotropic rectangular subregions overlap their neighbours by a single pixel. They sit in fixed positions, meaning that no orientation is favoured over another [23], [24].

Nevertheless, Kyprianidis et al. recovered that orientation in an anisotropic formulation that aligns the sampling window with local image structure, estimated through the structure tensor. This tensor is calculated directly from the RGB values of the input rather than from a luminance channel [22]. For an image $\mathbf{f}$ with colour-valued partial derivatives $\mathbf{f}_x$ and $\mathbf{f}_y$, the smoothed tensor is the 2×2 matrix

$$\mathbf{S} = G_\rho * \begin{bmatrix} \mathbf{f}_x \cdot \mathbf{f}_x & \mathbf{f}_x \cdot \mathbf{f}_y \\ \mathbf{f}_x \cdot \mathbf{f}_y & \mathbf{f}_y \cdot \mathbf{f}_y \end{bmatrix}, \tag{2.9}$$

where each dot product runs over those channels and the Gaussian G_ρ smooths every component so that the estimate reflects a neighbourhood rather than one pixel. Its eigenvectors give the dominant local orientation and the eigenvalues indicate how strong that orientation is. An elliptical kernel derived from the two aligns the brush marks with salient contours. The ellipse is partitioned into N sectors in place of the rectangles, and typical values are $N = 4$ or $N = 8$ [22]. That smoothing step is itself a convolution across the image, so it requires a render pass of its own.

3. Related Work

This chapter examines the prior work most relevant to the central research questions of this thesis. That examination spans three established areas. The first is the perceptual and affective consequences of avatar realism and the uncanny valley phenomenon. The second is trust, credibility and social presence in virtual agent interactions. The third is the technical design space of NPR for animated human figures and the viewer responses that different styles elicit. The chapter concludes by identifying the specific gap that this thesis addresses.

3.1. The Uncanny Valley and Avatar Realism

The uncanny valley hypothesis was first articulated by Mori. He proposed that human comfort towards an artificial figure increases with its human likeness. This rise holds only up to a critical threshold. Beyond it, subtle imperfections cause a sharp drop in comfort. The drop persists until the figure appears almost fully human. Only then is comfort recovered [4]. This relationship is captured in the curve shown in Figure 3.1.

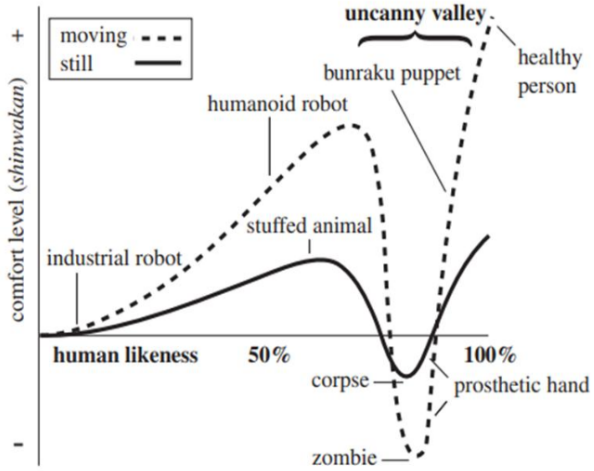


Figure 3.1.: The vertical axis represents comfort level (*shinwakan*). The horizontal axis represents degree of human likeness. The solid curve describes affective response to still figures, while the dashed curve describes figures in motion [4].

Mori plotted separate curves for still and moving figures. Both follow the same general shape, but the curve for moving figures drops far deeper into negative comfort. Moreover, the reference points that Mori placed inside the valley, such as a corpse, a zombie and a prosthetic hand, share a common characteristic. Each closely resembles a human at first glance. The eerie response arises only once the discrepancy registers [4]. Hence the place where that drop occurs is the most problematic region of the curve.

Mori had derived his curve from anecdote and no study had yet reproduced it with stimuli spanning the range of human likeness. MacDorman and Ishiguro closed that gap and advanced the uncanny valley from a qualitative hypothesis into an empirically tractable research programme. They constructed that range of stimuli photographically rather than with robots.

Morphed image sequences ran from a mechanical humanoid robot, through an android, to a photograph of the person the android was modelled on. An android in this sense is a robot designed to be indistinguishable from a human in appearance and behaviour, as shown in Figure 3.2. The 45 participants rated all 31 images on nine-point scales of human likeness and familiarity. They then marked the images that felt eerie and rated those on a ten-point scale [25]. Familiarity fell into a valley in both sequences and its trough lay between the humanoid robot and the android. Perceived human likeness rose steadily across the same span while eeriness peaked close to where familiarity was lowest. This study reproduced Mori's

Figure 3.2.: An android (left, Repliee R1) alongside a mechanical-looking humanoid robot (right, Eveliee P1) [25].

observations from measured data [25]. Consequently, any
near-human avatar risks the same response if its rendering places it in that trough.

This risk became the subject of direct empirical study. McDonnell et al. investigated the perceptual consequences of render style choice directly. Separate groups of participants rated still images and video of the same animated character, rendered across 10 distinct styles spanning the range from photorealistic to fully stylised (Figure 3.3). The results exhibited a valley-shaped pattern across appeal, reassurance and trustworthiness alike. Four of the most abstract styles and the two most realistic ones scored significantly higher on all three measures than styles in the mid realism range. The authors concluded that styles at both ends of that range are viable. Interestingly, cartoon styles were rated friendlier than realistic ones. However, appeal does not follow from abstraction alone, since one of their abstract styles ranked among the least appealing [2]. Taken together, these findings suggest that stylised rendering can avoid the perceptual and trust costs concentrated in the mid realism range without requiring full photorealistic fidelity.

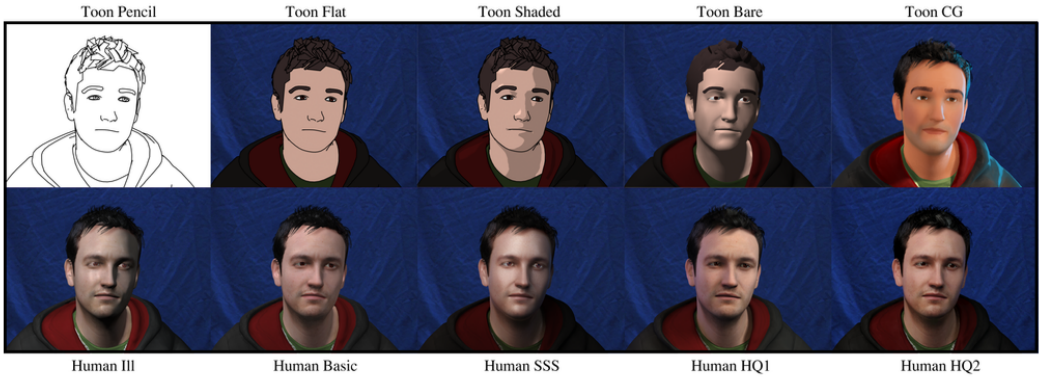


Figure 3.3.: The ten render styles compared by McDonnell et al., ranging from abstract to realistic. Top row, from left to right: Toon Pencil, Toon Flat, Toon Shaded, Toon Bare and Toon CG. Bottom row: Human Ill, Human Basic, Human SSS, Human HQ1 and Human HQ2 [2].

Furthermore, Dill et al. validated the uncanny valley response specifically for CG characters drawn from contemporary commercial films and games. Their familiarity ratings followed Mori's predicted curve, with the strongest negative response in the near-human range. Motion deepened the valley most visibly for two of the ten characters tested. Participants also named the eyes and the mouth as the facial regions producing the most discomfort [26]. This confirmation that the valley extends to digital characters from popular media, not only to physical robots or androids, motivates its consideration across all virtual avatar contexts. Modern avatar systems fall within this same problematic region of the uncanny valley curve. They achieve enough visual realism to raise viewers' expectations, but cannot fully deliver on them in motion timing, expressions or gaze behaviour. As a result, a photorealistic avatar may undermine the very trustworthiness it was designed to convey.

Conversely, Seymour et al. tested that prediction directly using a professionally produced, real-time photorealistic avatar in a field study at a conference. None of the 18 guests who interacted with it displayed an aversive response. Observers rated it as more trustworthy than a caricature counterpart, with greater affinity and stronger preference [27]. This result suggests that a sufficiently high-fidelity, real-time avatar can escape the uncanny valley entirely rather than merely offset its cost with other advantages.

A larger and more recent test of that possibility comes from a network meta-analysis by Tao et al. It pooled 13 independent studies and 2,343 participants across three realism levels. High realism avatars were rated significantly more trustworthy than low realism avatars, whereas medium realism differed significantly from neither. The cumulative ranking placed medium realism as the most eerie, although none of the eeriness comparisons reached significance [28]. The pooled evidence therefore favours realism for trust and leaves the eeriness pattern unresolved. Overall, the valley reproduces reliably across robots, morphed photographs and CG characters, yet these studies disagree on whether stylisation or full realism is the safer route around it.

3.2. Trust and Credibility in Virtual Agents

Trust in virtual avatars is multidimensional, not a single attitude. Mayer et al. established this structure by decomposing perceived trustworthiness into ability, benevolence and integrity. They further separated trustworthiness as an attribute of the trustee from trust as an attitude held by the trustor [10]. McKnight and Chervany subsequently carried that distinction into a typology separating trusting beliefs from trusting intentions in technology-mediated contexts [11]. Both frameworks are set out in Section 2.3. Their consequence for the present review is that trust should not be reduced to a single score, as rendering style could plausibly affect ability, benevolence and integrity differently.

Trust measurement has since been examined more broadly across the literature. Alipour et al. conducted a systematic review of 53 empirical studies to determine whether and how the uncanny valley effect influences trust. They found three different operationalisations of trust across this literature. The first is a direct rating of trust or trustworthiness. The second is a related construct such as competence or credibility. The third is a trust-related intention such as advice following. The reported relationship between the uncanny valley effect and trust was inconsistent across these studies, with some reporting a direct negative effect and others finding no relationship at all [1]. This inconsistency suggests that the effect of appearance on trust may depend on other perceptual factors.

One such factor is anthropomorphism. Nowak and Biocca manipulated both the anthropomorphism of a partner's virtual image and its perceived agency. Of these two factors, only anthropomorphism affected presence. Its effect also ran counter to their own prediction, because the less anthropomorphic image produced more copresence and social presence than the highly anthropomorphic one. The authors attributed this to expectation, since a more anthropomorphic image raises what the viewer anticipates and presence falls when the interaction does not meet it [29]. This pattern is structurally analogous to the uncanny valley, where affinity likewise ceases to scale linearly with human resemblance and inverts before recovering at full fidelity. Nowak and Rauh later carried this question into credibility. They asked 255 participants to rate a series of avatars. Avatars perceived as more anthropomorphic were also rated as more credible and were chosen more often for self-representation [9]. Therefore, avatar design choices function as meaningful social signals.

Canales et al. tested this signalling role directly by manipulating avatar stylisation and measuring its effect on multi-component trust. Participants were presented with virtual doctors rendered at three stylisation levels: realistic, semi-realistic and caricatured. Trust was measured after each doctor interview, both as a direct rating and across the ability, benevolence and integrity dimensions. In addition, participants were asked which doctor they would consult for a follow-up. This provided a behavioural indicator alongside the self-report scale. Together these span all three measurement approaches in the Alipour et al. taxonomy [1]. The results indicated that self-reported trust ratings were broadly equivalent across all three style levels. However, the authors found that the semi-realistic condition attracted the highest rate of behavioural selection [5]. This suggests that moderate stylisation can match or exceed the behavioural trust of full photorealism. Gao et al. compared a stylised agent against recorded video of the real speaker it was modelled on [30]. The two did not differ significantly on a direct trust rating, but the real speaker was rated higher on ability, benevolence and

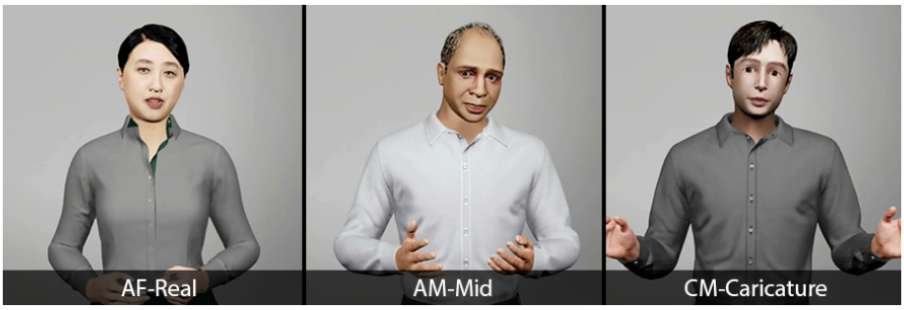


Figure 3.4.: The three avatar stylisation levels used by Canales et al.: realistic (AF-Real), semi-realistic (AM-Mid) and caricatured (CM-Caricature) [5].

integrity. Overall, these studies indicate that a single self-report rating does not fully capture how render style shapes trust.

Moreover, Dubosc et al. applied the same comparison to eeriness rather than trust. They conducted a VR face-to-face interaction study and found that stylised avatars were rated significantly less eerie than photorealistic avatars. Trustworthiness also trended higher for the stylised avatars, though this difference did not reach statistical significance [31]. This pattern suggests that stylisation reduces the uncanny valley response without compromising social credibility. The mechanism behind that response has itself been reviewed. Cihodaru-Ştefanache and Podina examined 29 empirical studies on embodied conversational agents and reported the perceptual mismatch hypothesis as its best supported explanation [32]. Uncanniness on that account arises when an agent's features are perceived at inconsistent levels of anthropomorphism.

Testing this pattern reliably requires measurement across more than a single trust rating. The Godspeed Questionnaire Series, developed by Bartneck et al., offers one such option [12]. Even though the series was originally developed for human-robot interaction, it has since been widely adapted for virtual avatar studies. Section 2.3 sets out its structure and the subscale adopted in this work.

On the other hand, none of the Godspeed subscales measures eeriness [12]. MacDorman and Ishiguro had measured that response directly, on a scale separate from human likeness and familiarity [25]. Ho and MacDorman later refined this into a validated instrument with four indices: perceived humanness, warmth, eeriness and attractiveness. Their validation established that eeriness and human likeness are orthogonal [13]. Separating the two matters for interpreting any reduction in eeriness, since a drop could reflect reduced discomfort or merely reduced human likeness.

3.3. NPR for Virtual Humans

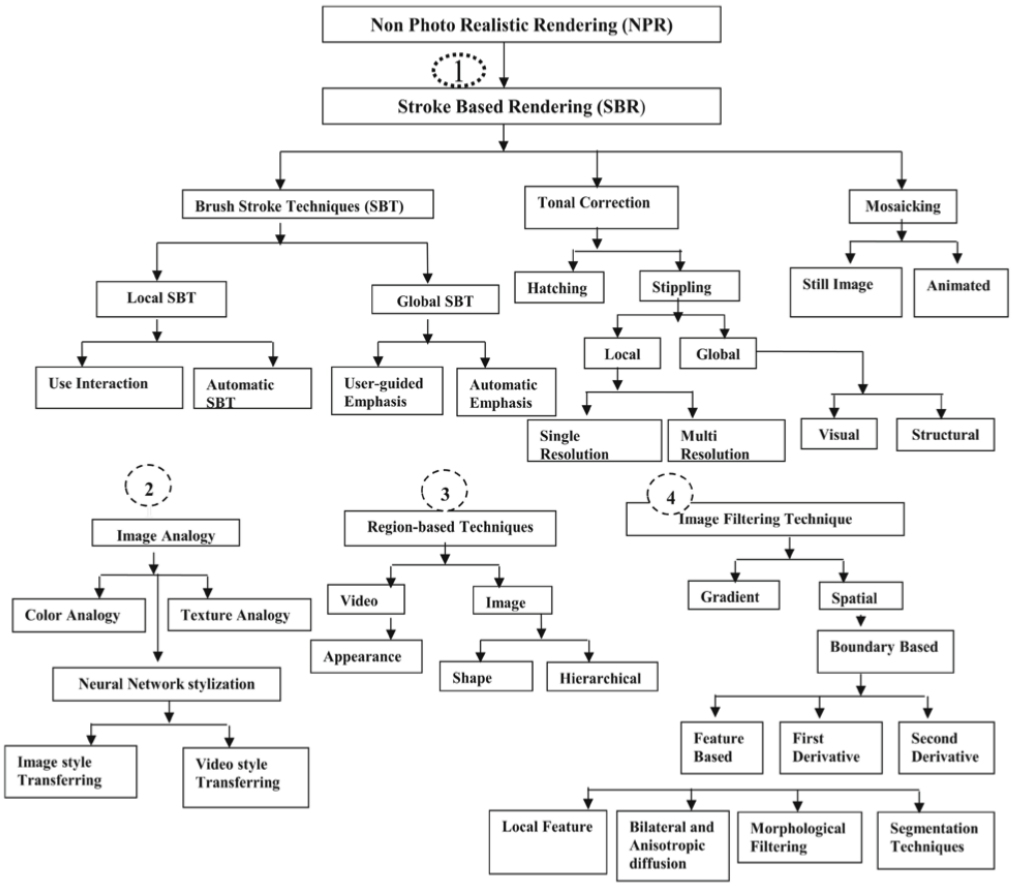
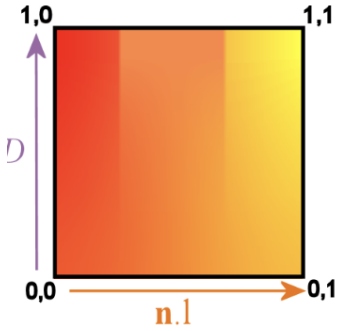


Figure 3.5.: The four-family classification of NPR techniques surveyed by Pavan Kumar et al.: stroke-based rendering, image analogy, region-based techniques and image filtering [33].

NPR transforms an image into abstracted, artistic or stylised output instead of reproducing it exactly. Pavan Kumar et al. surveyed these approaches under four established categories: stroke-based rendering, image analogy, region-based techniques and image filtering [33]. Stroke-based rendering builds an image from discrete brush strokes that vary in size, colour, texture and orientation. Placing these strokes typically requires a dedicated analysis pass to decide where and how each stroke should be oriented. That requirement is a drawback for pipelines with only one shading stage per pixel. Image analogy instead transfers colour and texture style from an example pair of source and target images onto a new image. This transfer depends on a precomputed exemplar or a patch-matching search rather than information available within a single frame. By comparison, region-based techniques segment an image into distinct regions before abstracting each one individually. Segmentation is an iterative process that considers the whole image. Its result cannot be computed within a single, independently executed per-pixel operation. Finally, image filtering applies local mathematical operations, such as gradients or bilateral smoothing, directly to the image to suppress detail while preserving structure [33]. These operations examine only a small neighbourhood around each pixel. As a result, they remain computable within a single shading pass. Figure 3.5 summarises this classification.

However, this survey's scope is limited to 2D image processing. Image filtering crosses that boundary where the other three do not, since each of them requires a pass that a single shading stage cannot provide. Other real-time techniques fall outside Pavan Kumar et al.'s four families, because they act on geometry and lighting rather than on a finished image. Toon shading is one such technique. It quantises the continuous diffuse lighting term into a small number of discrete bands. Consequently, this approach produces flat-shaded regions separated by hard edges and gives characters a cartoon appearance, as depicted in Figure 3.6a. That separation depends on a single input, the lighting term [34]. This dependency leaves the ramp unable to respond to any other surface property.

(a) Toon shading [34]



(b) The X-Toon ramp [35]

Figure 3.6.: (a) A cartoon character rendered with toon shading, showing flat tonal bands separated by hard edges. (b) The 2D X-Toon ramp, where the horizontal axis carries the diffuse lighting term $\mathbf{n} \cdot \mathbf{l}$ and the vertical axis carries the tone detail value D.

Barla et al. extended toon shading's quantisation into a richer parameter space with X-Toon. It retains the lighting term on one axis of the ramp and adds a second axis for tone detail, driven by a view-dependent attribute such as depth or surface orientation [35]. Figure 3.6b shows the resulting 2D ramp. This enables a 2D lookup that encodes artistic intent unavailable in a 1D quantisation step.

Silhouette rendering is a second such technique. It derives line structure from geometry rather than from the lighting term. Raskar and Cohen achieved this by processing the mesh twice from the same viewpoint. Initially, the front-facing polygons are drawn in the usual way. The back-facing polygons are then rendered and translated towards the camera. Thus, part of this back-facing geometry appears in front of the occluding front-facing surface and the visible rim left at this boundary forms the outline, as Figure 3.7 illustrates. The outline emerges from geometry alone and requires no precomputed adjacency information [36].

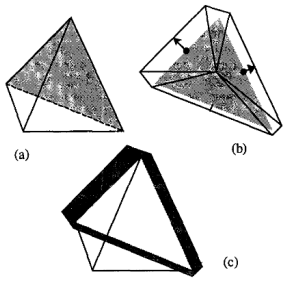


Figure 3.7.: Silhouette rendering: (a) front-facing polygons, (b) back-facing polygons translated towards the camera, (c) the resulting outline [36].

That outline is only one of several line types that convey shape. In order to detect those lines, Isenberg et al. surveyed the available algorithms and reported that the literature groups silhouette lines together with creases, borders and self-intersections under the heading of feature lines. The silhouette lines themselves split further, and the survey calls the interior ones internal silhouettes [37]. This distinction matters for avatar rendering, where expressive detail such as facial features, clothing folds and skin creases sits inside the outline. However, lines that appear only in the texture fall outside this classification, since each type it names is defined on the mesh geometry.

The same survey identifies a further family of algorithms that addresses this limitation by working on the rasterised output rather than on the mesh. These image-space methods, an instance of the image filtering category described above, extract discontinuities from the buffers that conventional rendering already produces and represent the result as features in a pixel matrix [37]. Saito and Takahashi introduced one such buffer, the G-buffer. It is a set of channels that preserves per-pixel geometric properties such as depth and surface normals. The authors recovered profile and internal edges from the depth channel. Sobel was recommended for the profile, and a second-order operator produced the internal edges [38]. Nienhaus and Döllner later applied the same analysis to the normal channels and detected crease edges from their discontinuities in real time [39]. Isenberg et al. favoured the combination of both channels on visual grounds [37].

However, neither channel registers a boundary that exists only as colour, since a change of pigment leaves depth and surface orientation alike untouched. Colour-buffer edges are treated in the survey as a liability, because shading and texture variation cause detections that are not silhouettes [37]. That liability becomes both an objective and a challenge once the target is a stylised avatar rather than a silhouette. Clothing patterns, facial contours and painted surface detail are the features that need marking here. Detecting them in the colour channel requires a gradient operator. The Sobel filter is preferable for that purpose, because its kernel suppresses noise better than the simpler alternatives [14]. Yet this operator responds to high-frequency content as strongly as to the boundaries it is meant to mark. Marr and Hildreth placed a Gaussian ahead of the derivative operator [18]. Therefore, prefiltering of this kind is the usual precaution against that sensitivity.

Nonetheless, a single gradient operator produces noisy, discontinuous edges even after prefiltering. A preprint by Lino Roshaan addressed this by splitting edge detection into three layers. Depth differences give the silhouette layer. The Scharr and Kirsch gradient operators trace the structure layer. An extended DoG (XDoG) operator recovers the texture layer. Max-pooling merges the three. Figure 3.8 shows each one alongside the composited result. A luminance-driven gain raises the structure layer's sensitivity in dark regions and holds line quality steady without manual retuning. The framework runs as a real-time post-processing shader and needs access to both the colour and the depth buffer [40].

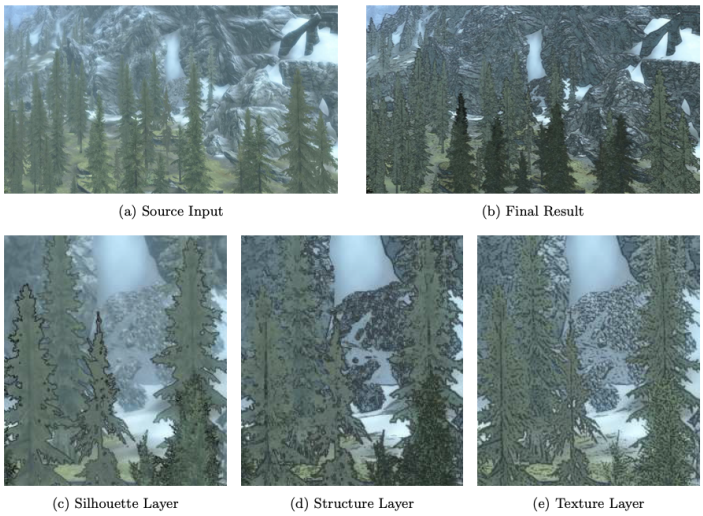


Figure 3.8.: The three-layer edge hierarchy of the AHEAD framework: (a) the source frame and (b) the composited result, above the three layers that produce it. (c) The silhouette layer comes from depth differences, (d) the structure layer from gradient operators and (e) the texture layer from the extended DoG operator [40].

However, neither NPR abstraction nor image filtering is confined to feature detection. Another class of operators, such as the Kuwahara filter, reshapes the texture itself rather than outlining the structures within it. Pavan Kumar et al. grouped this filter with anisotropic and bilateral filtering as NPR filtering techniques. It reduces shape, edge and colour detail in a single pass, and the prominent features of the image remain intact [33]. It is also described as the classical member of the edge-preserving filters used for image abstraction [23]. The anisotropic generalisation of Kyprianidis et al. shapes the sampling region according to local feature geometry. The flattening then follows those features instead of cutting across them, and the surface takes on a painted appearance [22]. Two artefact problems survived that generalisation, one in noisy regions and one across broad areas of even tone. A later paper by Kyprianidis addressed both. Thresholding the sector weights removes the first, and a multiscale scheme suppresses the second. That scheme carries orientation estimates and partial filtering results together through the levels of a smoothed pyramid, and it still runs fast enough for graphics hardware to apply it interactively [24]. Both formulations derive their kernel shape from a structure tensor, and that computation needs a render pass of its own [22], [24]. Figure 3.9 compares both variants against the unfiltered input.

(a) *Original image* (b) *Anisotropic Kuwahara filter* (c) *Proposed method*

Figure 3.9.: Painterly abstraction produced by Kuwahara filtering. The first image is the unfiltered input and the second is the single-scale anisotropic filter. The third is the multiscale variant [24].

Taken together, this technical catalogue establishes what NPR can render, but not whether the result changes how an avatar is perceived. Psychophysical and VR experiments have examined that question directly. Wallraven et al. compared brush, cartoon and illustrative stylisation on sequences of facial expression. Recognition accuracy held under the cartoon variant but fell under the other two when the sequences were recorded video. Computer-generated animations of the same expressions reversed that ordering, and the illustrative variant achieved marginally higher recognition than the cartoon one [41]. Hence, the outcome depends on which technique produces the abstraction, not on its degree alone. Wisessing et al. later demonstrated that toon shading reduces the perceived intensity of facial emotional expression while leaving recognition accuracy intact. The authors attribute that effect to the loss of shading information, and this may conceal facial form and features that emotion perception depends on [42]. Characters rendered in this style accordingly convey the correct emotional content without the perceptual amplification that smooth CG shading provides. A follow-up study identified lighting parameters as a further modulator of appeal and eeriness. Higher brightness made characters more appealing regardless of the emotion portrayed. Darker shadows reduced the appeal of cartoon characters and left the appeal of realistic ones unaffected. They also raised eeriness, though that effect came mainly from the male character [43]. Together these findings reveal that render style and lighting both shape how a character is received, independently of the emotion being performed.

Those studies measured how a character is read from the outside. Zibrek et al. instead placed observers inside the scene and ran a large between-group VR study with 622 participants. That study compared three render styles from photorealistic to sketch across three emotional scenarios (Figure 3.10). The results of that experiment showed that render style left social presence unchanged and produced no eeriness difference between any two individual styles. Photorealism shifted which emotions the character evoked rather than strengthening them [44]. This challenges the assumption that rendering quality scales linearly with social utility. Zibrek and McDonnell extended this investigation to embodied VR, where participants inhabited a virtual body rather than observing from outside. The results indicated that photorealism significantly raised

both self-reported social presence and place illusion, the sense of physically inhabiting the virtual space. A rendering change can therefore shift what people report without shifting what they do [45].

Figure 3.10.: The three render styles compared by Zibrek et al., from left to right: realistic, simple and sketch, applied to both the character and the room [44].

Figure 3.11.: Shape axis, s0 to s4 [46].

Render style is not the only dimension a designer controls. Zell et al. separated a character's shape from its material and varied each independently across five stylisation levels. Shape governed perceived realism, whereas material governed appeal and eeriness. Figure 3.11 shows the shape axis, which holds the material constant while the geometry moves from a cartoon head at s0 to a scanned likeness at s4. Furthermore, Figure 3.12 shows the material axis, where the geometry stays fixed and the shader and texture move across the same five levels. Strong mismatches between the two levels also produced very unappealing characters. A stylised material on a realistic shape was among those combinations [46]. The uncanny response depends in turn on the level of material stylisation and on its consistency with the shape.

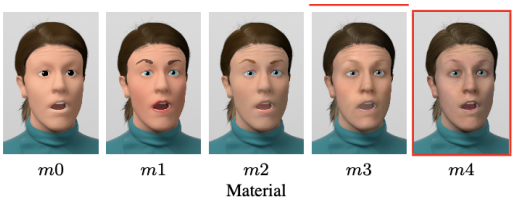


Figure 3.12.: Material axis, m0 to m4 [46].

Each of these studies varied how the character was rendered and arrived at a different verdict. Wisessing et al. observed that stylisation softened expressive intensity. Zibrek et al. found presence and eeriness untouched by render style, whereas Zell et al. traced eeriness to material stylisation. Those verdicts depend on which property was varied and which response was measured.

3.4. Research Gap

Prior work on both NPR and avatar trust perception has assumed open pipeline access. That work relied on custom assets and research engines rather than production SDK systems. No published work has been identified that builds the same technique set across the avatar platforms that supply consumer applications and differ in rendering architecture. What each architecture permits and how a technique must change to survive it remain unknown. This gap is addressed by the technical contribution of this thesis (RQ1).

Moreover, NPR techniques have been developed and compared extensively, but largely on images, scenes and generic 3D shapes rather than on virtual avatars. Even though Wisessing et al., Zibrek et al. and Zell et al. did study human figures, each varied coarse stylisation levels rather than specific operators [42], [44], [46]. Wallraven et al. did compare named techniques for expression recognition, but applied them to the rendered frames as a post-process rather than to the avatar's own shading [41]. That distinction matters most on the face, since Wisessing et al. showed that shading style alters the perceived intensity of an expression [42]. Equally, the discomfort that CG characters provoke is not uniform across the humanoid body. Dill et al. revealed that their participants traced it predominantly to the eyes and the mouth [26]. Therefore, which technique or combination best balances abstraction against expressive legibility on avatar faces remains largely unstudied. This gap is addressed by the iterative technique development and selection described in the following chapters (RQ2).

By contrast, whether NPR abstraction on virtual avatars helps or harms perceived trust has been tested directly, and the results disagree. McDonnell et al. found appeal and trust ratings lowest in the mid realism range, which stylisation avoids [2]. Canales et al. report no trust penalty for stylised avatars relative to photorealistic ones [5]. On the other hand, a network meta-analysis by Tao et al. shows high realism avatars receiving significantly higher trustworthiness ratings than low realism ones [28]. Thus, no study to date measures both trust and eeriness for an NPR pipeline applied to both the shading and the texture of one photorealistic avatar. The Canales study came closest, but it compared coarse style categories and did not treat eeriness as a dimension separate from overall stylisation [5]. Nor has any reviewed study set a real human speaker, a photorealistic avatar of that speaker and an NPR rendering of the same avatar against one another. This gap is addressed by the user study of this thesis (RQ3).

This thesis addresses all three. Eight NPR techniques were developed in sequence and integrated on Mixamo, the Meta Avatars SDK and Avaturn, platforms that expose different degrees of rendering control. A parameter adjustment study then selected the composite carried into the evaluation. A within-subjects video study measured what that composite did to perceived eeriness and trust.

4. Methodology

This chapter describes the research questions guiding this work, the three avatar platforms used as development and evaluation targets, the NPR stylisation techniques developed across those platforms and the design of the user study that evaluates their perceptual consequences.

4.1. Research Questions

Three research questions structure the contributions of this thesis.

RQ1: How can NPR stylisation techniques be adapted to avatar platforms with different rendering architectures and what capabilities and limitations does each platform impose?

This question is primarily technical. It asks how NPR techniques can be adapted for distinct platforms that differ substantially in mesh topology, shader accessibility and pipeline structure. It also asks what each platform permits and what it prevents, since the architectural constraints of each platform determine which NPR techniques are realisable and how they appear in practice.

RQ2: Which NPR technique or combination of techniques best balances visual abstraction against facial and expressive legibility under production rendering constraints?

This question drives the iterative technique development documented in this chapter. Each version introduces a different abstraction approach and addresses a limitation identified in its predecessor. The overall goal is a configuration that achieves a stable, readable stylised appearance without an excessive cost to facial and expressive legibility. The visual quality assessment and the parameter adjustment pilot study provide the empirical basis for this selection.

RQ3: How does NPR stylisation affect perceived eeriness and trustworthiness and can an NPR avatar reach the trustworthiness of a photorealistic avatar and of a real human speaker?

Uncanny valley theory places a real human speaker at the peak of the affinity curve, since maximal human likeness corresponds to maximal comfort [4]. A photorealistic avatar instead sits in the trough immediately before that peak, where negative affect is strongest. That negative affect is what eeriness measures, so how eerie a character feels indicates where on the curve it sits. The question spans both the discomfort a stylised avatar evokes and the trust it receives. If the composite selected under RQ2 relocates the avatar onto the stable rising slope below the trough, it should register as less eerie than the photorealistic avatar that remains in it. The resulting trust level may then approach that of a real human speaker, even though the avatar's human likeness remains lower. RQ3 tests this possibility directly, with the real speaker serving as the reference point against which both avatar renderings are compared. The hypotheses that formalise these comparisons are stated in Section 4.8.1.

4.2. User Study Overview

A within-subjects study was conducted to address RQ3. Participants viewed video stimuli of an avatar delivering a spoken recommendation across three everyday decision scenarios. Every participant saw three conditions: a real human speaker (C1), a photorealistic avatar (C2) and the same avatar rendered with an NPR composite (C3). These labels are used throughout the thesis. The study used prerecorded video rather than a live VR session, because C1 has no VR equivalent. Video additionally fixes a single recorded performance for every condition. The facial blendshape stream that animates C2 and C3 was captured from the real speaker while the C1 video was filmed, meaning that the same gestures play in all three.

Each advisor was rated on perceived eeriness, likeability and three dimensions of trust: ability, benevolence and integrity. These three follow the Ability-Benevolence-Integrity model introduced in Section 2.3. They were measured separately rather than combined into a single score, since a rendering change could plausibly

affect one dimension while leaving the others untouched. A forced ranking of the three advisors closed the session and supplied a behavioural measure alongside the self-report scales.

Likeability and eeriness came from two separate instruments. The Godspeed questionnaire supplied the likeability items [12]. Eeriness was taken from the instrument of Ho and MacDorman instead, since the Godspeed subscales do not separate cleanly from one another [13]. The full design, questionnaire structure and analysis plan are explained in the following sections.

4.3. Avatar Platforms and Rendering Constraints

The NPR techniques developed for this thesis were evaluated across three distinct avatar platforms, each offering a specific combination of capabilities and integration constraints relevant to NPR development. They span the range from unrestricted shader access to a closed injection interface. The sections below describe each in turn.

Mixamo.

The Mixamo platform provides an online library of rigged humanoid characters and skinned animations [47]. These are exported to Unity as standard FBX assets. Because Mixamo avatars are standard skinned meshes without proprietary shader restrictions, all NPR techniques can be implemented directly within the URP rendering pipeline. That freedom makes Mixamo the least constrained of the three platforms and the most appropriate environment for initial exploration and development before adaptation to the more restrictive platforms. However, Mixamo character exports do not include facial rigging and the avatar's face remains static throughout all renders generated on this platform. This absence makes the platform unsuitable for the user study. The Mixamo character (Figure 4.1) still served as the primary prototyping target during the initial development phase, where unrestricted shader access supported rapid iteration and refinement.

Figure 4.1.: Mixamo: baseline avatar for NPR shader development.

Meta Avatars SDK.

Figure 4.2.: Meta avatar: baseline avatar for NPR integration within the Meta Avatars SDK.

The Meta Avatars SDK is Meta's avatar framework for its VR systems, and it handles body tracking internally [48]. The techniques were to be assessed in a VR environment, so the Meta Avatars SDK was examined for the shader customisation capabilities it provides for NPR integration (Figure 4.2). Its built-in body tracking drives the avatar from the wearer's own tracked pose and removes the need for an external motion capture system. The proposed NPR shaders can thus be evaluated on an avatar animated live by the wearer, which neither of the other two platforms supports.

In contrast, custom NPR shaders cannot be integrated through standard URP material slots, because the SDK manages rendering internally. Instead, NPR rendering logic must be injected through hook functions exposed by the SDK at defined points in the shader pipeline. This

constrains both the range of achievable NPR effects and the structural requirements of the shader code. Furthermore, the SDK skins its avatars on the GPU and holds the animated geometry in internal buffers rather than in Unity's mesh data. Rendering stages outside its own shader consequently have no access to the posed mesh. The technical details of this integration are described in Chapter 5.

Avaturn.

Figure 4.3.: Avaturn: baseline avatar for the user study.

Avaturn is an avatar creation service that reconstructs a personalised rigged avatar with facial blendshape targets from three uploaded photographs (Figure 4.3) [49]. The service exports its output as `.glb` files, which implement the glTF 2.0 standard and bundle geometry, PBR materials and skeletal rigging within a single self-contained file [50]. Since Unity does not natively support this format at runtime, an additional import package is required to load the avatar. Once loaded, it functions as a standard skinned mesh with PBR materials and unrestricted access to the URP rendering pipeline. The proposed NPR shaders can thus be implemented directly. Accordingly, Avaturn served as the primary platform for the final user study, since it is the only platform among the three that combines unrestricted shader access with personalised facial reconstruction and full blendshape animation. This complemented the prototyping workflow established with Mixamo and the live tracked animation available on the Meta Avatars SDK.

Table 4.1.: Summary comparison of the three avatar platforms across dimensions relevant to NPR integration.

	Mixamo	**Meta Avatars SDK**	**Avaturn**
Shader access	Full URP, unrestricted	Hook injection only (SDK post lighting)	Full URP, unrestricted
Transparency	Explicit alpha cutout	SDK managed	Explicit alpha cutout
Facial animation	None	Requires Quest Pro hardware	Blendshape animation
Key challenge	No facial rigging	Shader manipulation	Initial Unity import
Strength	Full shader freedom	Native Quest 3 integration	High fidelity facial quality

4.4. NPR Technique Design

NPR in computer graphics encompasses a wide range of abstraction techniques and these vary in the level of abstraction they produce. Neither extreme of that range serves an avatar well, since intermediate realism lowers appeal and heavy abstraction weakens perceived expressive intensity [2], [41]. Selecting an appropriate degree therefore requires calibrating the balance between realism and visual expressiveness.

This calibration was performed through an iterative design process in which several NPR techniques were developed and evaluated. Each version addressed a specific limitation identified in the preceding version and the final selection was validated through a formal user study. Main techniques are labelled V1 through V6. Extended variants carry a decimal suffix, as in V2.2 and V4.2, indicating a directed modification of the preceding version.

Development proceeded in sequence across the three platforms described in the previous section. The SDK's shader hook architecture imposed constraints that shaped several design decisions. The final transfer to Avaturn then verified each technique's visual output across platforms. The following sections describe the

motivation, design rationale and visual characteristics of each approach. Full implementation details are provided in Chapter 5.

4.4.1. Shared Rendering Structure

Most proposed NPR techniques share a common two-pass rendering structure. The first pass generates the silhouette outline, and this method is applied consistently across all three platforms. The second pass computes the NPR surface effect for each technique within the forward rendering pass. Thus, the same architecture underlies the techniques described below. However, V5 and V6 depart from it and run as post-processing passes on the rendered image.

4.4.2. V1 — Geometric Silhouette Outline

Object silhouettes are a defining visual feature of nonphotorealistic rendering [37]. Accordingly, a reliable outline generation technique was required as the foundation for all NPR techniques developed in this thesis. The geometric silhouette method was selected since it produces outlines without an edge detection pass. It generates an outline shell by enlarging back-facing polygons and relies on GPU face culling to form the outline [36]. The method operates across two dedicated stages:

1. **Outline Pass.** The mesh is rendered with front-face culling enabled, displacing each vertex outward along its world-space surface normal by a configurable distance w:

$$\mathbf{P}' = \mathbf{P} + \mathbf{N} \cdot w \tag{4.1}$$

   where $\mathbf{P}$ is the original vertex position and $\mathbf{N}$ is the world-space surface normal. Only the back-facing geometry is rasterised, forming an enlarged shell around the mesh.

2. **Forward-Lit Pass.** The mesh is subsequently rendered with standard back-face culling. The front-facing surfaces overwrite the interior of the previously rendered shell, leaving only the protruding rim visible as a silhouette border with controllable thickness.

(a) Mixamo: geometric silhouette outline.

(b) Meta SDK: geometric silhouette outline.

(c) Avaturn: geometric silhouette outline.

Figure 4.4.: Geometric silhouette outline applied to all three avatar platforms. The slight floor penetration visible at the Meta Avatar's feet in (b) reflects a tracking boundary offset during capture and is not a property of the outline technique.

This approach is well suited to real-time NPR avatar rendering because it requires no offline mesh preprocessing [37]. It also executes entirely within the GPU vertex transformation stage. As a result, its computational cost scales predictably across assets of varying complexity. However, the expanded shell and the original surface may share similar depth values and this can introduce z-fighting artefacts along the silhouette boundary. A depth bias prevents this by shifting the outline geometry slightly towards the camera in clip space.

Despite its stability and low rendering cost, the geometric silhouette method has several limitations when applied to avatar rendering. Figure 4.4 shows the results across all three platforms. Surfaces with degenerate or coincident normals, such as flat planes, yield no visible outline because displacing vertices along identical surface normals does not generate an expanded shell. In addition, the method does not register interior surface boundaries. Consequently, creases between body regions, clothing seams and facial contours receive no edge marking. The expanded shell yields only the outer contour, which in the terminology of Isenberg et al. excludes the internal silhouettes lying inside it [37]. This limitation is shared by geometry-based outline methods and motivates the interior edge detection techniques introduced in V3.

Alpha Cutout for Transparency

Avatars on the Mixamo and Avaturn platforms contain alpha-masked elements such as eyelashes and hair that are represented as flat cards. Their visible shape is defined by discarding fragments below an opacity threshold. Without alpha testing in the outline pass, the expanded hull would be generated from the full geometric extent of each card. This produces solid rectangular borders rather than outlines that follow the visible hair silhouette, as shown in the left column of Figure 4.5. Therefore, alpha testing is applied in the outline pass. The GPU's built-in fragment clip operation discards fragments whose texture alpha value falls below a configurable cutoff threshold α_{cutoff}:

$$\text{if } \alpha_{\text{texture}} < \alpha_{\text{cutoff}} \quad \text{then discard} \tag{4.2}$$

This test executes in both the outline pass and the forward-lit pass to maintain silhouette consistency. Figure 4.5 shows the resulting outline with the threshold applied. Unlike alpha blending, this binary approach does not require depth sorting and introduces negligible additional fragment processing cost. Conversely, the Meta Avatars SDK handles hair and eyelash rendering through its internal pipeline. Transparent fragments are consequently already discarded internally by the SDK, meaning that manual alpha testing is not required during outline generation.

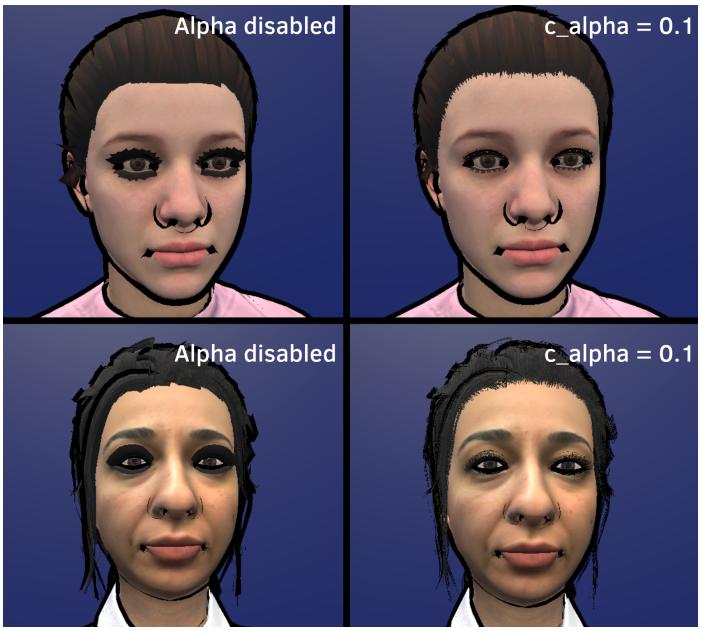


Figure 4.5.: Alpha cutout on the geometric silhouette outline pass (Mixamo, top; Avaturn, bottom). Left: alpha testing disabled. Right: alpha cutout enabled at $\alpha_{\text{cutoff}} = 0.1$.

4.4.3. V2 — Toon Shading

Toon shading achieves stylisation by simplifying the tonal representation of a surface rather than reproducing physically accurate illumination. Discretising the illumination response into uniform tonal regions produces the characteristic flat appearance associated with hand-drawn animation. Correspondingly, the approach is well suited to interactive character rendering [34]. Toon shading was therefore selected as the V2 baseline because it provides a controlled starting point from which each subsequent technique introduces a targeted increase in stylisation. However, implementing this stylisation across the three avatar platforms requires adaptations that account for differences in each platform's rendering architecture.

On Mixamo and Avaturn, the continuous illumination response is converted into discrete tonal regions using the following quantisation function:

$$t = \frac{\lfloor s \cdot n \rfloor}{n} \tag{4.3}$$

where s represents the surface lighting response and n defines the number of shading bands. This operation replaces continuous illumination gradients with discrete tonal transitions, producing the characteristic stepped appearance of toon shading [34]. Figure 4.6 shows the resulting appearance at the default band configuration used in this thesis, with the specific value and its selection criteria given in Section 5.4.2. The resulting tonal structure consists of highlight, midtone, shadow and deep shadow regions. Furthermore, a rim highlight was added to improve visual separation between the avatar and its background where flattened shading reduces depth cues (Figure 4.7). Finally, scene cast shadows were deliberately excluded because they introduce an additional visual variable that could interfere with evaluating the effect of tonal quantisation in isolation.

(a) Mixamo: four discrete shading bands with ambient and rim.

(b) Meta SDK: four discrete shading bands applied to the fully composited colour produced by the SDK renderer.

(c) Avaturn: four discrete shading bands with ambient and rim.

Figure 4.6.: V2 (Toon Shading) across all three avatar platforms.

Unlike Mixamo and Avaturn, where intermediate lighting values are directly accessible, the Meta Avatars SDK exposes only the final composited appearance after its internal rendering process. Consequently, quantisation is applied to the available colour output rather than to the surface lighting response itself. Despite the different implementation, both approaches achieve the same intended stylisation by replacing smooth illumination gradients with discrete tonal regions.

Across all three platforms, V2 successfully establishes a simplified graphic appearance through tonal quantisation. Nevertheless, V2 has two important limitations. First, the distribution of tonal regions remains closely coupled

to the lighting response. This leaves little independent artistic control over the tonal transitions. Second, V2 does not resolve the interior edge problem left open by V1, since it is driven purely by shading. These constraints motivate the subsequent development stages. V2.2 extends the tonal representation through X-Toon shading to provide greater control over the tonal distribution, while V3 introduces interior edge detection to enhance the visibility of local avatar features beyond what tonal quantisation alone can convey.

Figure 4.7.: Rim highlight effect in V2 toon shading.

4.4.4. V2.2 — X-Toon Shading

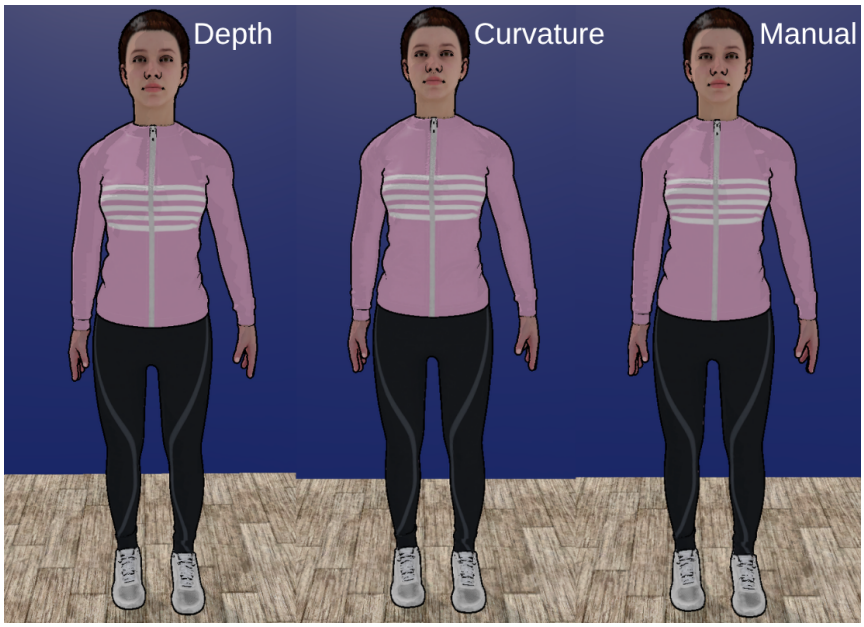
X-Toon shading extends conventional toon shading by replacing the 1D toon ramp with a 2D ramp texture, where the horizontal axis represents the lighting response and the vertical axis controls the abstraction level [35]. This second axis allows the surface appearance to be adjusted independently of the lighting conditions and this provides finer stylisation control. A higher abstraction level reduces local tonal contrast and softens transitions between shading regions towards a flatter, more uniform appearance. Figure 4.8 illustrates the ramp texture used in this work.

Figure 4.8.: 2D ramp texture used for X-Toon shading.

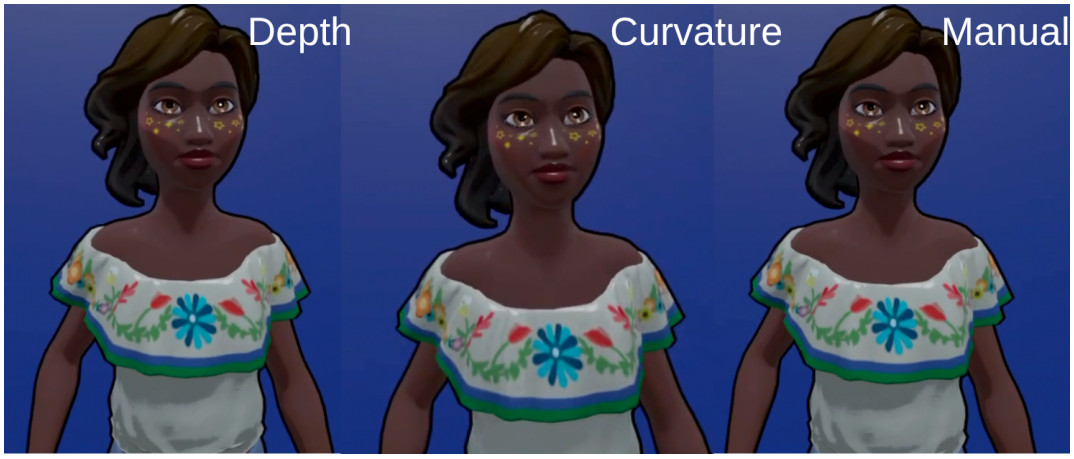
Barla et al. also described a shape detail extension. The extension blends the surface normals of the mesh with those of a simplified copy of the same shape [35]. This blending was incorporated into the Mixamo and Avaturn implementations. It reduced local surface variation and gave more coherent stylised shading.

Barla et al. further described a general attribute mapping mechanism that maps a chosen surface attribute onto the abstraction axis and they demonstrated it with depth-based and orientation-based maps [35]. Three driving modes were evaluated here: depth, curvature and manual control. Depth mode and manual mode coincide on the Meta Avatars SDK, since the avatar is framed head and shoulders and its visible surface spans too narrow a depth range for the depth-driven abstraction value to vary. Curvature mode stays distinct there, because it responds to local surface orientation rather than to distance from the camera. The user study used a fixed camera distance much closer to the avatar than the general platform comparison setup. At that distance the depth and curvature modes produce almost no variation, as Section 4.5 explains in detail. X-Toon produced a softer and more controllable stylised appearance than V2, since the stylisation level can be adjusted independently of the illumination response. It was consequently retained as the toon shading component in all later technique combinations.

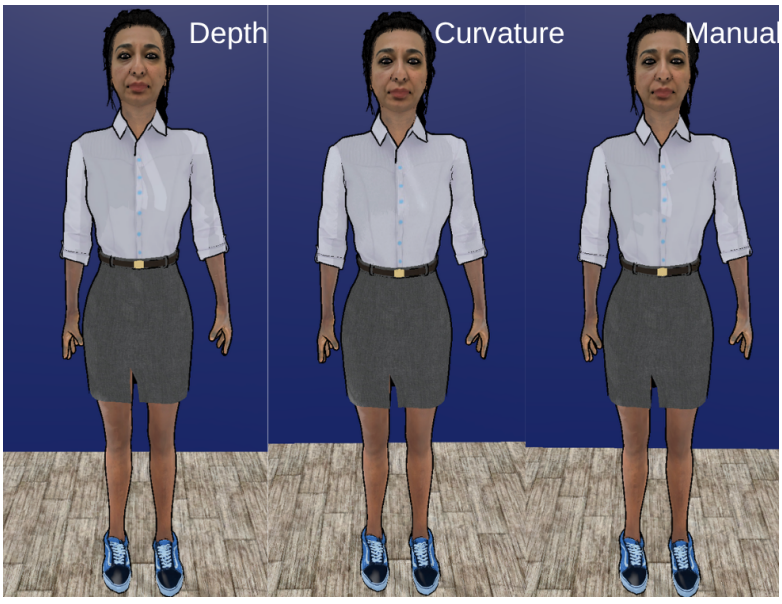
Although the second axis broadens this control, X-Toon still abstracts through shading alone. The interior edge problem inherited from V2 therefore stays open, and it motivates the edge-based techniques introduced in V3.



(a) Mixamo: depth, curvature and manual abstraction modes.



(b) Meta SDK: depth, curvature and manual abstraction modes.



(c) Avaturn: depth, curvature and manual abstraction modes.

Figure 4.9.: V2.2 (X-Toon) across the three avatar platforms. From left to right: depth-driven, curvature-driven and manually controlled abstraction.

4.4.5. V3 — Screen Space Normal Edge Detection

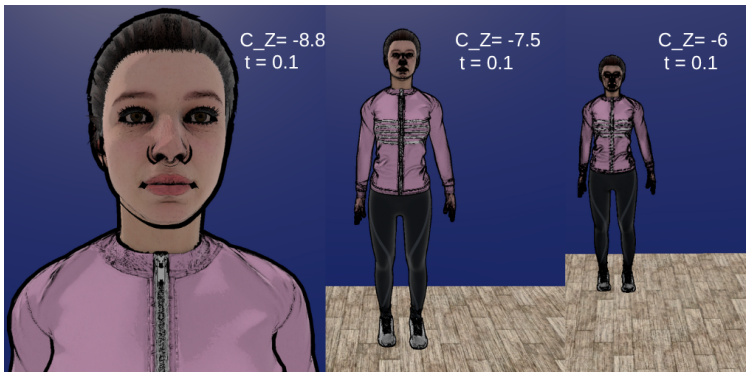
Silhouette extraction techniques can be classified by how they detect edges, into image space, object space and hybrid approaches [37]. Unlike the object-space outline method introduced in V1, image-space techniques analyse per-pixel information generated during rendering to identify visually significant boundaries. One commonly used cue is the surface normal, whose orientation changes abruptly at creases, folds and other geometric features. Access to that cue gives image-space techniques the interior geometric structure that silhouette methods alone leave undetected, as the work reviewed in Section 3.3 demonstrates.

V3 builds on this image-space principle and introduces the first interior edge signal into the proposed rendering pipeline. The signal is the normal gradient magnitude of Equation (2.6), and it measures how far the interpolated world-space surface normal turns between neighbouring screen pixels. The magnitude runs high along geometric creases and clothing folds, and it falls close to zero over smooth surfaces. Screen-space derivatives supply that difference within the fragment pass itself, meaning that neither a stored normal texture nor an additional rendering pass is required. A smoothstep function then converts this raw magnitude into a continuous edge response and this prevents abrupt temporal changes as animated surfaces cross the detection threshold. The resulting edge mask is blended with the X-Toon shaded surface using the same compositing formulation adopted throughout the remaining edge detection techniques. The geometric silhouette outline generated in the first pass is retained, so the final image carries the outline and the interior edges together.

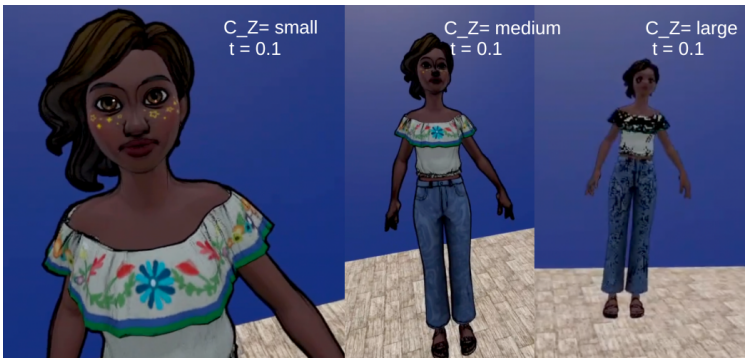
Across all three platforms, V3 complements the silhouette outline established in V1 by introducing interior contours derived from surface geometry. Figure 4.10 illustrates this across the three avatar platforms, where the edge output varies with both the normal information each platform exposes and the viewing distance. Mixamo and Avaturn both compute derivatives from shading normals. This produces a rich edge network that improves the readability of clothing folds, seams and other regions with pronounced surface curvature. By contrast, the Meta Avatars SDK exposes only interpolated geometric normals through the shader hook. This restricted input results in considerably sparser interior contours, most noticeably on clothing surfaces where finer surface curvature would otherwise be detected.

Even though V3 resolves the interior edge problem, two limitations remain. The first is the viewpoint dependence established in Section 2.4.2. Since that operator measures change per pixel rather than per unit of surface, interior edges thicken as the camera retreats. The second concerns the cue itself. Surface orientation is the only signal the method reads and texture boundaries carrying no normal discontinuity thus remain undetected.

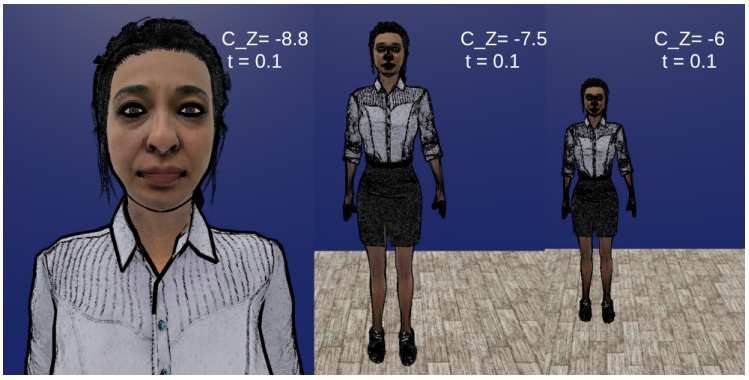
These limitations prevented V3 from being selected for the final rendering pipeline. Edge thickness scales with the viewing distance and this produces inconsistent results whenever a viewer moves relative to the avatar. A fixed camera distance avoids this specific problem. However, the reliance on surface orientation remains regardless of distance. That reliance leaves many perceptually important facial and material boundaries unmarked, since they are defined primarily by texture rather than geometry. Addressing this second limitation required an edge detector sensitive to colour boundaries instead of surface orientation. This need motivates the Sobel-based approach introduced in V4.



(a) Mixamo: normal edge detection at increasing viewing distances.



(b) Meta Avatars SDK: normal edge detection at increasing viewing distances.



(c) Avaturn: normal edge detection at increasing viewing distances.

Figure 4.10.: V3 (Screen Space Normal Edge Detection) across the three avatar platforms. C_Z: camera-to-avatar viewing distance; t: detection threshold.

4.4.6. V4 — Sobel Edge Detection

V4 applies the Sobel operator of Section 2.4.1 to the diffuse albedo texture rather than to the surface normal field. This shift targets colour boundaries in the surface appearance, including clothing patterns, facial details and painted surface structures. Additionally, it avoids the edge thickness growth with viewing distance observed in V3.

The gradient magnitude M follows from the two kernels of Equation (2.2). Pixels for which M exceeds a configurable threshold are classified as edge pixels. These are then composited over the X-Toon shaded surface to form the final stylised appearance.

Across the three avatar platforms, this output quality depends strongly on the available texture source. On Mixamo, the relatively clean painted albedo enables the detection of clothing boundaries and texture transitions with limited interference from unrelated intensity variations. In contrast, Avaturn uses photographic textures that contain illumination information captured during the reconstruction process. Consequently, the Sobel operator responds not only to material boundaries but also to baked lighting variations. As shown in Figure 4.11, this effect is particularly visible around the nose bridge, where the transition between highlight and shadow produces a strong response despite not representing a meaningful texture boundary. On the Meta Avatars SDK, the Sobel operator was applied to the available composited rendering output rather than the original diffuse texture. Although this introduces responses tied to scene lighting, the rendered texture contains less high-frequency photographic variation. Edge structures are thus cleaner and more consistent than on Avaturn, particularly along clothing boundaries. Mixamo remains the least affected of the three, though its painted albedo still carries compression artefacts.

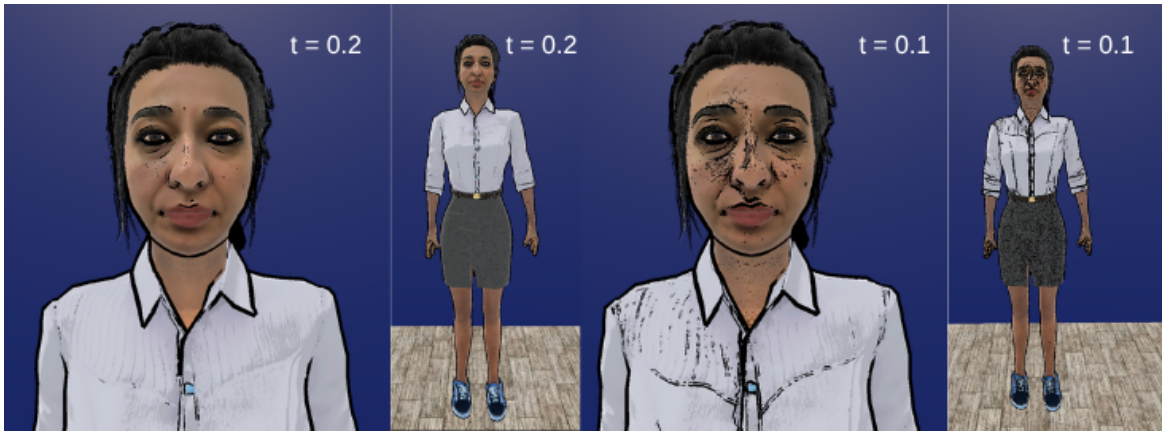
These input differences directly affect the threshold calibration required on each platform. Mixamo produced almost no unwanted edge responses at $t = 0.2$, whereas Avaturn yielded markedly more artefacts at the same threshold. This arose from its photographic texture content. Increasing the threshold on Avaturn suppressed these responses, but it also removed meaningful facial contour lines and eliminated segment boundaries present

at the lower setting, as Figure 4.11d shows. The Meta Avatars SDK requires lower threshold values because its composited output yields weaker gradients at material boundaries. This same property reduces sensitivity to fine texture variations and results in comparatively cleaner edge patterns. These observations confirmed that the Sobel response depends on the available appearance representation rather than solely on underlying geometry. The threshold must therefore be calibrated to each platform's signal characteristics and not treated as a fixed operator property.

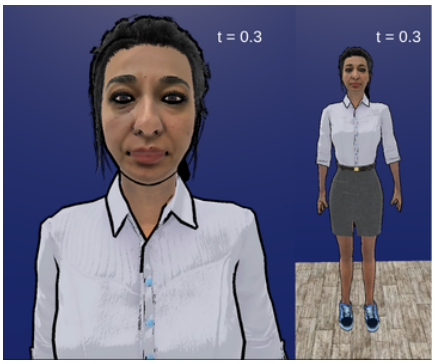
Compared with V3, Sobel edge detection successfully captures these texture-defined boundaries, which remain invisible to normal-based methods. However, the operator responds to all local intensity variations. Fine texture details, photographic illumination and compression artefacts consequently introduce unwanted responses alongside genuine structural boundaries. These false detections motivate V4.2, where Gaussian smoothing is introduced before gradient computation to reduce sensitivity to high-frequency variation.

(a) Mixamo: Sobel edge detection at two threshold values.

(b) Meta Avatars SDK: Sobel edge detection at two threshold values.



(c) Avaturn: Sobel edge detection at the baseline threshold ($t = 0.2$).



(d) Avaturn: Sobel edge detection at an increased threshold ($t = 0.3$).

Figure 4.11.: V4 (Sobel Edge Detection) across the three avatar platforms. t: Sobel threshold.

4.4.7. V4.2 — Gaussian Prefiltered Sobel

V4.2 inserts the low-pass stage that Section 2.4.3 identifies as a prerequisite for stable gradient estimation. A discrete 3×3 Gaussian kernel smooths the texture before the Sobel operator computes the image gradients. This addition targets the noise sensitivity that limited V4 rather than any deficiency in the operator itself.

Gaussian prefiltering was intended to produce cleaner edge maps than V4, but the outcome differed markedly by texture source (Figure 4.12). On Mixamo, Gaussian smoothing reduced edge responses caused by compression artefacts and subtle texture variation while preserving the major clothing boundaries and facial features. On the Meta Avatars SDK, prefiltering similarly attenuated high-frequency artefacts in the composited rendering output. Edge structures were consequently cleaner, particularly on clothing regions. Avaturn remained the most challenging platform because its photographic albedo contained baked illumination that introduced strong intensity gradients unrelated to surface structure. Although Gaussian smoothing attenuated some of these responses, many illumination gradients remained sufficiently strong to influence the Sobel output.

Overall, Gaussian prefiltering reduced unwanted texture responses on the painted and composited inputs. However, on the photographic albedo it did not: sharpening the surviving gradients amplified baked-illumination responses alongside genuine boundaries. Even where the reduction did occur, it came at the cost of weaker structural contours, particularly along features with limited contrast. Gaussian smoothing attenuated both noise and genuine edges. Any reduction in false detections therefore removed some meaningful structural information alongside them. Ultimately, a compromise between contour preservation and noise reduction remained unavoidable.

Among the evaluated platforms, Avaturn exhibited the weakest performance under both V4 and V4.2. The baked illumination present in the photographic texture prevented reliable separation of structural boundaries from lighting variations. Gaussian prefiltering was expected to resolve this by suppressing noise on the skin texture enough to extract facial segments such as the lips and eyebrows. It reduced some of this noise, but not enough to recover these segments reliably. This shortfall motivated the texture preprocessing pipeline described in Section 4.6 instead, where the source texture was edited to reduce gradients introduced by illumination before edge extraction. Once this preprocessing was applied, the basic Sobel operator already produced reliable structural edge detection, without the added cost of Gaussian prefiltering. Section 4.5 reports this outcome in detail.

Although Gaussian prefiltering improves the robustness of texture-based edge detection, its effectiveness remains dependent on the quality of the input texture. The following techniques therefore investigate post-processing stylisation approaches that operate on the rendered image rather than directly on the texture. This reduces the influence of albedo quality on edge detection and yields greater control over the final stylised appearance.

(a) Mixamo: Gaussian prefiltered Sobel edge detection.

(b) Meta Avatars SDK: Gaussian prefiltered Sobel edge detection.

(c) Avaturn: Gaussian prefiltered Sobel edge detection.

Figure 4.12.: V4.2 (Gaussian Prefiltered Sobel) across the three avatar platforms. t: Sobel threshold.

4.4.8. V5 — Hierarchical Edge Detection with Multiple Cues

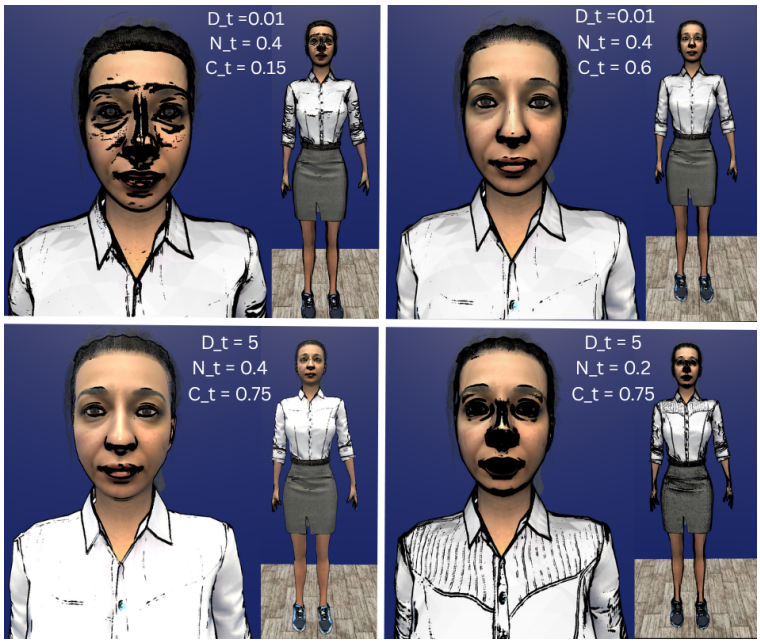
Hierarchical edge detection combines multiple complementary boundary cues into a single, unified representation rather than relying on any one signal alone [40]. V5 adopts this approach as a post-processing architecture that operates on the rendered image rather than on the source texture. Depth discontinuities recover object silhouettes and surface normal discontinuities reveal geometric structure. Neither cue reads surface colour, and this reduces the dependence on albedo quality identified in V4.2. Colour boundaries that leave no geometric trace require a third cue, and the colour layer uses the Roberts Cross operator for it. The original formulation derives its structure layer from gradient operators instead of from surface normals [40]. V5 substitutes the normal cue there, because baked illumination in the source texture would otherwise register as geometric detail. Each layer carries an independent weight and the responses are blended through adjustable parameters so that the contribution of each cue can be controlled interactively. Since a gradient operator produces weak responses in dark regions that fall below the detection threshold, a further mechanism from the same framework scales the edge response inversely with local luminance [40]. V5 applies this gain on the Meta Avatars SDK, where the composited output is darker than the raw albedo the other two platforms read.

Although the same hierarchical formulation was evaluated across all three avatar platforms, the rendering architecture determined which cues could be computed. On Mixamo and Avaturn, V5 sampled the depth, normal and colour buffers provided by the URP.

Figure 4.13 compares the results across the three avatar platforms. V5 detects a broader range of structural boundaries on the Mixamo and Avaturn rendered frames. The depth threshold has little visual influence, since the geometric silhouette outline already provides the silhouette. The normal and colour thresholds instead control the density of interior contours. These contours trace geometric curvature, appearing along the underside of the nose and the outer lip edges, but do not produce clear boundaries between adjacent facial segments. Furthermore, the colour gradient layer occasionally responds to illumination gradients in the rendered image, particularly around the nose bridge where shading transitions are interpreted as texture edges. Although combining the remaining cues still produces richer interior detail than the previous techniques, the absence of screen-space depth prevents silhouette detection on the Meta Avatars SDK. The composited SDK output contains baked lighting gradients absent from the raw albedo, making the texture layer more prone to false responses on this platform. The colour layer therefore replaces the Roberts Cross computation with the XDoG operator of Winnemöller et al. on this platform [20], [21]. This reduced these false responses, as Figure 4.13(b) shows.

(**a**) Mixamo: hierarchical edge detection at varying C_t (top row) and N_t (bottom row).

(**b**) Meta Avatars SDK: hierarchical edge detection at varying threshold combinations with XDoG pre-blur enabled and disabled.



(c) Avaturn: hierarchical edge detection at varying C_t (top row) and N_t (bottom row).

Figure 4.13.: V5 (Hierarchical Edge Detection with Multiple Cues) across the three avatar platforms. D_t, N_t and C_t are the depth, normal and colour thresholds. Only N_t and C_t are varied, since the outline already fixes the silhouette on Mixamo and Avaturn and the Meta Avatars SDK exposes no depth buffer.

4.4.9. V6 — Kuwahara Painterly Filter

The anisotropic Kuwahara formulation aligns an elliptical sampling window with local image structure rather than with the fixed rectangular subregions of the classical filter [22]. V6 adopts this formulation, set out in Section 2.5. Orientation-aligned marks were the property required here, since brushwork ignoring surface direction reads as a uniform blur, not as a painted surface.

The filter operates on the rendered avatar surface and produces a form of abstraction qualitatively different from the contour techniques of V3 through V5. The result is analogous to a painted or illustrated representation. A post-processing architecture is adopted on Mixamo and Avaturn because the anisotropic variant requires multiple rendering passes. The Meta Avatars SDK single-pass constraint does not permit this.

Figure 4.14 compares the results across the three avatar platforms. On Mixamo and Avaturn, the full anisotropic implementation produced coherent painterly regions with well-oriented brushstrokes that follow the contours of clothing and facial features. On the Meta Avatars SDK, the noisy tensor caused the inverse-variance weighting to favour the most uniform colour patch in each neighbourhood rather than following feature orientation. Coherent painterly regions were consequently absent and local colour saturation was amplified instead.

However, achieving a clearly painterly effect comes at a cost. The painted regions only become perceptually distinct at a large kernel radius. Fine facial features such as eyebrows, lip boundaries and iris details are then subsumed into the surrounding skin region. Reducing the kernel radius preserves these details but diminishes the painted quality of the result. This trade-off between abstraction strength and feature legibility suggested that the Kuwahara filter is most effective when combined with a contour detection layer that restores structural boundaries.

(a) Mixamo: standard render (left) and Kuwahara painterly filter (right).

(b) Meta Avatars SDK: standard render (left) and Kuwahara painterly filter (right).

(c) Avaturn: standard render (left) and Kuwahara painterly filter (right).

Figure 4.14.: V6 (Kuwahara Painterly Filter) across the three avatar platforms.

4.5. Platform and NPR Technique Selection for the User Study

A valid perceptual trust comparison required three properties of the platform:

- **Unrestricted URP shader access**, so that the NPR techniques could be applied directly.
- **Full blendshape animation**, so that facial expression could be animated in the video stimuli.
- **Photorealistic facial reconstruction**, so that the avatar could be compared against the real human speaker.

On this basis, Avaturn was selected as the platform, since it is the only candidate among those evaluated in Section 4.3 that satisfies all three requirements. The Avaturn avatar used for both conditions C2 and C3 was reconstructed from photographs of the same individual who served as the real human speaker in C1. Therefore, rendering style was the primary variable distinguishing C2 from C3. Further preprocessing was required to support reliable edge detection and the procedure is described in Section 4.6.

The NPR composite for condition C3 was assembled from five candidate components: a geometric silhouette outline, Sobel edge detection and X-Toon tonal shading, alongside two post-processing passes, a Kuwahara painterly filter and a hierarchical depth edge layer. Parameter settings for all five were determined through a parameter adjustment study and full details are in Section 4.7. In addition, it established whether the two post-processing passes should be included in the final composite.

Among the techniques evaluated in Section 4.4, three were set aside before the composite was assembled for that study. Basic toon shading (V2) was replaced by X-Toon (V2.2), since the 2D ramp provides independent control over the abstraction level that procedural quantisation cannot offer. However, this control depends on how the abstraction level is driven. The depth and curvature modes described in Section 4.4.4 respond to camera distance and surface curvature respectively. Since the camera stays fixed and close to the avatar throughout the study recordings, this leaves little room for either quantity to vary across the visible surface. Manual mode avoids this problem entirely. It sets the abstraction value directly rather than computing it from camera distance or surface geometry. Therefore, this mode was the natural choice for the C3 composite. The study then fixed this value, referred to there as the detail bias.

Screen-space normal edge detection (V3) was excluded primarily because it cannot recover facial features defined by colour or texture contrast instead of geometric curvature. Smooth skin produces minimal normal discontinuities and the eyebrow boundary, lip contour, nostril edge and eyelid margin consequently remain invisible to the technique. These are precisely the landmarks that determine facial legibility in a conversational advisor context. V3 detects clothing folds and seams reasonably well through normal map discontinuities, but the Sobel operator subsumes this capability entirely. V4 captures the same clothing contours and additionally identifies edges at boundaries defined by colour contrast, a category that V3 cannot detect.

Furthermore, Gaussian prefiltered Sobel (V4.2) was excluded. Its additional sampling cost no longer produced a detectable improvement over the basic Sobel operator once the head albedo was preprocessed to reduce baked illumination gradients (see Section 4.6). The hierarchical pass was the only candidate to introduce a depth cue, but its two remaining layers replicate detection strategies already set aside earlier in this section. The surface normal layer fails on smooth facial skin for the same reasons that excluded V3. The colour gradient layer adds no information beyond what the Sobel operator already provides. Both passes were then evaluated in the same study. Full details of the visual assessment and the study results are described in Chapter 6.

4.6. Avaturn Avatar Preprocessing

The head albedo texture was preprocessed in Blender in order to reduce the edge detection artefacts caused by the reconstructed albedo texture identified in Section 4.4.7. The avatar was then re-exported as a GLB file. The preprocessing addressed two distinct problems in the head albedo: baked illumination variation that introduced spurious edge responses at boundaries unrelated to surface structure and the absence of a strong tonal cue at the lip boundary.

The first of these is a known limitation of reading edges from the colour buffer rather than from geometry [37]. Blurring and skin tone unification removed the illumination variation. Increased saturation across the mouth and an added lipline then supplied the missing tonal cue. Figure 4.15 shows the result against the unmodified avatar.

This preprocessed avatar served as the basis for the C3 stimuli. C2 used the unmodified Avaturn avatar, so the preprocessing is part of the NPR treatment rather than a property shared by both avatar conditions. Extending it to C2 would have removed the photorealistic baseline that the comparison requires, since the modified texture visibly departs from a photographic surface.

Figure 4.15.: Left: the default Avaturn avatar with no preprocessing and no NPR applied. Right: the preprocessed head texture, also with no shader applied.

4.7. Parameter Adjustment Study

A parameter adjustment study established both the parameter settings for all five components and whether the two post-processing passes should be included in the final composite. Participants ordered rendered images of the avatar by perceived trustworthiness as an advisor and the result of each round fixed one parameter group for all subsequent rounds. The study covered the parameter groups with the greatest expected influence on perceived edge density, tonal balance or abstraction strength. Parameters with comparatively minor expected visual impact, including rim light presence, Kuwahara kernel size, Gaussian blur strength, sector count and normal edge strength, were instead set through direct visual inspection during development. This kept the study at a manageable length for participants.

One parameter group was varied while all remaining settings were held at the level fixed by the preceding test in each round. Table 4.2 lists all stages and the values examined at each.

Table 4.2.: Parameter groups and values tested in each round of the parameter adjustment study. R4A and R5A carry forward the R1E configuration as the no-treatment baseline for their respective rounds.

Round	Parameter group	Values tested (A, B, C, . . . in order)
R1	Inner line threshold	0.01, 0.02, 0.03, 0.04, 0.05
R2	Outline width and depth offset	width: 0.003, 0.005, 0.007; offset: on / off
R3	Shadow strength and detail bias	(1.0, 0.5); (0.0, 0.5); (0.5, 0.5); (0.5, 1.0); (0.5, 0.0) as (strength, bias)
R4	Kuwahara hardness and sharpness	none (= R1E); (1, 18); (9, 9); (18, 1)
R5	Hierarchical depth threshold	none (= R1E), 0.01, 0.02, 0.03, 0.04, 0.05

Participants were required to assign a unique rank to every image in the round. Any ballot with a missing rank or a tied rank was excluded. Images were converted to Borda-count points in each round, where the image

ranked last received 1 point and the image ranked first received points equal to the number of options. The totals were then summed across participants. Demšar recommends the Friedman test with post-hoc comparison for more than two conditions and the Wilcoxon signed-rank test for a single pair [51]. Therefore, an omnibus Friedman test established whether the ranking within a round differed from chance. A pairwise Wilcoxon signed-rank test then compared the configuration ranked first against its closest competitor. Section 6.2.1 reports the outcome of each round and the reasoning that fixed each value.

The resulting configuration is listed in Table 4.3.

Table 4.3.: Final C3 composite parameter set established by the parameter adjustment study.

Parameter	Value	Source
Outline width	0.003	R2 Borda winner
Depth offset	On	R2 Borda winner
Inner line threshold	0.05	R1 appeal tiebreaker
Shadow strength	0.0	R3 appeal tiebreaker
Detail bias	0.5	R3 appeal tiebreaker
Kuwahara filter	Off	R4 excluded (significant)
Hierarchical depth layer	Off	R5 excluded (significant)

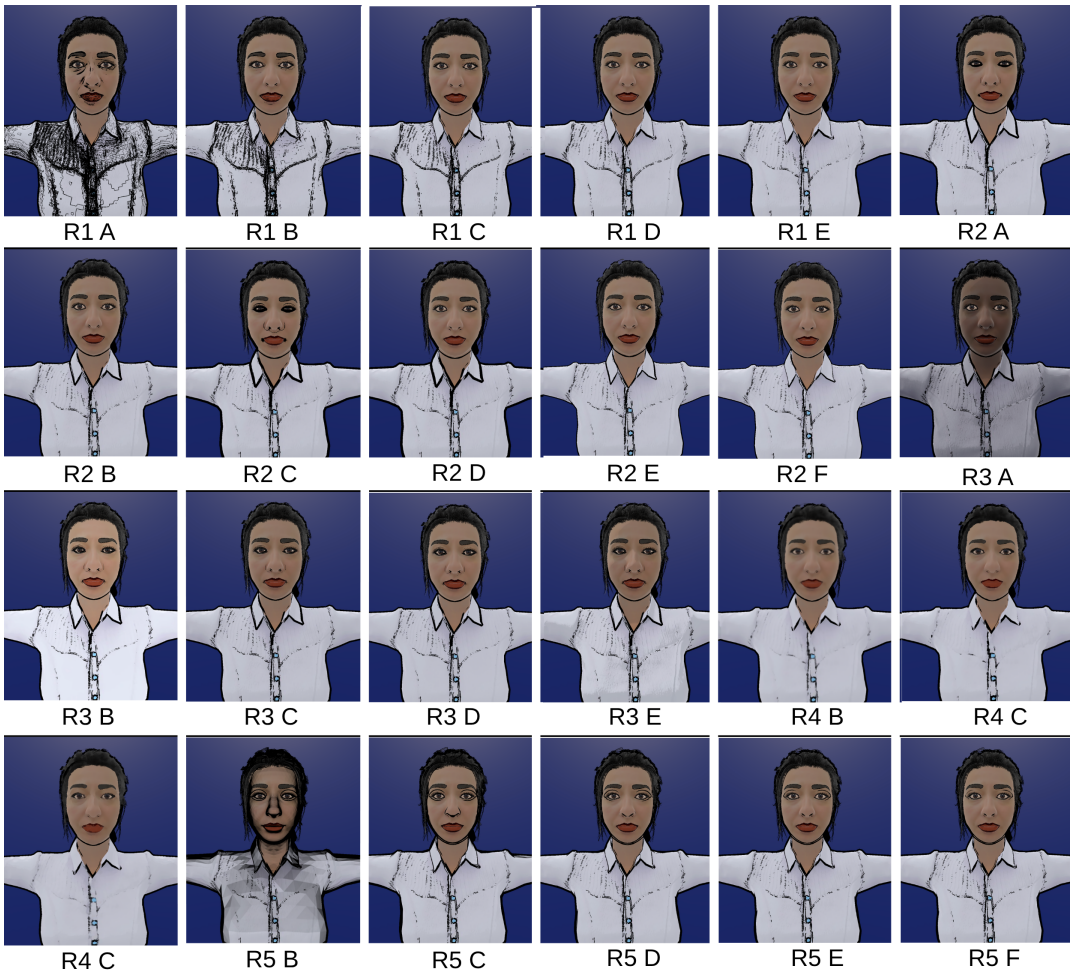


Figure 4.16.: All rendered variants presented during the parameter adjustment study. Labels follow the convention R*n*X, where *n* is the round number and X is the variant letter in the order listed in Table 4.2. R4A and R5A are identical to R1E and are not shown as separate images.

4.8. User Study Design

4.8.1. Objective and Hypotheses

The user study addresses RQ3 by measuring how the visual representation of a speaker affects perceived eeriness and trustworthiness in a concrete decision situation. Four hypotheses are tested.

H1: The NPR avatar (C3) will produce lower eeriness ratings than the photorealistic avatar (C2).

Dubosc et al. found that rendering the same avatar in stylised and semi-realistic form produced significantly lower eeriness ratings than the realistic version [31]. Thus, H1 tests whether the NPR composite reproduces that reduction between C2 and C3.

H2: The NPR avatar (C3) will not produce significantly lower perceived trust than the photorealistic avatar (C2).

Prior work varied stylisation level within a fixed advisor role and scenario and found no significant effect on self-reported trust [5]. Benevolence was the one partial exception and there the realistic style trended lowest rather than highest. H2 tests whether this pattern holds when rendering style is the primary variable that changes between conditions.

H3: Eeriness will negatively predict trust ratings across all conditions.

Eeriness is a distinct negative affective response rather than general dislike [13]. Additionally, a systematic review by Alipour et al. found that the relationship between eeriness and trust is not uniform [1]. The review coded direct, conditional and absent effects among the studies it covered. H3 tests whether this relationship is detectable across all three conditions within each participant.

H4: The NPR avatar (C3) will not produce significantly lower trust ratings than the real human speaker (C1).

Gao et al. found a real speaker rated higher than a stylised agent modelled on that same speaker across all three trust dimensions [30]. However, they did not differ significantly on a direct trust rating. H4 tests whether the abstraction responsible for H1's predicted reduction in eeriness closes the trust gap enough to reach this reference point, rather than assuming a real human speaker remains categorically more trusted than any avatar.

4.8.2. Conditions

The study compares three presentation conditions:

C1 — Real Person: A video recording of the same human speaker whose photograph was used to generate the Avaturn avatar, so that face identity, voice and demographic appearance are held constant across all three conditions. C1 serves as the trust ceiling baseline against which both avatar conditions are compared.

C2 — Photorealistic Avatar: The same speaker scenario rendered using a photorealistic Avaturn avatar with standard PBR shading. Facial expressions are driven by blendshape animation captured in the same recording session as C1, so that expression timing and facial gestures are identical across all three conditions.

C3 — NPR Avatar: The Avaturn avatar rendered with the NPR technique selected through the parameter adjustment study on the basis of its balance between abstraction and expressiveness.

Using the same Avaturn geometry for both C2 and C3 holds avatar identity, facial structure and animation constant across the two avatar conditions. Render style is consequently the primary independent variable between them. Comparable work crossed stylisation level with speaker identity, and identity produced the stronger effect throughout those results [5]. Holding the speaker constant removes that source of variance from the present comparison. The theoretically critical comparison is C2 versus C3. Since animation is held constant between these two conditions, any observed difference in trust or eeriness ratings is attributable primarily to the change in rendering style, with one material-level exception described in Section 4.6. C1 provides a second point of comparison for H4. This test asks whether C3 reaches the real-person reference point overall rather

than isolating rendering style specifically, as animation naturalness and speaker identity also differ between C1 and both avatar conditions.

4.8.3. Scenarios

Three video scenarios present naturalistic situations in which each speaker delivers a recommendation that participants must decide whether to follow. Each scenario has two plausible options and the speaker recommends one with explicit reasoning. Several design principles govern the scenario construction.

Ecological validity. The study was expected to recruit largely from a university population aged roughly 20 to 35. The three scenarios are accordingly drawn from everyday situations that such participants are likely to encounter in their social and professional lives: a career decision between two job offers, route planning in an unfamiliar city and selection of a learning resource for career development. Grounding the scenarios in contexts drawn from daily life ensures that trust judgements reflect responses to the advisor rather than the effort of interpreting an unfamiliar situation.

Vignette anchoring. Before each video, participants read a short context paragraph that anchors them in a specific, concrete situation with defined personal circumstances. Without this anchoring, trust ratings would reflect individual differences in life experience, not responses to the speaker. The vignette eliminates this source of variance across participants and every participant enters each video from the same defined role.

Vignette structure. Each vignette follows a fixed four-step layout:

1. Establish the participant's role and the decision to be made.
2. State the constraints that bound the decision, such as the time available or the budget.
3. Present the first option with its advantages and its single cost or limitation.
4. Present the second option in the same form, with the trade-off reversed.

Both options receive equal detail, so that neither reads as the obviously correct choice.

Recommendation alternation. The three scenarios are designed so that the recommended option alternates across videos: Scenario 1 recommends Option A, Scenario 2 recommends Option B and Scenario 3 recommends Option A. This prevents participants from detecting a systematic preference for either option label across the study.

Presentation order alternation. Within each script, the recommended option is not always discussed last. Discussing the recommended option last in every video would introduce a recency effect, where participants associate recommendation with position rather than with the speaker's reasoning. Scenario 1 and Scenario 3 discuss the recommended option first. Scenario 2 discusses the recommended option last. All scripts conclude with an explicit recommendation statement regardless of discussion order.

Coverage of trust dimensions. Each scenario is built around a different component of the trust model of Mayer et al. Rendering style may affect ability, benevolence and integrity unequally. A design that activated only one of them could not separate a general effect on trust from one confined to a single dimension. Scenario 1 turns on the advisor's ability, Scenario 2 on benevolence alongside ability and Scenario 3 on integrity.

Script argumentative structure. Each spoken recommendation follows a fixed five-step structure, applied uniformly across all three scenarios:

1. State the recommendation explicitly.
2. Acknowledge the primary drawback of the recommended option.
3. Introduce the key advantage of the recommended option with an explicit contrastive marker.
4. Name the central weakness of the alternative option.
5. Close with a concise restatement.

This structure ensures the advisor engages visibly with trade-offs before concluding, which holds rhetorical strategy constant across scenarios. Any difference in trust ratings between conditions therefore reflects the visual presentation of the speaker rather than variation in argumentation pattern between scenarios.

Table 4.4 sets out the three scenarios. Each row gives the vignette text participants read before the video and the recommendation the advisor then delivered.

Table 4.4.: The three scenarios as administered. Each entry gives the vignette shown on screen before the video and the recommendation the advisor delivered in it. The trust dimension in italics is the one the scenario is designed to activate.

Scenario 1 — Career Choice *Ability*	
Vignette	You recently completed your master's degree and have received two job offers. Both start in one month. You have five days to decide. **Job A** is a generalist role at a smaller company. You would define your own work area, your contributions would be directly visible to the team, and the skills you build would transfer across industries. The starting salary is below the industry average for your field. **Job B** is a specialist role at a large, well-known company. The employer name carries weight on a CV and the role is in a high-demand field. However, the position involves deep specialisation within a large team, with limited individual visibility. The starting salary is above the industry average for your field.
Advisor	"I'd go with Job A. Yes, the salary is lower right now, but the generalist role gives you a range of experience that is genuinely hard to get later. Salary gaps can be closed in the future. But once you specialise, switching back is costly. Take the pivot while you can."
Scenario 2 — Travel Route *Benevolence and ability*	
Vignette	You have just arrived by train in a city you have never visited. You need to reach your hotel, where colleagues are expecting you in one hour. **Route A** is a direct route on a single line with no transfers. It is straightforward to navigate in an unfamiliar city. If the journey runs on time, you would arrive approximately 10 minutes before your meeting begins. **Route B** involves one transfer at a mid-size station you have not used before. If the connection runs on time, you would arrive approximately 40 minutes before your meeting begins. However, the transfer station is unfamiliar and a missed connection could leave you with very little time.
Advisor	"I'd take Route B. Yes, it has a transfer at a station you have never used. But it gets you there forty minutes early. If something goes wrong, you still have enough time to recover. Route A is direct but leaves only ten minutes. That may not be enough. Take the buffer."
Scenario 3 — Learning Resource *Integrity*	
Vignette	You want to develop a professional skill that would strengthen your CV and improve your career prospects. You have a few weeks available and you are a student with a limited budget. **Option A** is a structured online course from a well-known learning platform. It follows a clear curriculum and awards a recognised certificate you can add to your CV. It costs EUR 350. **Option B** is a video tutorial series by a practitioner with a strong reputation. It covers the same material at no cost and you set your own pace. However, there is no certificate and no structured curriculum.
Advisor	"I would take Option A. Yes, 350 euros is a real cost. But the certificate is what matters in here. It is not just about what you learn, it is about what you can show. A recruiter cannot verify a free tutorial series. That is worth it."

Scenario order is held constant across all participants: Career in Video 1, Travel in Video 2 and Learning in Video 3. The three conditions are counterbalanced across those fixed positions using a 3×3 Latin square (Table 4.5), with one study version per row.

Table 4.5.: Latin square counterbalancing across three study versions. Each condition appears exactly once per version, with scenario positions held fixed.

Version	Video 1 (Career)	Video 2 (Travel)	Video 3 (Learning)
A	C2 — Photorealistic Avatar	C1 — Real Person	C3 — NPR Avatar
B	C3 — NPR Avatar	C2 — Photorealistic Avatar	C1 — Real Person
C	C1 — Real Person	C3 — NPR Avatar	C2 — Photorealistic Avatar

Each appears exactly once per presentation slot and this controls for any systematic advantage conferred by video order. Sequential carryover between specific condition pairs is not addressed by this design and is examined in Section 6.2.6. In the questionnaire, the three video slots are named Video 1, Video 2 and Video 3 so that participants cannot identify the rendering style. Each version's condition mapping is stored separately and applied during data analysis. The complete questionnaire is reproduced in Appendix A.

4.8.4. Procedure and Measures

Procedure

The study was conducted entirely online via Google Forms. Each participant completed a consent and demographic questionnaire, then watched three video blocks in the order determined by their assigned study version. Each block followed four sequential steps:

1. Read the vignette establishing the scenario.
2. Record an initial option preference before watching (this pre-video baseline was used to compute preference shift as a secondary exploratory measure).
3. Watch the 23 second recommendation video.
4. Complete the per-block questionnaire.

After all three blocks, participants completed a post-study section. The entire session took approximately 10 to 15 minutes.

Per-Block Questionnaire

The per-block questionnaire was administered after each video. Parts B and C used a five point semantic differential scale. Parts D through G used a seven point Likert scale.

Part A — Comprehension check. Two gating items verified that participants watched the video and identified the correct recommendation. In addition, an embedded attention check item was included in two of the three blocks (Section 4.8.4). Since these checks were not present identically across every block, a participant who failed any gating item in any block was excluded from the analysis entirely rather than only for that block. This rule prevented any participant from contributing a mix of validated and unvalidated data.

Parts B, C — Semantic differential (5 point). Part B (likeability) was administered before Part C (eeriness) so that participants formed a general social impression before attending to uncanny affect. A reversed order would prime observers towards negative perceptions and risk inflating eeriness scores before the trust battery. Part B used the five-item Godspeed likeability subscale of Bartneck et al. Part C captured eeriness through eight items from the validated instrument of Ho and MacDorman. Their evaluation of the Godspeed questionnaire found four of its subscales too strongly intercorrelated to measure distinct constructs: anthropomorphism, animacy, likeability and perceived intelligence. It also contains no subscale corresponding specifically to eeriness [13]. Each polar pair was preceded by a short definition to support participants unfamiliar with the terms. This instrument treats eeriness as a distinct affective response from likeability and H3 can consequently be examined independently.

Parts D, E, F — Trust battery (7 point Likert). Nine items operationalised the Ability-Benevolence-Integrity model of Mayer et al. within the trusting beliefs framework of McKnight and Chervany, introduced in Section 2.3 [10], [11]. Parts D, E and F covered Ability, Benevolence and Integrity respectively, with three items each, measured independently rather than as a composite. These three exclude predictability, the fourth trusting belief in the framework. Judging predictability requires observing an advisor's behaviour repeatedly, whereas each participant saw a given advisor deliver a single scripted recommendation.

Part G — Response to recommendation. Part G recorded agreement with the recommendation and the participant's option choice after watching. This post-video choice was compared with the pre-video baseline to compute a preference shift score.

An embedded attention check item was included in the first and third blocks, with a different target response in each to prevent pattern detection.

Post-Study Section

Three separate ranking questions each asked participants to order all three advisors: from most to least wanted for consultation (P1), from most to least trustworthy overall and from most to least unsettling overall. P1 was the primary behavioural measure, adapted from the post-study advisor selection paradigm of Canales et al. [5]. Parts D through F record trusting beliefs, that is, what a participant holds to be true of the advisor. P1 instead records a trusting intention, a willingness to depend on that advisor. The two need not agree, and measuring both addresses the methodological gap identified by Alipour et al. [1]. The trustworthiness and unsettlingness rankings provided behavioural analogues to the corresponding Likert measures, Parts D through F and Part C respectively. Ranked and rated judgements can therefore be compared directly for both dimensions. A rank order was used for all three questions rather than a forced binary choice, since both avatar conditions were expected to score below C1. Limiting participants to two options would concentrate responses on C1 and leave insufficient variance to distinguish C2 from C3.

A visual style manipulation check (Part H) asked participants to rate the perceived visual stylisation of each advisor simultaneously. Its placement after all video blocks prevented awareness of rendering differences from priming trust ratings during data collection. The post-study section closed with two open-ended questions, each administered once rather than per block to limit response fatigue. The reflection question (Part I) asked participants to describe the most important factor in their trust judgement. A final open question asked what they believed the study was investigating and this serves as a check for demand characteristics.

The demographics section, completed before the first video block, also included an item measuring dispositional trust (seven point Likert: *In general, I tend to trust people I meet for the first time*) as an individual-level covariate. Mayer et al. describe this disposition as a trait that is stable across situations. They expect it to explain variance in trust when included alongside a fuller set of variables [10].

Analysis Plan

The primary analysis was a one-way repeated-measures analysis of variance (ANOVA) per subscale composite (likeability, eeriness, ability, benevolence and integrity), with condition (C1, C2, C3) as the within-subjects factor. Every hypothesis concerns an individual subscale rather than a linear combination of them and the subscales were accordingly analysed separately and not through a multivariate omnibus test. Pairwise comparisons between the three conditions followed, corrected by Holm's step-down procedure. It gives greater power than a single-step Bonferroni adjustment and adds no further assumptions about the hypotheses [51]. Effect sizes were reported as partial η^2. Comparable rating studies of avatar appearance have analysed within-participant conditions with the repeated-measures ANOVA [2], [46], [52].

Normality was checked with the Shapiro-Wilk test and sphericity with Mauchly's test. The Greenhouse-Geisser correction was applied where sphericity was violated. Both instruments report Cronbach's α per subscale and both treat .70 as the minimum acceptable value [12], [13]. The same was done here before the composites were formed.

The theoretically critical pairwise comparison is C2 versus C3. H2 predicts equivalence rather than a difference and this leads to a non-significant pairwise result providing insufficient evidence on its own. Lakens sets out a practical procedure for equivalence testing with two one-sided tests (TOST) [53]. Therefore, a TOST equivalence test was applied to the C2 versus C3 trust composite. Its bound was pre-specified at Cohen's $d = 0.3$, the smallest difference in trust treated as practically meaningful in this study. H4 predicts the same kind of equivalence against a different reference point and the identical procedure with the same bound was applied to the C1 versus C3 trust composite. Comparable work has inferred an absence of stylisation effects on trust from non-significant results alone [5]. That inference cannot distinguish a genuine null from an insensitive test.

For H3, person-centred multilevel regression was used, with eeriness as the within-participant predictor of trust and a random intercept per participant. This isolates the within-participant effect from stable differences in how trusting individual participants are. Ratings pooled across observers confound the two.

Three post-study ranking questions, covering P1, trustworthiness and unsettlingness, were each analysed with a Friedman test comparing mean advisor ranks across conditions. Pairwise Wilcoxon signed-rank tests were then applied to each pair, with Holm correction to control the family-wise error across the comparisons. C2 versus C3 is the critical pairwise comparison in each case. Rank responses carry only ordinal information. The parametric assumptions above thus do not apply. The parameter adjustment study used the same procedure (Section 4.7), meaning both rest on one statistical basis.

The Latin square assumes that a video's position in the sequence carries no effect of its own. Two checks tested that assumption. Position was entered as a within-subjects factor in place of condition and study version was tested as a between-subjects factor on each participant's mean across the three blocks. Because every participant saw each condition exactly once, that mean is free of condition effects. If the three versions differ, the cause could be the order in which the conditions appeared. It could also be that the three groups of participants differ from one another.

Ethics and Data Protection

Participation was voluntary throughout. A consent screen preceded the first video block and described the task, the approximate duration and the right to withdraw at any point without giving a reason. No names, email addresses or contact details were recorded and no response carries an identifier that links it to a participant. The demographic items were limited to age, gender, prior exposure to avatars and dispositional trust, none of which identifies an individual within a sample of this size. Only aggregate results are reported.

The speaker recorded for C1 is identifiable and photographs of the same person were used to reconstruct the Avaturn avatar shown in C2 and C3. That speaker is the author of this thesis and no third party's likeness was recorded or reconstructed for the study materials. That allocation was deliberate. Reconstructing a third party's face and replaying their recorded expressions across two rendered conditions would extend the use of their likeness well beyond the session itself. Using the author's own likeness avoids that requirement. Its consequence for the C1 ratings is examined in Section 6.3.3.

Sample Size

A power analysis was conducted for a three-condition within-subjects ANOVA. It targeted a medium effect size of partial $\eta_p^2 = 0.06$, $\alpha = .05$ and power = 0.90. The minimum required sample is $n = 38$ participants. The recruitment target was set to $n = 48$, with sixteen participants per study version across three versions. This number accommodated anticipated attrition and exclusions from the comprehension check and it preserved balance across all three counterbalancing patterns.

Equivalence testing carries its own power requirement. A paired TOST at the $d = 0.3$ bound reaches a power of 0.90 only near $n = 120$. Its power stays below 0.40 at $n = 48$ even when the true difference is zero. The recruitment target followed the ANOVA requirement, so the equivalence components of H2 and H4 were underpowered by design. Their outcome is reported as inconclusive rather than as evidence against equivalence.

5. Implementation

This chapter documents the technical realisation of the NPR avatar system. It covers the development environment, the integration path for each avatar platform, the shared rendering structure and the HLSL implementation details of each stylisation technique. Runtime tooling built to support interactive parameter exploration and evaluation capture is described near the end of the chapter. A final section covers the animation and recording pipeline used to produce the user study video stimuli.

5.1. Development Environment

The system was developed in Unity 6 (version 6000.0.60f1) using the Universal Render Pipeline (URP). The target hardware for technique development was the Meta Quest 3 standalone VR headset. Version 40.0.1 of the Meta Avatars SDK was used. All shader code is written in HLSL and targets the URP forward rendering path. Rendering cost was not an object of this evaluation, as the study stimuli were rendered offline to video and no reported result depends on frame rate.

5.2. Avatar Platform Integration

5.2.1. Mixamo

Mixamo characters were exported as FBX assets and imported into Unity as standard skinned mesh renderers. This SDK-free structure, introduced in Section 4.3, allows every URP material to be replaced directly with a custom NPR shader.

5.2.2. Meta Avatars SDK

The Meta Avatars SDK generates its shader source from a set of packaged templates managed through the SDK's shader manager component. This generated source is not intended to be edited directly. Custom logic is instead injected through the hook function that the SDK exposes at a defined point in the shader pipeline.

The hook used in this thesis was `AppSpecificPostManipulation`, which executes per fragment after all SDK PBR lighting (subsurface scattering, eye glints, hair shading) has been computed. It receives the accumulated lit colour and returns a modified colour. This grants full control over the final fragment output without disrupting any SDK rendering logic. Nonetheless, that control is confined to the fragment stage, and screen-space post-processing cannot be added alongside it. The generated shader declares no depth-normals pass, meaning that the avatar is absent from URP's screen-space normal buffer. Equally, an avatar mask is out of reach. A pass drawn with an override material would rasterise the bind pose, the undeformed posture in which the mesh is authored.

5.2.3. Avaturn

Since Avaturn exports its avatars as `.glb`, introduced in Section 4.3, the GLTFast package is required to handle the additional import step. It produces a standard Unity skinned mesh hierarchy with the glTF PBR materials in place. Therefore, custom NPR materials are applied by direct replacement after load, as on Mixamo. Additionally, the blendshape animation targets exported by Avaturn are preserved through this process.

5.2.4. Common Scene Environment for Avatar Platform Comparison

The Mixamo avatar, Avaturn and the Meta Avatars SDK avatar each occupy a separate scene, because every platform requires its own configuration. The same room asset is reused across all three: an enclosure with a tiled floor, brick walls and one dark blue accent wall, plus a controlled skybox and no emissive surfaces. Camera position and framing are matched as closely as possible so that comparison renders share an equivalent viewpoint. The Mixamo avatar and Avaturn additionally share a directional light. It is elevated $-30°$ and rotated $-30°$ horizontally relative to the avatar forward vector. This places it above and towards the front left of the avatar. Its intensity is fixed at 1.0 lux with a neutral white colour. Moreover, the Meta Avatars SDK scene carries additional lighting beyond this configuration in order to improve visibility of the applied technique during the comparison. Nevertheless, ambient intensity and the global illumination contribution, baked from the shared skybox, align across all three platforms.

5.3. Shared Rendering Structure

Each NPR technique on the Mixamo and Avaturn platforms is a standalone multi-pass shader and technique switching is performed by replacing the active material set. By contrast, all techniques on the Meta SDK platform share a single shader that dispatches to the corresponding technique based on an active keyword. Runtime switching then requires only toggling this keyword without recompilation.

On Mixamo and Avaturn, each avatar mesh is further split across several submesh materials, four on Mixamo and six on Avaturn. This lets technique parameters be tuned separately for regions such as skin, hair and eyelashes. The vertex shaders of these materials also carry a vertex fetch macro shared with the Meta Avatars SDK codebase. This macro has no effect on Mixamo and Avaturn, since standard Unity vertex attributes are already available directly. It is kept only for consistency across all three platforms' shaders.

The automatic level of detail (LOD) management system on the Meta SDK platform replaces avatar materials as camera distance changes and this reverts any custom shader assignment made at load time. A dedicated runtime script restores the assignment. It performs an initial material replacement across the different detail levels once the avatar finishes loading, then polls the active materials on each renderer at a configurable interval.

5.4. NPR Technique Implementation

The visual output of each technique across the three avatar platforms is presented with the technique itself in Section 4.4. This section covers only how each one is built.

All shader parameters described in the following subsections can be adjusted in two ways, as can the choice of which NPR technique is active. The Unity Inspector serves during offline development and a custom user interface (UI) panel serves inside the running application. Changes made either way take effect immediately without shader recompilation.

5.4.1. V1 — Geometric Silhouette Outline Pass

The outline pass implements the geometric silhouette by applying the vertex displacement of Equation (4.1). Its structure is identical across all three platforms. In the Mixamo and Avaturn shaders the pass culls front faces and writes depth using a standard depth test. The Meta SDK shader applies the same front-face culling. However, it disables depth writing and uses a looser depth test instead to avoid conflicting with the SDK's depth compositing. The fragment outputs a flat outline colour. A configurable depth bias shifts the outline towards the camera in clip space. This prevents z-fighting where the expanded shell meets the original surface.

Alpha Cutout Handling

Alpha cutout is controlled by a toggle flag. When set, both the outline pass and the forward-lit pass sample the base texture alpha and clip against the cutoff of Equation (4.2) before any displacement or shading is applied. A cutoff of $\alpha_{\text{cutoff}} = 0.1$ was selected visually as the lowest setting that reliably removes fully transparent regions without clipping the visible hair silhouette (Figure 4.5).

5.4.2. V2 — Toon Shading

This technique is realised as two shader implementations across the three avatar platforms. Mixamo and Avaturn share the toon quantisation code, while the Meta Avatars SDK requires a separate implementation. The outline pass carries over unchanged from Section 5.4.1 across all three platforms.

Mixamo and Avaturn

The toon pass quantises the Lambertian dot product $\mathbf{N} \cdot \mathbf{L}$ per step using a per-band smoothstep, shared without modification between the two platforms:

```
float steps  = max(1.0, _ToonSteps); // number of shading bands
float scaled = NdotL * steps; // map into band index
float band   = floor(scaled); // current discrete band
float frac   = scaled - band; // offset within the band
float blend  = smoothstep(1.0 - _ToonSmoothness, 1.0, frac); // edge blend
float toon   = saturate((band + blend) / steps); // combined 0-1 value
toon = lerp(1.0 - _ShadowStrength, 1.0, toon); // remap to shadow range
```

Listing 5.1: V2 toon quantisation (Mixamo and Avaturn, unified).

The default is four shading bands. This value was selected because visual assessment confirmed that it provides an effective balance between shape legibility and graphic simplification (Figure 4.6). Moreover, each band boundary receives its own narrow blend. In its absence, small normal changes during animation would flip an entire surface patch to the next band at once. An additive ambient term and a rim highlight are then added to the quantised diffuse.

Meta Avatars SDK

This implementation runs inside the hook and receives the fully composited PBR colour produced by the SDK's renderer. The quantisation therefore operates on RGB directly and preserves hue while stepping luminance:

```
float  bands      = max(2.0, _ToonColorBands); // number of colour bands
float3 posterized = floor(color.rgb * bands + 0.5) / bands; // round to band
color.rgb = lerp(color.rgb, posterized, _ToonPosterizeStrength); // blend

float lum  = dot(color.rgb, float3(0.2126, 0.7152, 0.0722)); // luminance
color.rgb  = lerp(float3(lum, lum, lum), color.rgb, _ToonSaturation); // desat
```

Listing 5.2: V2 posterisation (Meta SDK).

A plain floor operation always rounds down and this introduces a darkening bias. Rounding to the nearest band avoids this bias and distributes the quantisation error symmetrically. Saturation adjustment runs after posterisation to prevent clamping artefacts. The input already contains Meta's subsurface scattering, anisotropic hair and eye glints. Posterisation therefore stylises an image that is already physically correct, rather than rebuilding the lighting from scratch.

5.4.3. V2.2 — X-Toon Shading

The same split into two implementations as V2 carries over: Mixamo and Avaturn share the forward-lit pass, while the Meta Avatars SDK uses a separate implementation. Outline generation remains unchanged from V2 across all three platforms. The forward-lit pass replaces the procedural quantisation with a 2D ramp texture lookup, shown in Figure 4.8. Moreover, this adds an abstraction compression stage that is driven by a choice of three modes: depth, curvature or manual.

Mixamo and Avaturn

The forward-lit pass begins by optionally modifying the surface normal via Normal Field Abstraction, which is described in Section 4.4.4. The abstract normal map is a blurred version of the surface normal map and it is sampled when this option is active. It is then transformed from tangent space to world space through the TBN matrix. The result is linearly interpolated with the geometry normal.

The U coordinate for the ramp starts from the same Lambertian dot product used in V2 toon shading. The result is then remapped and interpolated towards the ramp midpoint to control light sensitivity.

The V coordinate (abstraction level) is computed differently in each of the three modes, selected at compile time via `shader_feature_local` keywords:

- **Depth mode** uses the fragment's world-space camera distance.
- **Curvature mode** uses the screen-space normal derivative magnitude.
- **Manual mode** uses a constant.

The ramp is then sampled at `(rampU, rampV)`. An abstraction compression stage then flattens the lighting bands as the abstraction level rises. It pulls the ramp coordinate towards the midpoint and widens the band blend, so boundaries soften rather than sharpen.

The resulting soft threshold blends the ramp-tinted albedo towards a configurable shadow colour. An overall shadow strength parameter controls how much of the effect applies over the plain albedo. A Blinn-Phong specular highlight is added after the shadow stage. A configurable smoothness parameter controls the transition width and values near zero produce a sharp specular highlight.

Meta Avatars SDK

The Meta implementation runs inside the hook. Unlike the Mixamo and Avaturn shaders, this hook does not own the vertex stage. World-space position is instead forwarded from the SDK vertex output as an explicit fifth argument to the effect function. This makes the same U axis computation as the standalone platforms possible. Thus, `GetMainLight` is called with a per-fragment shadow coordinate derived from that world-space position. The shadow-attenuated NdotL is then computed using the same approach as on the standalone platforms:

```
float4 shadowCoord = TransformWorldToShadowCoord(positionWS); // per-fragment
    shadow coordinate
Light mainLight = GetMainLight(shadowCoord); // main light
float3 lightDir = normalize(mainLight.direction); // light dir
float NdotL = dot(normalWS, lightDir) * mainLight.shadowAttenuation; // shadow-
    attenuated Lambertian term
float rampU = lerp(0.5, saturate(NdotL * 0.5 + 0.5), _XToonLightSensitivity); //
     remap towards ramp midpoint
```

Listing 5.3: U axis (shadow-attenuated NdotL), Meta SDK.

The surface tangent frame is not available in the hook and this leaves Normal Field Abstraction via an abstract normal map unsupported. Only the geometric smoothing path is available on this platform. The abstraction compression and dynamic smoothing stages match those on the standalone platforms. The same `lightDir` and shadow attenuation from the U axis computation are reused for the specular term.

5.4.4. V3 — Screen-Space Normal Edge Detection

The normal gradient magnitude M_N (Equation (2.6)) maps to an edge weight through

$$E = \text{smoothstep}(\tau - \sigma,\ \tau + \sigma,\ M_N) \cdot \alpha_N \tag{5.1}$$

where τ is the threshold centre, σ is the threshold half-width and α_N is the edge weight scale. Both `ddx` and `ddy` operate on screen-space pixel derivatives rather than object-space geometry and the resulting edge width therefore scales with viewing distance rather than object scale, as discussed in Section 4.4.5. These three symbols are exposed as shader properties on all three platforms and can be adjusted at runtime through the parameter tuning interface described in Section 5.5.1:

- `_NormalEdgeThreshold` (τ)
- `_NormalEdgeSmoothness` (σ)
- `_NormalEdgeStrength` (α_N)

This edge mask is composited over the shaded surface by linear interpolation towards a configurable line colour.

Mixamo and Avaturn

The normal must be decoded from the PBR normal map before the derivative is computed. The vertex shader builds the tangent frame from the mesh tangent attribute and the fragment shader then unpacks the normal map and transforms it into world space:

```
// Mixamo, decoded by the Unity texture importer:
float3 normalTS = UnpackNormalScale(
                      SAMPLE_TEXTURE2D(_BumpMap, sampler_BumpMap, uv),
    _BumpScale);
// Avaturn, loaded by GLTFast without that importer pass:
float3 normalTS = SAMPLE_TEXTURE2D(_BumpMap, sampler_BumpMap, uv).rgb * 2.0 -
    1.0;
float3x3 TBN = float3x3(tangentWS, bitangentWS, geomNWS); // tangent frame
float3 nWS = normalize(mul(normalTS, TBN)); // rotate normal into world space
```

Listing 5.4: Normal decode via TBN (Mixamo and Avaturn).

When the normal map is disabled, `nWS` falls back instead to the bare interpolated geometry normal. The resulting `nWS` then feeds into the same gradient magnitude and threshold computation shown in Listing 5.5 for the Meta SDK platform, with `nWS` in place of `worldNormal`.

Meta Avatars SDK

On the Meta SDK platform, the world-space normal supplied to `ddx`/`ddy` is the SDK's fully composited fragment normal. This already incorporates the SDK's internal normal-map decode and no additional texture read is thus required. The derivative and threshold computation itself is identical across all three platforms:

```
float3 dNdx = ddx((float3)worldNormal); // horizontal screen-space derivative
float3 dNdy = ddy((float3)worldNormal); // vertical screen-space derivative
float M = sqrt(dot(dNdx, dNdx) + dot(dNdy, dNdy)); // gradient magnitude
float nMin = _NormalEdgeThreshold - _NormalEdgeSmoothness; // lower threshold
    bound
float nMax = _NormalEdgeThreshold + _NormalEdgeSmoothness; // upper threshold
    bound
float edge = smoothstep(nMin, nMax, M) * _NormalEdgeStrength; // final edge
    weight
```

Listing 5.5: V3 normal gradient and edge weight, Meta SDK.

5.4.5. V4 — Sobel Edge Detection

The edge source shifts from geometry to the diffuse albedo texture. The eight neighbouring positions of the 3×3 neighbourhood are sampled and converted to a scalar luminance using the BT.709 coefficients defined in Equation (2.5). These coefficients match the sRGB primaries of the rendered textures and the URP colour pipeline. Matching them reduces colour channel cross-talk. This would otherwise create false edges at hue boundaries, where the RGB channels shift in opposite directions but the luminance stays constant. The Sobel kernels G_x and G_y are applied to these eight values together with the centre sample. Subsequently, the resulting gradient magnitude is compared against a threshold to produce the edge mask.

Mixamo and Avaturn

These platforms read the raw diffuse albedo texture directly. As a result, edge detection responds to surface material boundaries rather than to lighting:

```
float3 L = float3(0.2126, 0.7152, 0.0722); // BT.709 luma weights
float off = _InnerLineBlur * 0.001; // UV-space sample offset

float tl = dot(SAMPLE_TEXTURE2D(_MainTex, sampler_MainTex, uv+float2(-off, off))
    .rgb, L); // top-left tap
float t  = dot(SAMPLE_TEXTURE2D(_MainTex, sampler_MainTex, uv+float2(0, off)).
    rgb, L); // top tap
// ...the remaining six neighbour samples (tr, l, r, bl, b, br) follow the same
    pattern

float Gx = (tr + 2*r + br) - (tl + 2*l + bl); // horizontal gradient
float Gy = (tl + 2*t + tr) - (bl + 2*b + br); // vertical gradient
float edgeMag = sqrt(Gx*Gx + Gy*Gy); // gradient magnitude
float edge = step(_InnerLineThreshold, edgeMag) * _InnerLineStrength; //
    threshold and scale
color.rgb = lerp(color.rgb, _InnerLineColor.rgb, edge); // composite line colour
```

Listing 5.6: Eight-tap luminance sampling and Sobel gradient.

Additionally, the gradient magnitude is compared directly against `_InnerLineThreshold` to produce the edge mask.

Meta Avatars SDK

The Meta SDK hook receives only the composited PBR output rather than a raw albedo texture. The Sobel therefore responds to shadow boundaries and specular gradients in addition to material transitions. This reduces selectivity, but it still adds visible edges along clothing boundaries and broad facial contour lines. The Meta Avatar's stylised, illustrated texture nonetheless contains less high-frequency variation than the photographic albedo used on Avaturn and this yields a comparatively cleaner edge response, as discussed in Section 4.4.6. Only the sample source differs from Listing 5.6. The gradient and magnitude computation is otherwise identical:

```
float offset = _SobelSampleDist * 0.001; // UV-space sample offset
float tl = dot(tex2D(u_BaseColorSampler, uv+float2(-offset, offset)).rgb, L); //
    top-left tap
// ...the remaining seven taps follow the same pattern
```

Listing 5.7: Composited-colour tap replacing the albedo tap, Meta SDK.

Compositing the PBR output reintroduces the UV seam artefacts. This artefact is absent on the other two platforms, where sampling occurs on the pre-lighting albedo instead. The colour difference across a seam registers as a strong edge despite the absence of any real visual boundary. Two bounds suppress these false detections:

```
float lMin = min(min(min(tl,t),min(tr,l)), min(min(r,bl),min(b,br))); // darkest
     sample
float lMax = max(max(max(tl,t),max(tr,l)), max(max(r,bl),max(b,br))); //
    brightest sample

float seamMask = step(lMax - lMin, _SobelSeamLimit); // reject seam spikes
float inBand = step(_SobelThreshold, edgeMag)
             * step(edgeMag, _SobelMax); // accept only valid range

float edge = inBand * seamMask * _SobelStrength; // combine gates and scale
color.rgb = lerp(color.rgb, _InnerLineColor.rgb, edge); // composite line colour
```

Listing 5.8: Seam suppression and magnitude band gate, Meta SDK.

5.4.6. V4.2 — Gaussian Prefiltered Sobel

This method carries over the V4 Sobel gradient computation across the same eight positions, but each luminance sample is now drawn through a prefilter rather than a single point sample. At each call, one centre sample, four axis-aligned cardinal samples and four diagonal corner samples are drawn at a configurable offset from the given UV position. The three weight groups are normalised to unity before use so that the output is always a valid luminance estimate.

This prefiltering is implemented as a shared helper function, applied identically on all three platforms except for the texture it reads:

```
// samples a 9-tap weighted average of luminance around uv
float SampleLuminanceBlurred(float2 uv, float blurR, float cW, float cardW,
    float diagW)
{
    float3 L = float3(0.2126, 0.7152, 0.0722); // BT.709 luma weights
    float  v = 0.0; // accumulated luminance
    v += dot(SAMPLE_TEXTURE2D(_MainTex, sampler_MainTex, uv).rgb, L) * cW; //
    centre tap
    v += dot(SAMPLE_TEXTURE2D(_MainTex, sampler_MainTex, uv + float2(blurR, 0)).
    rgb, L) * cardW; // cardinal tap
    // ...three more cardinal taps and four diagonal taps follow the same
    pattern
    return v; // blurred luminance estimate
}

// Weight normalisation: computed once before the eight Sobel positions are
    sampled
float totalW = _GaussianCenterWeight + 4.0*_GaussianCardinalWeight + 4.0*
    _GaussianDiagonalWeight;
float cW    = _GaussianCenterWeight   / totalW; // centre weight share
float cardW = _GaussianCardinalWeight / totalW; // cardinal weight share
float diagW = _GaussianDiagonalWeight / totalW; // diagonal weight share
```

Listing 5.9: 9-tap Gaussian luminance helper with weight normalisation (unified across platforms).

This helper reads the raw diffuse albedo texture on Mixamo and Avaturn and the composited PBR output on the Meta SDK, with the rest of the logic unchanged. It is called once at each of the eight Sobel neighbourhood positions and the resulting values feed the same gradient magnitude computation used in V4.

Additionally, a configurable threshold band, applied identically on all three platforms, gates this gradient magnitude. A lower bound rejects weak gradients as noise and an upper bound suppresses detections with an unusually large magnitude, such as UV seam spikes. This band gate replaces the luma range seam gate used in V4.

A series of four progressive smoothstep passes then sharpens the thresholded magnitude into a crisp edge mask. The Meta SDK variant shown below always evaluates all four passes and drives them from a single tightness scalar, rather than from the individually toggleable flags used on Mixamo and Avaturn:

```
float hw1 = lerp(0.5, 0.03, _GSobelTightness); // pass 1 half-width
edge = smoothstep(0.5 - hw1, 0.5 + hw1, edge); // sharpen pass 1
float hw2 = lerp(0.5, 0.15, _GSobelTightness); // pass 2 half-width
edge = smoothstep(0.5 - hw2, 0.5 + hw2, edge); // sharpen pass 2
// ...two further passes repeat this pattern with hw3 = 0.25 and hw4 = 0.35
edge = pow(edge, _GSobelPowerCurve); // contrast curve
edge *= _GSobelStrength; // final edge weight
```

Listing 5.10: 4-pass progressive smoothstep sharpening and power curve, Meta SDK.

The four passes together compose a sigmoid whose steepness is set by a single tightness value, and they reduce to a linear identity at zero. A power curve above 1 suppresses soft halos, while values below 1 amplify thin lines instead.

Furthermore, three preset configurations, Light, Moderate and Aggressive, are offered on Mixamo and Avaturn. Table 5.1 lists the parameter values assigned by each. The default on all three platforms is the Light configuration, and that is what the V4.2 figures show.

Table 5.1.: V4.2 preset parameter values (Mixamo and Avaturn).

Parameter	Light	Moderate	Aggressive
Blur radius multiplier (`_BlurRadiusMultiplier`)	0.8	1.0	1.2
Centre weight (`_GaussianCenterWeight`)	0.25	0.30	0.25
Cardinal weight (`_GaussianCardinalWeight`)	0.125	0.12	0.125
Diagonal weight (`_GaussianDiagonalWeight`)	0.0625	0.05	0.0625
Threshold min multiplier (`_ThresholdMinMultiplier`)	0.2	0.4	0.85
Threshold max multiplier (`_ThresholdMaxMultiplier`)	2.0	1.6	1.15
Smoothstep tightness (`_SmoothstepTightness`)	0.2	0.3	1.0
Power curve (`_PowerCurve`)	1.5	2.0	3.0

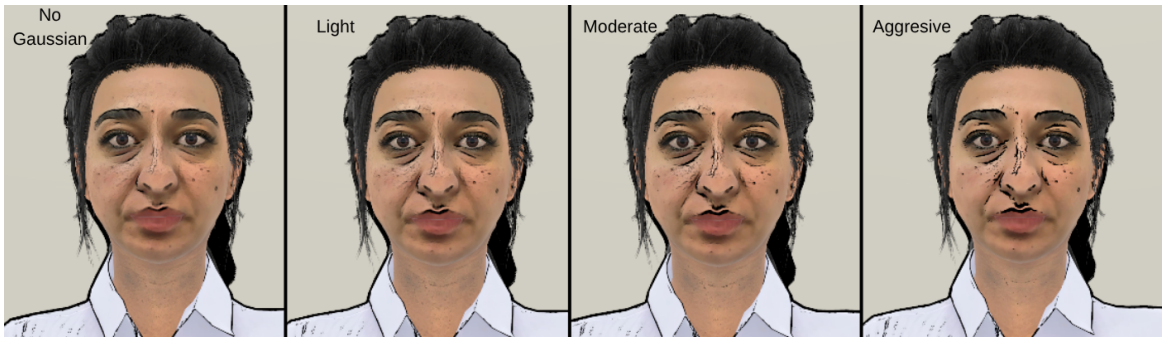


Figure 5.1.: V4.2 preset configurations compared on the Avaturn avatar. From left to right: no Gaussian prefilter, then the Light, Moderate and Aggressive presets of Table 5.1.

The presets sharpen more than they blur. Each step narrows the threshold band and steepens the power curve. The gradients that survive are therefore rendered as harder lines. This amplified the baked-illumination responses around the nose and eye sockets on the Avaturn albedo. The presets sharpened those responses alongside the genuine feature boundaries rather than suppressing them.

5.4.7. V5 — Hierarchical Edge Detection

V5 combines three independent edge cues into a unified boundary representation. Each cue is thresholded into a line signal and scaled by its own weight and the three weighted signals are then fused by max-pooling rather than addition. Thus, the strongest cue at a fragment dominates rather than being diluted by the others. However, the three input layers are computed differently on each platform. Mixamo and Avaturn provide direct access to the depth, normal and colour buffers. On the other hand, the Meta SDK exposes only the per-fragment hook.

Mixamo and Avaturn

Mixamo and Avaturn run V5 as a URP renderer feature: a post-processing pass that samples Unity's scene depth, normal and colour buffers directly. Each layer applies the Roberts Cross operator to a 2×2 diagonal sample pattern, as defined in Section 2.4.1:

```
float d_tl = GetLinearEyeDepth(uv + float2(-off.x,  off.y)); // top-left depth
    tap
// ...remaining three depth taps (tr, bl, br) follow the same pattern
float depthEdge = RobertsCross(d_tl, d_tr, d_bl, d_br)
                / max(GetLinearEyeDepth(uv), 0.01) * _DepthScale; // normalised
    , scaled edge
float depthLine = smoothstep(_DepthThreshold - 0.01, _DepthThreshold + 0.01,
    depthEdge); // to line signal
```

Listing 5.11: Roberts Cross depth layer (Mixamo and Avaturn).

The normal layer applies the same Roberts Cross pattern to the screen-space normal buffer. The colour layer applies it to the rendered colour buffer's luminance and an added chromatic term also responds to boundaries defined by colour alone, with no luminance difference. An adaptive factor derived from local brightness reduces the depth and colour layers' sensitivity in dark image regions, where noise would otherwise dominate the response. The normal layer receives only half this reduction, as its response depends less on scene brightness.

Since this post-processing pass operates on the entire screen, background geometry outside the avatar would also register spurious edges without correction. A dedicated avatar mask therefore restricts the composite to the avatar's screen-space silhouette. This mask is captured in a separate render pass, filtered to a Unity rendering layer assigned exclusively to the avatar. Consequently, only avatar fragments are drawn into it. The resulting mask value is then multiplied into the edge blend factor.

Meta Avatars SDK

The Meta SDK's single fragment hook has no access to a dedicated depth or normal buffer. The depth layer consequently estimates a distance gradient computed from the fragment's world-space distance to the camera. This gradient is normalised for perspective:

```
float depth = length(worldPos - _WorldSpaceCameraPos); // fragment-to-camera
    distance
float dDepthX = ddx(depth); // horizontal depth derivative
float dDepthY = ddy(depth); // vertical depth derivative
float depthGrad = sqrt(dDepthX*dDepthX + dDepthY*dDepthY) / max(depth, 0.01) *
    _DepthScale; // normalised, scaled gradient
float depthLine = smoothstep(_DepthThreshold - 0.005,
    _DepthThreshold + 0.005, depthGrad); // threshold to line signal
```

Listing 5.12: Depth layer, Meta SDK.

This proxy detects distance discontinuities within the avatar mesh. However, it cannot separate the avatar from the background. The background lies entirely outside the per-fragment computation. The normal layer reuses the world-space normal derivative computation from V3 and an AHEAD-style adaptive gain then amplifies this layer specifically in darker image regions [40]:

```
float luma = dot(color.rgb, float3(0.2126, 0.7152, 0.0722)); // perceptual
    luminance
float adapt = min(1.0 / (3.0 * luma + 0.1), 4.0); // inverse-linear gain, capped
     at 4x
normalLine *= lerp(1.0, adapt, _AdaptiveStrength); // blend gain by strength
```

Listing 5.13: AHEAD adaptive gain on the normal layer, Meta SDK.

The composited PBR output also contains baked lighting gradients that produce false Roberts Cross responses on this platform. The colour layer accordingly replaces Roberts Cross with XDoG [20], [21]: the difference between two Gaussian-blurred luminance samples at different radii, passed through a tanh soft threshold:

```
float g1 = GaussianLuma(uv, sigma1); // fine blur
float g2 = GaussianLuma(uv, sigma1 * _XDoGK); // coarse blur
float D = g1 - _XDoGTau * g2; // difference of Gaussians
float colorLine = D < -_ColorThreshold
    ? saturate(-tanh(_XDoGPhi * (D + _ColorThreshold))) : 0.0; // sharpened line
     signal
```

Listing 5.14: XDoG colour layer, Meta SDK.

Subtracting the coarser blur from the finer one cancels out broad, slowly varying illumination gradients. The sharper transitions that mark genuine material boundaries remain unaffected. This directly targets the baked lighting artefact described above, without requiring any change to the underlying composited input. The Meta SDK then applies one additional smoothstep pass to sharpen the fused edge value, while Mixamo and Avaturn composite it directly.

5.4.8. V6 — Kuwahara Painterly Filter

All three platforms use the anisotropic Kuwahara filter of Kyprianidis et al. [22]. This filter relies on a structure tensor to estimate the local feature orientation and anisotropy. The sampling neighbourhood then becomes an ellipse aligned with that orientation rather than a fixed square.

Mixamo and Avaturn

The structure tensor is computed as a Sobel gradient over luminance in a dedicated post-processing pass on these platforms. A second pass subsequently smooths the tensor with a 5-tap separable Gaussian blur before the main filter pass reads it:

```
float gx = -tl - 2.0*l - bl + tr + 2.0*r + br; // Sobel horizontal
float gy = -tl - 2.0*t - tr + bl + 2.0*b + br; // Sobel vertical
return float4(gx*gx, gx*gy, gy*gy, 1.0); // pack tensor components (E, F, G)

// Second pass: 5-tap separable Gaussian blur over the packed tensor
float4 result = SAMPLE_TEXTURE2D_X_LOD(_BlitTexture, sampler_LinearClamp, uv, 0)
     * 0.375; // centre tap
// ...four more taps at +/-offset and +/-2*offset, weighted 0.25 and 0.0625,
    complete the blur
```

Listing 5.15: Structure tensor pass and Gaussian blur pass (Mixamo and Avaturn).

Meta Avatars SDK

The Meta SDK cannot perform a blur pass inside its single fragment hook. The tensor is therefore estimated directly from `ddx`/`ddy` on the already-shaded luminance and this produces a noisier, unblurred orientation estimate. The shaded input adds further noise, since its gradients carry lighting variation as well as material variation:

```
float lum = dot(color.rgb, float3(0.2126, 0.7152, 0.0722)); // BT.709 luma
    weights
float gx = ddx(lum); // horizontal luminance derivative
float gy = ddy(lum); // vertical luminance derivative
float E = gx * gx; // tensor component E
float F = gx * gy; // tensor component F
float G = gy * gy; // tensor component G
```

Listing 5.16: Structure tensor, Meta SDK (unblurred).

All three platforms then derive the dominant orientation angle and the anisotropy ratio from the same eigenvalue decomposition of this tensor, given by

$$\lambda_{1,2} = \frac{E+G}{2} \pm \sqrt{\left(\frac{E-G}{2}\right)^2 + F^2}, \tag{5.2}$$

$$\phi = \frac{1}{2}\arctan\left(\frac{2F}{E-G}\right), \tag{5.3}$$

$$A = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2}, \tag{5.4}$$

where λ_1 and λ_2 are the tensor eigenvalues, ϕ is the dominant orientation and A is the anisotropy ratio.

The anisotropy A scales the ellipse's major and minor axes, elongating the sampling neighbourhood along the feature direction and narrowing it across. This axis scaling is a clamped linear blend between the isotropic and fully anisotropic extremes on Mixamo and Avaturn. The Meta SDK instead applies the ratio formula from Kyprianidis et al. directly, driven by an additional eccentricity parameter with no equivalent on the other platforms.

Eight overlapping elliptical sectors are then sampled around this ellipse rather than four fixed rectangular subregions, though the sampling strategy itself differs by platform.

Mixamo and Avaturn scan every pixel within the kernel radius and accumulate it into whichever sector it falls into, and the sample count grows with the radius. The Meta SDK instead takes a fixed three samples per sector at set radii, plus one sample shared across all sectors, for a constant total regardless of radius. Each sector accumulates a mean colour and a variance. These are then blended by an inverse-variance weight, so that the smoothest sector dominates the result. That approach follows Kyprianidis et al., who weight each sector inversely to its standard deviation [22]. The Meta SDK applies that form directly. Mixamo and Avaturn instead raise the variance to half the exponent, which is algebraically identical and avoids a square root on every sector. The accumulation that follows the weight is shared across all three platforms:

```
// weight expression, Meta SDK
w_i = 1.0 / pow(max(tau, sigma_i), q);
// weight expression, Mixamo and Avaturn
w_i = 1.0 / (1.0 + pow(variance_i * 1000.0, 0.5 * _Sharpness));

// accumulation, shared across all three platforms
result += mean_i * w_i; // accumulate weighted colour
totalWeight += w_i; // accumulate normalisation factor
```

Listing 5.17: Inverse-variance sector weight and the shared accumulation step.

The final colour on both platforms is obtained by normalising the accumulated weighted colour by the total weight. Mixamo and Avaturn then composite this result over the original shaded colour through the same avatar mask pass used in V5, since V6 also runs as a post-processing effect on these platforms. The Meta SDK instead blends by a strength parameter.

Table 5.2 collects the divergences described across the preceding subsections. One pattern runs through all of them. Every divergence traces back to a stage that the SDK either resolves internally before the hook executes or does not expose to it at all.

Table 5.2.: Principal points at which the Meta Avatars SDK integration diverges from the Mixamo and Avaturn implementations.

Technique	Aspect	Mixamo / Avaturn	Meta SDK
V1	Outline depth	Depth written, standard test	Depth write disabled, looser test
	Alpha cutout	Explicit clip in both passes	Resolved by the SDK before the hook
V2	Quantisation input	Lambertian dot product	Composited RGB output
	Rim and shadow	Computed within the pass	Pre-composited by the SDK
V2.2	Normal abstraction	Abstract normal map or geometric smoothing	Geometric smoothing only, no tangent frame
	Shadow coordinate	Per-vertex, interpolated	Per-fragment
V3	Normal source	Decoded normal map, one extra sample	SDK composited normal, no extra sample
V4	Gradient input	Raw diffuse albedo	Composited PBR colour
	Threshold gate	Single lower threshold	Band gate plus seam rejection
V4.2	Sharpening passes	Four, individually toggleable	Four, driven by one tightness scalar
V5	Architecture	URP renderer feature, post-processing pass	Per-fragment hook
	Depth layer	Roberts Cross on the scene depth texture	Camera-distance derivative proxy
	Colour layer	Roberts Cross on luminance plus a chromatic term	XDoG
	Avatar isolation	Dedicated mask pass	Not required
V6	Tensor smoothing	Separable Gaussian blur pass	None, unblurred
	Compositing	Masked composite via the mask pass	Strength-parameter blend

5.5. Runtime Tooling

Two runtime utilities were developed to address constraints that cannot be resolved at shader compilation time. The first enables interactive parameter exploration within the VR headset during the visual assessment phase. The second freezes the avatar's pose for evaluation screenshots.

5.5.1. In-VR Parameter Tuning Interface

Shader parameters can also be adjusted through a custom world-space UI panel during live headset sessions, reproducing the same controls available in the Unity Inspector. This allowed values to be explored and compared while the avatar remained visible in the headset at correct VR scale. This panel was central to the visual assessment process used to select the technique for the user study.

5.5.2. Pose Freeze for Evaluation

A dedicated script freezes the avatar's body-tracking pose at a chosen instant, preserving a specific expression and posture for evaluation screenshots. Since the Meta Avatars SDK exposes no public API for pausing tracking, the freeze is implemented through reflection. A snapshot provider that returns a fixed pose state is substituted for the SDK's live input provider.

5.6. Animation and Recording Pipeline for the User Study

Facial animation for the Avaturn avatar is driven using Apple ARKit. The Live Link Face application on an iPhone streams all 52 ARKit blendshape coefficients and head rotation to Unity. It sends them over the user datagram protocol (UDP). A custom component there maps each coefficient to the corresponding Avaturn

blendshape target. This stream drives the blendshape weights at runtime. As a result, the avatar's expression follows the iPhone performer directly with no intermediate animation clip. The same captured stream was then replayed for both the photorealistic and the NPR render. Any visible difference between C2 and C3 therefore originates from the shader rather than from the animation data.

Moreover, the camera in the recording scene used a reduced field of view of 20°, compared with the 60° default used in the general platform comparison scenes described in Section 5.2.4. A wide angle exaggerates the facial features nearest the lens through perspective distortion, most visibly the nose. This effect is strongest at the close camera distance required for a talking-head framing. Narrowing this angle suppresses the distortion while preserving an equivalent framing of the avatar.

6. Evaluation

This chapter evaluates the NPR avatar system developed in this thesis. The first part documents the visual quality assessment of the eight NPR techniques across the three avatar platforms, establishing which technique was carried into the user study and why. The second part presents the user study that tests whether NPR stylisation affects how participants perceive trust in a speaking avatar. It then interprets each result in relation to the study hypotheses.

6.1. Visual Quality Assessment

6.1.1. Assessment Methodology

Each NPR technique was assessed qualitatively inside the Unity Editor and on the Meta Quest 3 headset, using the world-space UI panel described in Section 5.5.1. The avatar was placed in the shared studio environment described in Section 5.2.4 and animated with a repeating idle loop. It was observed from several distances, including the close conversational framing of approximately 1 m used to record the user study videos. Parameters were adjusted interactively through this panel until the visual output satisfied four criteria:

1. **Facial readability.** Facial features (eyes, mouth boundary, nose bridge and jaw line) must remain individually legible under stylisation.
2. **Temporal stability.** Edges and stylisation effects must remain stable during animation. Screen-space techniques are particularly susceptible to flickering when the avatar translates across the screen, because the pixel-space neighbourhood shifts while the underlying geometry does not.
3. **Artefact suppression.** Visible UV seam lines, rectangular outlines around hair cards, z-fighting at the outline hull and false edge detections at skin-tone boundaries must be suppressed to a level that is not noticeable under normal observation.
4. **Expressive abstraction.** The stylisation should be visually legible as an intentional artistic choice, not as a rendering error. A technique that produces only isolated noise marks instead of coherent contours would not read as deliberate stylisation.

Evaluation proceeded iteratively for each technique. Parameters were first tuned on the Mixamo avatar, where unrestricted shader access made per-pass debugging straightforward. They were then verified on Avaturn and the Meta SDK avatar. Techniques that satisfied all four criteria on at least two of the three platforms were considered viable candidates for the user study.

6.1.2. Per-Technique Assessment

V1 — Geometric Silhouette Outline. As the shared silhouette pass underlying every other technique, V1 was also assessed on its own as the minimal abstraction reference in the study materials. Temporal stability holds without additional tuning, as the outline geometry is generated in the vertex stage rather than from a screen-space pixel neighbourhood. The alpha cutout and depth bias described there remove rectangular hair-card borders and silhouette z-fighting. Artefact suppression is thereby satisfied. Facial readability holds as well, since the pass never alters the underlying texture or geometry. However, expressive abstraction does not: the method marks only the external silhouette. Creases, clothing seams and facial contours consequently remain undelineated, consistent with its interior boundary limitation. V1 was retained as the foundational outline pass and lowest abstraction reference, but not as a standalone candidate for the composite.

V2 — Toon Shading. The quantised diffuse term yields a consistent, low-frequency tonal structure that satisfies both facial readability and temporal stability without flicker. Clothing seams and facial features carry no dedicated edge cue beyond the local tonal contrast that the shading bands happen to produce. These

accordingly stay unmarked at the 1 m framing used in the study, matching the interior line limitation already noted in Section 4.4.3. Nevertheless, unlike V1, the coherent tonal bands still read as a deliberate stylistic choice and not an incomplete or erroneous render. Expressive abstraction remains satisfied despite the missing interior detail.

V2.2 — X-Toon Shading. The abstraction level axis described in Section 4.4.4 is driven by camera distance or surface curvature. At the wider framing used for general platform comparison, the two modes differ only marginally and Avaturn exhibits slightly greater variation than the other platforms. However, neither mode registers a meaningful signal at the 1 m conversational framing used in the study. The depth range across the avatar's torso and face spans only a few centimetres, too narrow relative to the configured depth window. Correspondingly, the screen-space normal derivatives produced by curvature mode across the smooth head and body geometry remain close to zero, too faint to shift the ramp lookup. Ultimately, neither dynamic mode yields a strongly different outcome at this fixed conversational distance. This finding is why manual mode was selected as the driving mode for the composite assembled for the user study, as Section 4.5 explains. Because manual mode holds a fixed constant without recomputing a value per fragment, it carries none of the flicker risk that affects the screen-space techniques assessed elsewhere in this section. Facial readability and temporal stability are both satisfied without qualification. Expressive abstraction is likewise satisfied. The fixed detail bias visibly compresses the tonal bands towards the ramp midpoint, even though the parameter adjustment study found no significant preference across the tested bias values (Section 4.7).

V3 — Screen-Space Normal Edge Detection. The camera distance stays fixed throughout each study recording. This is why the distance-dependent edge thickening identified in Section 4.4.5 remains mild here rather than growing as it would under a varying viewing distance. Facial readability is nonetheless not satisfied, since the eyebrow, lip and eyelid boundaries remain undetected. This is the primary reason V3 was not selected for the composite (Section 4.5). Despite this exclusion, V3 adds only a single texture sample for the normal map. This makes it the least costly technique after V2.

V4 — Sobel Edge Detection. Baked-illumination artefacts, such as the nose bridge artefact identified in Section 4.4.6, are suppressed on the Avaturn avatar through the Blender texture preprocessing described in Section 4.6 and a raised threshold on the head material. The skin-to-clothing boundary generates the most visually significant lines. Eyebrow and lip boundaries emerge at moderate threshold settings. The step function used for thresholding on Mixamo and Avaturn (Section 5.4.5) can toggle a pixel between frames as lighting varies slightly during animation. Temporal stability required the same threshold tuning to keep lines at material boundaries from flickering.

V4.2 — Gaussian Prefiltered Sobel. The facial segment recovery shortfall on Avaturn and the resulting shift to texture preprocessing are described in Section 4.4.7. This came at a cost of up to 72 texture samples per fragment. The preprocessing pipeline made the basic Sobel operator sufficient on its own. The added sampling cost of Gaussian prefiltering was thus no longer justified for the composite assembled for the user study.

V5 — Hierarchical Multi-Cue Edge Detection. V5 was a candidate addition to the C3 composite, assessed together with the Kuwahara filter through the parameter adjustment study. The interior contours do not resolve clear boundaries between adjacent facial segments, as Section 4.4.8 describes. This leaves facial readability unsatisfied. The same limitation extends to artefact suppression. The colour layer's response to illumination gradients around the nose bridge is left unaddressed on Avaturn, unlike on the Meta Avatars SDK. This candidate addition was subsequently excluded from the final composite following that study.

V6 — Kuwahara Painterly Filter. V6 does not satisfy facial readability on its own at radius values large enough to read as clearly painterly. This matches the trade-off between abstraction and legibility identified in Section 4.4.9. The trade-off motivated testing V6 as an addition to the C3 composite. The ranking outcome thereafter excluded it from the final configuration.

Table 6.1.: Summary of visual quality assessment results across the eight NPR techniques on the Avaturn platform. Criteria are rated qualitatively: ✓= satisfactory without tuning, (✓) = satisfactory after tuning, × = limitation observed.

Technique	Facial readability	Temporal stability	Artefact suppression	Expressive abstraction
V1 — Outline	✓	✓	(✓) alpha cutout, depth bias	× not in isolation
V2 — Toon	✓	✓	✓	(✓) no interior lines
V2.2 — X-Toon	✓	✓	✓	(✓) manual mode chosen for flexibility
V3 — Normal edge	× facial landmarks undetected	✓	✓	✓
V4 — Sobel	(✓) moderate threshold	(✓)	(✓) preprocessing	✓
V4.2 — Gaussian Sobel	× insufficient for facial segments	(✓)	✓	✓
V5 — Hierarchical	× segment boundaries unclear	✓	× nose bridge artefact	✓
V6 — Kuwahara	× at large r	✓	✓	✓

6.1.3. Technique Comparison Summary

6.1.4. Technique Selection for the User Study

The eight techniques assessed were narrowed to five candidates. The reasons for setting aside basic toon shading, screen-space normal edge detection and Gaussian prefiltered Sobel are given in Section 4.5.

The parameter adjustment study then determined which of the five components to retain and how to configure them. Only the geometric silhouette outline, X-Toon tonal shading and Sobel edge detection were carried forward into the final composite. Section 6.2.1 reports that process in detail. The combination uses the submesh material split established for the Avaturn avatar (Section 5.3). This lets the Blender-preprocessed texture suppress baked-illumination artefacts on the head material alone while facial features stay legible. The combination also reads clearly as an intentional stylisation rather than a rendering artefact. The manipulation check embedded in the study questionnaire (Part H) depends on that clarity. A technique that produces only subtle modifications to the photorealistic baseline could fail the check and reduce usable data.

6.2. User Study Results

6.2.1. Parameter Adjustment Study Results

The study was distributed through personal networks, a university mailing list and social media and it ran for approximately two weeks. A total of 45 participants took part. Each round was analysed separately. Any ranking that did not assign every rank exactly once was excluded from that round alone. Between 40 and 44 valid ballots per round remained.

The parameter adjustment study described in Section 4.7 fixed the parameter values used in the C3 composite and produced the empirical basis for excluding two candidate components (Figure 6.1). Table 6.2 reports each round.

Two outcomes determine the final composite. In R1, threshold 0.05 received the most first-place votes (20 of 43 participants), yet threshold 0.03 accumulated the higher cumulative Borda score (160 vs. 149 points), since it was ranked more consistently near the top across all participants. The pairwise Wilcoxon signed-rank test found no significant difference between the two ($p = .31$), but the omnibus Friedman result confirmed a clear directional trend. The two lowest thresholds were significantly disfavoured. The top two configurations were accordingly treated as equivalent. In R2, the narrower outline width (0.003, depth offset on) accumulated the highest Borda total but was not statistically distinguished from the standard width (0.005, depth offset on, $p = .82$). The narrower setting was therefore retained since no significant preference justified the wider

Table 6.2.: Statistical test results for the parameter adjustment study by round.

Round	Borda winner	Friedman	Runner-up test
R1	threshold 0.03	$\chi^2_F(4) = 36.48$, $p < .001$	$p = .31$ vs. 0.05 (not distinguished)
R2	width 0.003, offset on	$\chi^2_F(5) = 126.19$, $p < .001$	$p = .82$ vs. width 0.005, offset on (not distinguished)
R3	shadow 0.5, bias 1.0	$\chi^2_F(4) = 3.18$, $p = .53$	no configuration differed significantly
R4	none (no Kuwahara)	$\chi^2_F(3) = 38.97$, $p < .001$	$p = .003$ vs. best Kuwahara variant (significant)
R5	none (no hierarchical layer)	$\chi^2_F(5) = 117.20$, $p < .001$	$p < .001$ vs. best threshold (significant)

option. In R3, no significant differences emerged across any configuration (p = .53) and the Borda winner was shadow strength 0.5, detail bias 1.0, with no configuration statistically distinguishable from another. However, in both R4 and R5 the baseline without any post-processing was significantly preferred over every tested variant of the Kuwahara filter and the hierarchical depth layer respectively. Because trustworthiness is the primary outcome measure of the main study, both passes were excluded from the C3 composite on this basis, consistent with Section 4.5. The per-round ranking distributions for all configurations are provided in Appendix A.

None of R1, R2 and R3 produced a statistically distinguishable winner. R2 was settled by retaining its Borda winner for the reason given above. A final question asking which avatar participants found most visually appealing served as the tiebreaker for the other two rounds. R1E (threshold 0.05) received the most nominations overall with seven and R3B (shadow strength 0.0, detail bias 0.5) followed with six. Both were thus selected for their respective rounds.

This pattern suggests that perceived trustworthiness in this ranking paradigm was more sensitive to whether an additional abstraction layer was present at all than to fine calibration within a layer already established.

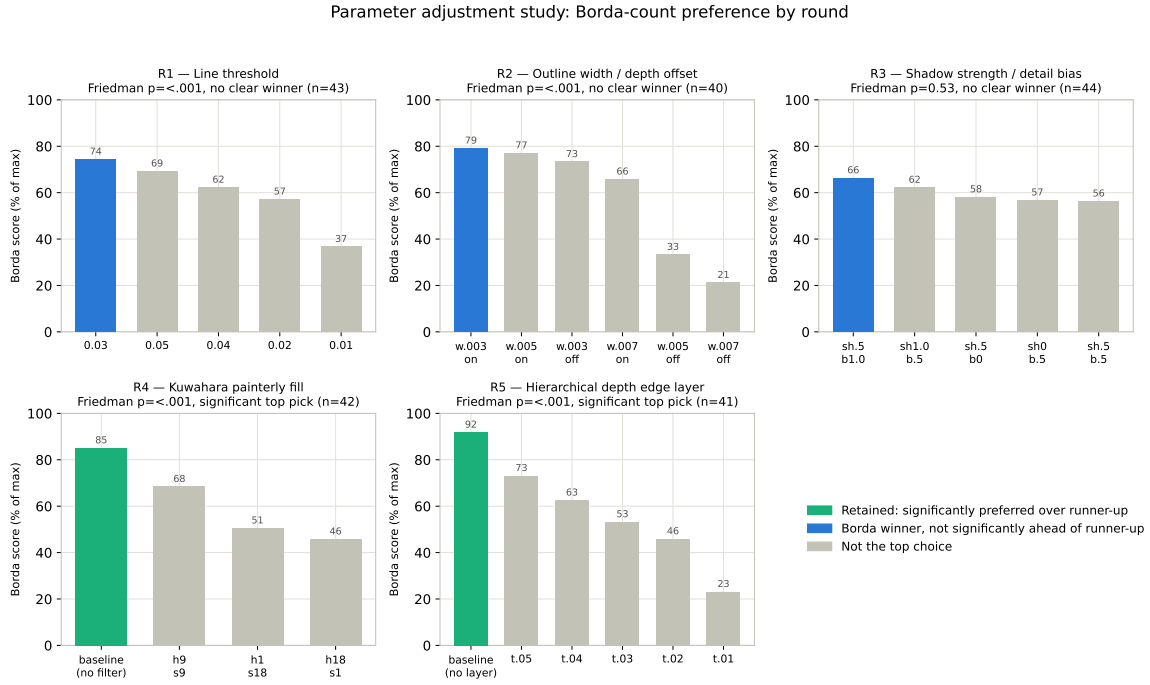


Figure 6.1.: Borda-count preference score by round, as a percentage of the maximum possible score. Green marks the baseline configuration retained after a significant preference in R4 and R5. Blue marks the Borda winner in R1–R3, where the top choice was not significantly ahead of the runner-up.

6.2.2. Participants

Participants were recruited through the same channels as the parameter adjustment study and over a comparable two-week window. The two samples were not identical, but an unknown share of participants took part in both. Neither round recorded identifying data, so the overlap cannot be quantified. Anyone who ranked stylised variants earlier may have arrived with some awareness of the manipulation. The suspicion check reported below bears on that possibility directly.

Inclusion in the study required an age of 18 or older and sufficient language proficiency to follow the vignette scenarios. A total of 52 participants took part and 11 were removed under the exclusion rule of Section 4.8.4. Thus, every test reported below draws on the remaining 41 participants, each with valid data in all three conditions. An implementation error altered the recap thumbnails in one study version. Those responses were relabelled from the available evidence instead of being excluded and a residual risk that the correction misses a participant's intended response remains.

Table 6.3.: Participant demographics and study version distribution.

Parameter	**Value**
Total enrolled	52
Excluded (failed a gating check)	11
Final analytic sample	41
Age: mean (SD)	27.6 (5.4) years
Gender	23 women, 17 men, 1 non-binary
Avatar/CGI interaction frequency: mean (SD)	2.88 (1.35), 1–5 scale
General trust in new people: mean (SD)	3.54 (1.61), 1–7 scale
Study version (A–C)	15 / 12 / 14

The closing suspicion check asked participants what they thought the study investigated and 24 of the 41 named avatar realism, stylisation or trust. Participants who inferred the manipulation could rate conditions to match what they assumed was expected, not their genuine reaction. Accordingly, the H1 and H2 tests were repeated on the remaining 17 and the pattern of results did not change. This suggests participants answered according to their own honest impressions.

6.2.3. Measure Reliability

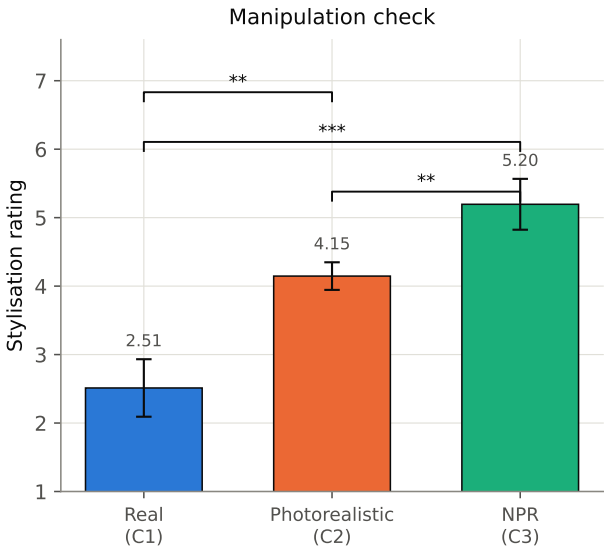
Each multi-item subscale was averaged into a single composite score for the tests reported below. Internal consistency was checked first, as Section 4.8.4 specifies, to establish that the items within a subscale measure one construct. Cronbach's α exceeded the conventional .70 threshold on every subscale, both overall and within each condition separately (Table 6.4). Averaging the items is therefore justified for all five composites and no subscale required item removal.

Table 6.4.: Cronbach's α per subscale, computed across all blocks and separately per condition. k = number of items, n = 41 participants contributing 123 blocks.

Subscale	k	**Overall**	**C1**	**C2**	**C3**
Likeability	5	.94	.90	.94	.93
Eeriness	8	.87	.87	.80	.86
Ability	3	.96	.96	.96	.94
Benevolence	3	.95	.95	.95	.94
Integrity	3	.92	.81	.93	.94

6.2.4. Manipulation Check

Part H asked participants to rate how visually stylised each advisor appeared, from "completely realistic" (1) to "very stylised" (7). A successful manipulation requires C3 to score significantly higher than C2, with C1 lowest.



Comparison	Result
Omnibus (condition)	$F(2, 80) = 15.45$ $p < .001$, $\eta_p^2 = .28$
C1 vs C2	$p = .001$, $g = 0.87$
C1 vs C3	$p < .001$, $g = 1.23$
C2 vs C3	$p = .004$, $g = 0.55$

Figure 6.2.: Left: mean visual stylisation rating by condition, Part H of the post-study questionnaire ($n = 41$), with within-subject corrected SEM error bars. Right: Greenhouse-Geisser corrected repeated-measures ANOVA (sphericity violated, $W = .53$, $p < .001$) and Holm-corrected pairwise comparisons, reported as Hedges' g.

Mean stylisation ratings increased monotonically from Real through Photorealistic to NPR (Figure 6.2). A repeated-measures ANOVA confirmed a significant effect of condition and Holm-corrected pairwise comparisons separated all three conditions from each other. The manipulation thus succeeded. This means any null result on eeriness or trust below cannot be attributed to an unnoticed rendering difference.

6.2.5. Hypothesis Tests

All tests below follow the analysis plan pre-specified in Section 4.8.4.

H1 — NPR Reduces Eeriness

H1 predicted that the NPR composite in C3 would lower scores on the eeriness subscale relative to the photorealistic avatar in C2.

Table 6.5.: Repeated-measures ANOVA and Holm-corrected pairwise comparisons (Hedges' g) per subscale, $n = 41$.

Subscale	Omnibus	C1 vs C2	C1 vs C3	C2 vs C3
Likeability	$F(2, 80) = 26.85$, $p < .001$, $\eta_p^2 = .40$	$p < .001$, $g = 0.92$	$p < .001$, $g = 1.02$	$p = .54$, $g = 0.08$
Eeriness	$F(2, 80) = 20.39$, $p < .001$, $\eta_p^2 = .34$	$p < .001$, $g = 0.98$	$p < .001$, $g = 0.98$	$p = .66$, $g = 0.07$
Ability	$F(2, 80) = 10.91$, $p < .001$, $\eta_p^2 = .21$	$p = .004$, $g = 0.53$	$p < .001$, $g = 0.75$	$p = .21$, $g = 0.20$
Benevolence	$F(2, 80) = 14.31$, $p < .001$, $\eta_p^2 = .26$	$p < .001$, $g = 0.59$	$p < .001$, $g = 0.79$	$p = .18$, $g = 0.19$
Integrity	$F(2, 80) = 12.47$, $p < .001$, $\eta_p^2 = .24$	$p = .001$, $g = 0.66$	$p < .001$, $g = 0.80$	$p = .27$, $g = 0.16$

Results indicated a pattern opposite to what H1 predicted. Eeriness rose from C1 through C2 to C3 (Figure 6.3, Table 6.5). This trend was significant in the omnibus test. The follow-up pairwise comparisons showed that C1 differed from both avatar conditions. By contrast, C2 and C3 were statistically indistinguishable from one another.

According to these test results, H1 is not supported. This is not for lack of a perceptible difference between C2 and C3, since the manipulation check already established that participants distinguished the two conditions on perceived stylisation (Section 6.2.4). Stylisation and eeriness instead appear to be separate effects, a dissociation that Section 6.2.5 explains further.

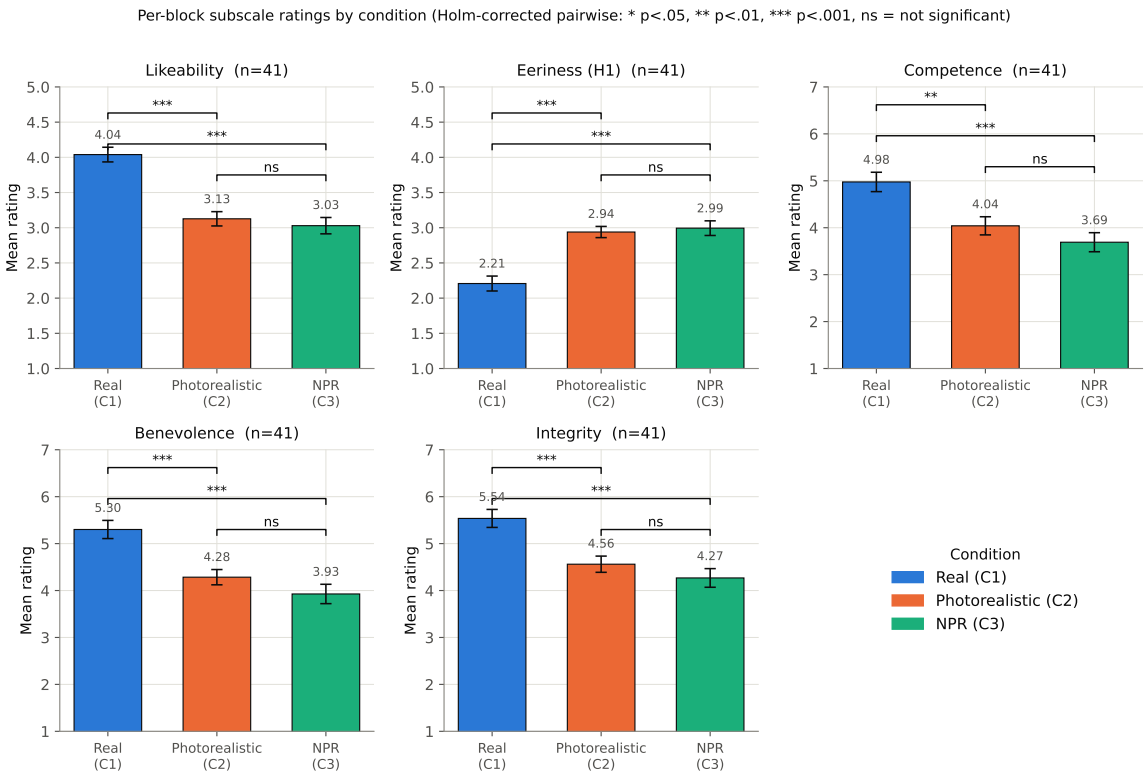
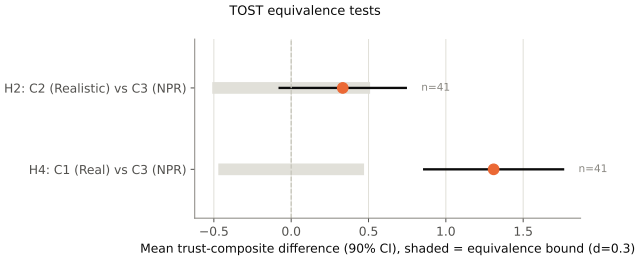


Figure 6.3.: Mean per-block rating by condition for the five subscales retained in the final questionnaire. Error bars show the within-subject corrected standard error of the mean. Brackets mark Holm-corrected pairwise comparisons: *** $p < .001$, ** $p < .01$, ns = not significant. $n = 41$.

H2 — NPR Does Not Reduce Trust

H2 predicted that the NPR composite in C3 would not significantly reduce scores on the Ability, Benevolence or Integrity subscales relative to the photorealistic baseline in C2.

The trust subscales followed a pattern consistent with H2. C1 scored highest on Ability, Benevolence and Integrity alike, while C2 and C3 did not differ significantly from each other on any of the three (Figure 6.3, Table 6.5). None of the three subscales pointed to a specific trust dimension being affected. A non-significant pairwise result alone remains insufficient evidence for equivalence and the TOST procedure was accordingly applied to the combined trust composite.



	Diff.	Bound	TOST p
H2: C2 vs C3	0.33	±0.51	.240
H4: C1 vs C3	1.31	±0.47	.998

Table 6.6.: TOST results, trust composite (n = 41). Bound = Cohen's d = 0.3 in raw scale points (Chapter 4).

Figure 6.4.: TOST equivalence tests on the combined trust composite (the mean of Ability, Benevolence and Integrity). Points show the mean difference between conditions, black lines the 90% confidence interval and the shaded band the pre-specified equivalence bound. Equivalence requires the full confidence interval to fall inside the shaded band.

H2 is therefore neither confirmed nor disconfirmed by the pre-specified criteria. The pairwise comparisons showed no detectable difference, but the confidence interval on the trust composite still spans both a small negative and a moderate positive difference (Figure 6.4). Consequently, the sample is not large enough to rule out a difference as large as the pre-specified bound. This outcome was anticipated in the power analysis

of Section 4.8.4. A paired TOST at the $d = 0.3$ bound requires roughly $n = 120$ to reach a power of 0.90. The realised sample of 41 leaves it near 0.20 even when the true difference is exactly zero. Separating a genuine null effect from an insensitive test would thus require close to three times the participants recruited here.

H3 — Eeriness Predicts Trust

H3 predicted a negative within-participant relationship between eeriness and trust across all three conditions. This was tested via a person-centred multilevel regression fitted for each trust subscale composite in turn, with the eeriness composite as the within-participant predictor and a random intercept per participant.

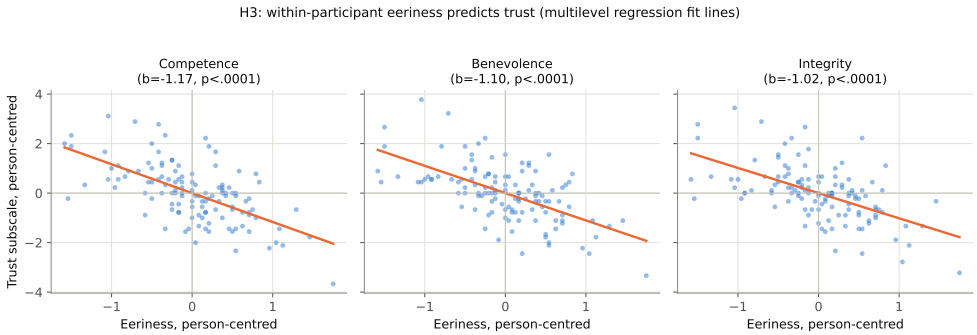


Figure 6.5.: Person-centred relationship between eeriness and each trust subscale. Both axes are expressed as each participant's deviation from their own mean across the three conditions, isolating the within-participant association tested by H3 from between-participant differences in overall trust or eeriness. Lines show the fixed-effect slope from the corresponding multilevel model ($n = 123$ valid blocks from 41 participants).

Table 6.7.: Fixed-effect slopes from the person-centred multilevel model shown in Figure 6.5. All slopes $p < .001$.

Subscale	b	**95% CI**
Ability	−1.17	[−1.49, −0.84]
Benevolence	−1.10	[−1.42, −0.78]
Integrity	−1.02	[−1.34, −0.69]

The person-centred multilevel model confirmed a negative within-participant association between eeriness and every trust subscale (Figure 6.5, Table 6.7). A one-point increase in a participant's own eeriness rating, relative to their own mean across the three conditions, predicted roughly a one-point drop on each trust subscale. The intraclass correlations (.48 to .57) indicated that roughly half of the total variance in each trust subscale sat between participants rather than within them. This is consistent with substantial individual differences in how trusting participants are overall. Hence, H3 is confirmed.

This result matches the direct negative relationship between eeriness and trust that Alipour et al. reported in 11 of the 24 studies that measured trust directly [1]. The near-identical slopes across ability, benevolence and integrity indicate that eeriness suppresses trust broadly rather than through a pathway specific to benevolence.

The eeriness-trust relationship established above also explains why H1 and H2 both returned null results. H1 predicted lower eeriness for C3 than C2, which would have raised trust through the same pathway. However, C2 and C3 did not differ in eeriness. Trust in turn had no reduction to act on. The H1 and H2 nulls therefore stem from the same missing ingredient: a genuine reduction in eeriness. The NPR composite may not have moved the avatar's position on the uncanny valley curve relative to the photorealistic baseline far enough to produce a trust benefit.

H4 — NPR Matches Real-Human Trust

H4 predicted that the NPR composite in C3 would not significantly reduce trust scores relative to the real human speaker in C1. It extended the same null logic as H2 to a different reference point. Nonetheless, residual differences in human likeness, animation naturalness and speaker identity remain between these two conditions.

H4 draws on the same subscale comparisons reported for H2, now restricted to the C1 versus C3 contrast. Ability, benevolence and integrity were all significantly higher for C1 (Table 6.5). The TOST test on the C1 versus C3 trust composite fell far outside the equivalence bound, in the direction of C1 (Table 6.6). This means H4 is not supported.

In summary, the NPR composite did not close the trust gap with the real speaker and the shortfall is substantial rather than marginal. Two explanations remain open. The uncanny valley account predicts this gap. Recognition of the speaker by some participants would produce it equally well. The current design cannot separate them.

6.2.6. Order Effects

The counterbalancing scheme assumes that a video's position in the sequence carries no effect of its own (Section 4.8.4). Position produced no significant main effect on any subscale, although likeability came close to the threshold ($F(2, 80) = 3.04$, $p = .054$) and eeriness showed a weaker trend ($F(2, 80) = 2.47$, $p = .091$). The three trust subscales fell well short of it, with p between .13 and .27.

The three study version groups did differ on likeability ($F(2, 38) = 7.07$, $p = .002$) and eeriness ($F(2, 38) = 5.57$, $p = .008$), with no such difference on any trust subscale ($p \geq .14$). However, this second comparison is between subjects and the groups comprise 15, 12 and 14 different people. A difference of that kind is equally consistent with a sequence effect and with the groups differing in how eerie they find avatars generally. The present design cannot separate the two.

Condition and position are perfectly aliased within each version, since a participant assigned to one version always meets the same condition in the same slot. Carryover between a specific pair of conditions is therefore not estimable from a single Latin square. The condition contrasts themselves remain unaffected either way, because every participant supplies all three conditions and each condition occupies every slot equally often across the three versions.

6.2.7. Post-Study Rankings

The three post-study ranking questions, covering advisor preference (P1), trustworthiness and unsettlingness, were analysed as specified in Section 4.8.4. Mean ranks per condition (1 = most preferred, 3 = least preferred) were compared with a Friedman test per question, followed by pairwise Wilcoxon signed-rank tests with Holm correction. C2 versus C3 is the critical comparison in each case.

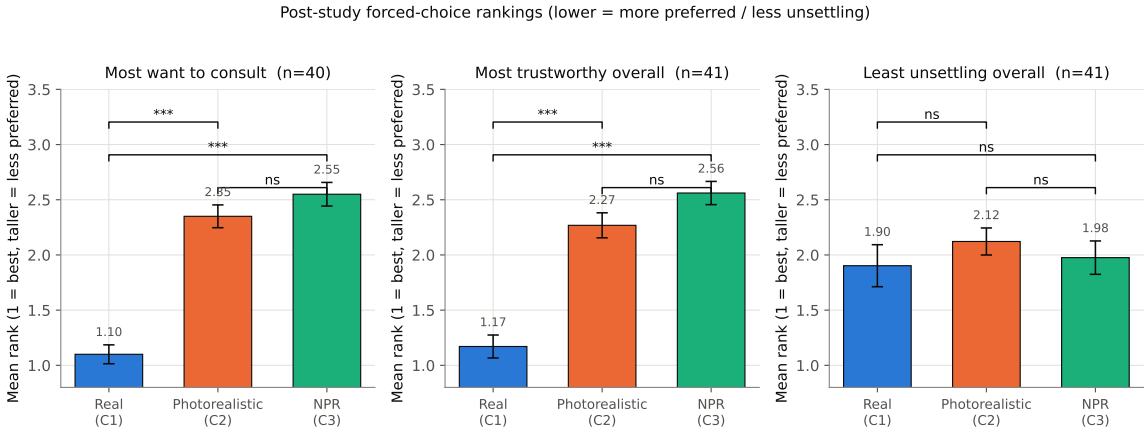


Figure 6.6.: Mean forced-choice rank by condition for the three post-study comparison questions. Error bars show the within-subject corrected standard error of the mean. Brackets mark Holm-corrected pairwise Wilcoxon comparisons: *** $p < .001$, ns = not significant.

Two of the three rankings reproduced the ordering of the Likert measures (Figure 6.6, Table 6.8). Participants ranked C1 first on both the primary behavioural measure and the trustworthiness ranking, well ahead of C2 and C3. Those two did not differ significantly from each other on either question. The unsettlingness ranking showed no discernible pattern on any comparison.

Table 6.8.: Friedman tests and Holm-corrected pairwise Wilcoxon comparisons, post-study rankings.

Question	Friedman	C1 vs C2	C1 vs C3	C2 vs C3
Most want to consult ($n = 40$)	$\chi^2_F(2) = 49.40$, $p < .001$	$p < .001$	$p < .001$	$p = .21$
Most trustworthy ($n = 41$)	$\chi^2_F(2) = 44.05$, $p < .001$	$p < .001$	$p < .001$	$p = .07$
Least unsettling ($n = 41$)	$\chi^2_F(2) = 1.02$, $p = .60$	$p = .97$	$p = .97$	$p = .97$

Canales et al. found a divergence between explicit and behavioural measures in their comparable study [5]. The preference and trustworthiness rankings here converged with the Likert measures instead. C3 attracted no more behavioural endorsement than C2 did, meaning that the forced ordering surfaced nothing that the graded ratings had obscured. C1's dominance is the most consistent finding in the study. It holds on every subscale and on both rankings that produced a significant result.

6.2.8. Secondary Measures

Part G's item on the advisor's recommendation was analysed as a secondary exploratory measure, separately from the preference shift reported in Section 6.2.9. Recommendation agreement did not differ significantly across conditions (Table 6.9). Participants evaluated the substance of each recommendation largely independently of how the advisor was rendered. The scenario design held argumentative content constant across conditions, so this outcome was expected.

Table 6.9.: Secondary measures, $n = 41$.

Recommendation agreement, M (SD)	**C1**	**C2**	**C3**
	4.07 (2.20)	4.68 (1.93)	3.85 (1.97)
Omnibus: $F(2, 80) = 2.10$, $p = .13$			
Trust propensity × trust composite: $r(39) = -.10$, $p = .52$			

The dispositional trust covariate was examined as an individual difference moderator. It showed no meaningful association with the mean trust composite (Table 6.9). A quartile-based moderation analysis was not pursued, because the sample is too small to detect an interaction reliably. Canales et al. similarly found no significant relationship between propensity and trust ratings in their comparable manipulation [5].

6.2.9. Pre-Video Baseline Shift

Preference shift was computed per participant per block as the change from the pre-video option choice to the post-video choice recorded in Part G (Section 4.8.4). The comparison isolated the persuasive effect of the advisor's recommendation from the participant's prior inclination.

Preference shift towards the advisor's recommendation was infrequent across all three conditions. Four of 41 participants (9.8%) shifted towards the recommendation after C1, five of 41 (12.2%) after C2 and one of 41 (2.4%) after C3. These counts are too small to support a formal test of condition differences. The numerically higher shift rate for C2 should be read as descriptive rather than as evidence of a persuasive advantage for the photorealistic avatar. The low rate across all conditions suggests that the scenarios kept both alternatives genuinely plausible, as the design in Section 4.8.3 intended. Most participants entered each video already inclined towards one option and were not easily moved by a short recommendation regardless of how the advisor was rendered.

6.3. Discussion

Two patterns in the study outcome call for an explanation. The real speaker separated sharply from both avatar conditions on every subscale and on both rankings that produced a significant result. The manipulation check confirmed that participants registered the two avatar conditions as differently stylised. They nonetheless stayed

statistically indistinguishable on every outcome measure. A further result constrains any explanation of these two. Eeriness predicted trust within participants, meaning that the route from appearance to trust was open. The composite changed what participants saw without travelling it. The explanations below move outward from the stimulus itself to the viewers watching it.

6.3.1. What the Stylisation Left Unchanged

The composite stylises the avatar's material while leaving its shape untouched and this split may be why eeriness did not fall. Material governs eeriness in the account of Zell et al. and this alone predicts a reduction for C3 [46]. Their mismatch result points the other way. C3 sets a stylised material on a photogrammetrically reconstructed shape. Zell et al. also reported that the influence of material stylisation on perceived realism grows as shape realism rises, and this places C3 at the point where a mismatch would be most visible [46]. The abstraction may consequently have removed eeriness from the material only to reintroduce it through the gap between the two.

The features that the participants of Dill et al. blamed for their discomfort, the eyes and the mouth, were largely preserved [26]. The preprocessing altered the mouth texture in C3, but the geometry of both regions is identical across the two conditions. Additionally, both conditions inherit the same gaze and lip animation from one recording session. A shading operator cannot reach that animation and any mismatch in its timing thus survives the stylisation.

Consistency across the whole presentation offers a third reading. Cihodaru-Ştefanache and Podina reported the perceptual mismatch hypothesis, under which uncanniness arises from features perceived at inconsistent levels of anthropomorphism, as the best supported explanation of the effect [32]. C3 exhibits precisely that inconsistency, since a recorded human voice accompanies a visibly abstracted face. The stylisation on this reading introduced a new source of uncanniness even as it removed another.

6.3.2. The Comparison Set

Because every participant saw all three conditions, the real speaker was available as a reference point for the other two. C1 ranked first on advisor preference and on trustworthiness and it was rated most likeable, least eerie and highest on all three trust dimensions. Once a genuine human occupies the top of the scale, the judgement left to the participant reduces to how far each avatar falls below it.

That comparison compresses the range left for the two avatar conditions. Both were recognisably CG and both fell short of C1 on the attribute participants weighted most heavily. Therefore, a difference of rendering style competes against a categorical distinction that is far more salient. In particular, the measured difference between C2 and C3 on likeability and eeriness was negligible and this fits that reading.

The sequence sharpens this further. C3 never followed C2 in any version and whenever C3 was not the first video it followed C1 directly. C2 was never preceded by C1. The critical comparison accordingly places C3 immediately after the real speaker in two versions out of three, while C2 never occupies that position. Any contrast effect from the real human falls on C3 alone. The present design cannot establish whether it did.

Furthermore, withdrawing the real speaker would change the comparison rather than merely narrow it. Abstraction becomes the sole axis on which the stimuli can be ordered with only two avatars in view. A study presenting C2 and C3 without C1 would test this directly.

6.3.3. Familiarity with the Real Speaker

The speaker in C1 is the researcher who conducted the study. Hence, some participants may have recognised them. Such acquaintance supplies an existing basis for trust that no rendering can match. C1 would then sit above the level an unfamiliar human speaker would reach. The gap both avatar conditions are measured against widens accordingly. The questionnaire included no recognition item and the share of participants affected cannot be established.

6.3.4. Viewer Expectations of Generative Video

A participant who had completed the study asked afterwards which AI system had been used to generate the videos. The question indicates that at least one viewer filed the material under generative AI output rather than avatar rendering. The question also concerned the videos as a set, not any one condition.

Generative video is routinely encountered outside any research setting and viewers arrive with expectations formed by that exposure. A viewer who takes an avatar to be an AI-generated artefact may apply a general scepticism towards media produced by machines. This would depress trust for both avatar conditions equally rather than separating them.

Nothing in the study materials ruled that assumption out. Neither the consent text nor the block instructions mentioned how the recordings had been produced and each participant was left to decide for themselves. Stating at the outset that the videos were not AI-generated would have settled the question.

6.3.5. The Video-Based Setting

Participants watched prerecorded video and never interacted with the speaker. The setting supplied no stereoscopic depth, no motion parallax and no sense of sharing space with the avatar. A comparable VR study held shape and animation constant and varied only render style. Photorealism significantly raised both place illusion and social presence [45]. A rendering difference that registers weakly on a flat screen may consequently register more strongly once the avatar occupies the viewer's own space.

The absence of interaction works in the same direction. An advisor who cannot respond, hold eye contact or adapt to the participant offers fewer occasions for a mismatch between appearance and behaviour to surface. These results accordingly describe a speaking head in a browser window. An interactive setting such as VR may separate C2 and C3 where the present design did not.

6.4. Limitations

Both the parameter adjustment study and the main study were designed to take approximately 10 to 15 minutes, as longer sessions were expected to make participant recruitment more difficult. Achieving this target was challenging because each measure in the main study was completed three times, once for each video. Consequently, the available session time constrained both the number of measures that could be included and the breadth of the experimental design. The main study used a single NPR composite, one avatar identity, one platform and a fixed set of three scenarios. The findings should therefore be interpreted as characterising this particular configuration rather than NPR stylisation as a category. Other rendering techniques, levels of abstraction, avatars or decision contexts may produce different effects on trust.

The same time constraint also limited the scope of the parameter adjustment study. Only the parameter groups expected to have the greatest visual impact were systematically tested, while the remainder were fixed through direct inspection during development. As a result, the final NPR composite should not be treated as globally optimal across the full parameter space. It is better understood as a configuration judged suitable within the limits of the study.

Furthermore, the fixed thesis submission deadline imposed a second constraint on the study. It determined the data collection period, which coincided with the university break. One of the three recruitment channels was a university mailing list, but many students were away during this period. Thus, recruitment had to close before a larger sample could be obtained. Although the resulting sample met the minimum size indicated by the power analysis, it was insufficient for the equivalence tests, which need more participants to provide informative evidence. Their results remained inconclusive.

The limited sample size also affected the counterbalancing strategy. A full design with three experimental conditions would require six possible presentation orders. Achieving adequate group sizes across all six would have demanded approximately twice the available sample. Accordingly, only three orders were used.

7. Conclusion

7.1. Summary of Findings

Rendering a virtual human at high fidelity can lower rather than raise the confidence of users who interact with it. NPR offers a route around that penalty, but the platforms that supply avatars to consumer applications expose narrow and unequal control over their own rendering. This thesis developed NPR stylisation for Mixamo, the Meta Avatars SDK and Avaturn. It then measured what the resulting abstraction did to perceived trust. Eight techniques were built in sequence, ranging from a geometry-based outline through several edge detection operators to a painterly filter. Three of them make up the composite carried into the user study: a geometric silhouette outline, X-Toon tonal shading and Sobel edge detection over a preprocessed head texture.

Every technique was integrated on all three platforms to answer RQ1. Building the same operators three times exposed systematic differences in the signal each pipeline supplied. Specifically, the most restrictive of the three offers a single fragment hook and no additional passes. An operator assuming a separate buffer or pass hence had to be rebuilt around a substitute. The hierarchical depth layer reads a camera-distance derivative instead of a depth texture and the Kuwahara structure tensor loses its Gaussian smoothing altogether. Identical parameter values likewise produced visibly different output from one host to the next. Neither constraint ruled a technique out entirely. The dividing line arises from whether an operator needs new parameter values or a new input signal.

RQ2 asked which combination best balances abstraction against facial legibility. The answer came from calibration, not visual inspection. Each version addressed a limitation identified in its predecessor and five of the eight became candidates for the composite on that basis. A parameter adjustment study then asked 45 participants to rank rendered variants by perceived trustworthiness across five parameter groups. Friedman and Wilcoxon tests removed both the Kuwahara filter and the hierarchical edge layer from the composite. Three of the seven final values came from a separate preference question, as no configuration in those rounds proved statistically distinguishable. However, no technique in the set reached that balance on the Avaturn avatar by rendering alone. Its photogrammetric albedo carries baked illumination and gradient operators read that illumination as structure. Facial features became legible only after the head texture was preprocessed. The answer to RQ2 is thus not a technique on its own. It is the three-component composite paired with texture preprocessing.

RQ3 was addressed through a within-subjects study with 41 participants. The same speaker appeared as a real person (C1), as a photorealistic Avaturn avatar (C2) and as that avatar under the selected composite (C3). The questionnaire paired validated likeability and eeriness instruments with trust subscales for ability, benevolence and integrity. A forced ranking measure followed the three video blocks. Three decision scenarios were written so that each targeted a different trust dimension. The recommended option and the order of discussion alternated across those scenarios and this kept their content from confounding the comparison of rendering styles.

The outcomes qualify RQ3 without answering it affirmatively. H1 expected the stylised avatar to be rated less eerie than the photorealistic one. The two did not differ. H1 is not supported. Participants registered the two conditions as differently stylised, meaning that an unnoticed rendering difference cannot explain the null result. H2 predicted that stylisation costs no trust relative to that same baseline. Neither pre-specified criterion settled it. The pairwise comparison detected no trust difference between the two avatar conditions. Nevertheless, the equivalence test could not exclude a difference as large as the bound it was set against.

H3 held that eeriness suppresses trust. Eeriness predicted every trust subscale negatively within participants. Even though the composite failed to lower it, the route from appearance to trust stayed open. H3 is confirmed. H4 extended the same claim of no trust cost from H2 to the real speaker. Trust in the stylised avatar remained well below trust in the real speaker. H4 is accordingly not supported.

Although the composite did not reduce eeriness relative to the photorealistic baseline, it produced no observed trust penalty either. The available evidence therefore does not indicate that stylisation reduced trust, but it is not sufficient to establish that the two avatar conditions are equivalent. The practical implication is that a developer choosing between the two styles need not assume an inherent disadvantage for the stylised version. The question of equivalence itself stays open. What remains is the distance to the real speaker and neither avatar condition closed any part of it. Reaching the credibility of an actual human is consequently a harder problem than selecting a rendering style. It is also the problem that decides whether virtual advisors become genuinely trusted participants rather than a passing novelty.

7.2. Future Work

The clearest next step is a larger version of the study reported here. H2 was left unresolved because the equivalence test, with only 41 participants, could not separate a genuine null from an insensitive one. Recruitment had to close before more could be gathered due to the fixed submission deadline. The retained sample met the requirement for detecting a difference, not the far larger one that equivalence demands. Repeating the C2 versus C3 comparison with roughly 120 participants would settle it. A result inside the equivalence bound confirms that stylisation costs no trust. A result outside it exposes a cost too small for this sample to detect.

The competing explanations for the null results point to three adjustments in the study design. The first extends abstraction to the geometry, so that proportion and feature scale shift alongside the shading. This variant tests whether a stylised material on a photogrammetric shape is what held eeriness constant. The second drops the real speaker from the comparison set. Removing that anchor releases the range it compressed and reveals whether the two avatars separate when no human reference is in view. The third states at the outset that the videos were not generated by AI. Nothing in the present materials ruled out the belief that they were. Such a belief may have lowered the trust ratings for both avatar conditions.

A further study should vary abstraction strength continuously instead of testing one setting against a photorealistic baseline. The composite fixes abstraction at the single value the parameter adjustment study selected. Sweeping that value across its range would replace the point measured here with a curve. Such a curve traces how eeriness and trust move together as abstraction rises. It can also reveal whether an optimum sits between full realism and full abstraction.

An interaction with a virtual human normally involves reciprocal behaviour and shared spatial presence. Participants here watched prerecorded videos. The advisor thus never responded to their questions or returned their gaze. Furthermore, they viewed the avatar on a flat screen instead of within the same physical environment. Restoring both in a follow-up design would show whether they change the effect of stylisation on trust. A viewer who shares that environment with the avatar, as in VR, may pay closer attention to subtle variations in its appearance. A rendering difference that was too weak to influence trust in this study could consequently produce a measurable effect. This would expose a gap between C2 and C3 that the current design was unable to detect.

Moreover, the comparison needs repeating across multiple avatar identities and platforms. The study used only one of each. The avatar was reconstructed from photographs of a single individual and it carries that person's specific features. However, the effect of stylisation may depend on the identity and appearance of the character being rendered. For example, the same degree of abstraction may be perceived differently when applied to a character who appears male, elderly or non-human. Only a wider sample of identities can separate two readings of the null result reported here. Either stylisation leaves trust unchanged in general or that outcome is specific to this avatar rendered with this particular composite.

8. Literature

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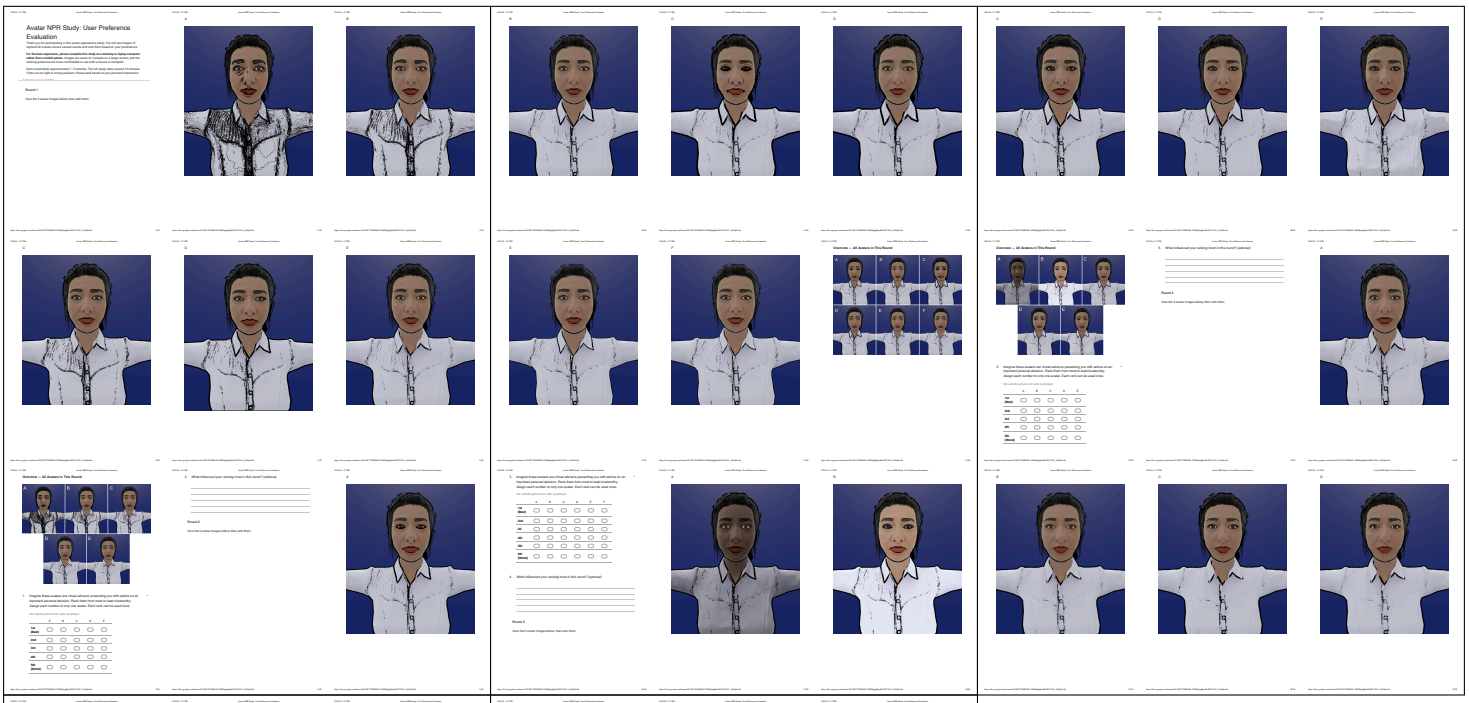
A. Appendix

Parameter Adjustment Study Questionnaire

The following pages reproduce the parameter adjustment study questionnaire administered before the main study. The resulting ranking data and aggregated scores are reported at the end of this appendix.


Avatar NPR Study: User Preference Evaluation
Round 1
Round 2
Round 3
Round 4
Round 5
Overview — All Avatars in This Round
Grand Impression
Thank you for completing this evaluation

Online Questionnaire

The following pages reproduce the main study questionnaire as administered to participants via Google Forms.

Evaluation of Video Advisors

Welcome, and thank you for taking part.

In this study you will watch three short videos of advisors giving their opinion on everyday decisions. After each video you will answer brief questions about your impression of the advisor. At the end you will be asked to compare all three advisors overall, so please watch each video carefully and pay attention to each advisor.

Please be aware that the visual presentation of the advisors in this study varies across videos. Some advisors appear more realistic, while others are more stylised or abstracted. This variation is a deliberate aspect of the study design. You are asked to focus on your overall impression of each advisor rather than on the realism of the animation.

The study takes approximately 20 minutes — please complete it in one sitting with sound on. A desktop or laptop is strongly recommended.

Before you begin:

- Participation is voluntary and you may stop at any time.
- No personally identifying information is collected.
- Your anonymised data will be used solely for academic research at the University of Konstanz.
- Questions? Contact: nidanur.guenay@uni-konstanz.de

* Indicates required question

1. I have read the information above, I am 18 years or older, * and I agree to participate in this study.

Check all that apply.

- [ ] I consent

Demographics

2. Please indicate the gender you identify with *

Mark only one oval.

- ◯ Man
- ◯ Woman
- ◯ Non-binary
- ◯ Prefer not to say
- ◯ Other: ____

3. Please indicate your age *

_____

4. How often do you interact with virtual avatars or computer-generated characters * (in games, VR, video calls, social media, etc.)?

Mark only one oval.

	1	2	3	4	5	
Nevı	◯	◯	◯	◯	◯	Very often

5. In general, I tend to trust people I meet for the first time. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

Round 1

Read the scenario below carefully, then answer one quick question before watching the video.

Situation

You recently completed your master's degree and have received two job offers. Both start in one month. You have five days to decide.

JOB A — Generalist Role (Small Company)

Job A is a generalist role at a smaller company. You would define your own work area, your contributions would be directly visible to the team, and the skills you build would transfer across industries. The starting salary is below the industry average for your field.

JOB B — Specialist Role (Large Company)

Job B is a specialist role at a large, well-known company. The employer name carries weight on a CV and the role is in a high-demand field. However, the position involves deep specialisation within a large team, with limited individual visibility. The starting salary is above the industry average for your field.

6. Round 1: Before watching, which job would you choose? *

Mark only one oval.

- ◯ Job A
- ◯ Job B

Video 1

Please watch the video below in full, then continue to the next page.
Make sure your sound is on.

Advisor 1

http://youtube.com/watch?v=68sg8fSFH1E

The advisor said
"I'd go with Job A. Yes, the salary is lower right now, but the generalist role gives you a range of experience that is genuinely hard to get later. Salary gaps can be closed in the future. But once you specialise, switching back is costly. Take the pivot while you can."

7. I have watched the full video. *

Check all that apply.

- [ ] I have watched the video

Round 1 Questionaire

8. Which job did the advisor recommend? *

Mark only one oval.

- ◯ Job A
- ◯ Job B
- ◯ I am not sure

Warmth and Appeal

Below you will see pairs of opposite words. Select the number that best matches your impression of this advisor. A lower number means you lean toward the word on the left. A higher number means you lean toward the word on the right. If you feel neither or somewhere in between, select 3.

9. *

Mark only one oval.

	1	2	3	4	5	
Disli	◯	◯	◯	◯	◯	Like

10. *

Mark only one oval.

	1	2	3	4	5	
Unfr	◯	◯	◯	◯	◯	Friendly

11. *

Mark only one oval.

	1	2	3	4	5	
Unki	◯	◯	◯	◯	◯	Kind

12. *

Mark only one oval.

	1	2	3	4	5	
Unp	◯	◯	◯	◯	◯	Pleasant

13. *

Mark only one oval.

	1	2	3	4	5	
Awfi	◯	◯	◯	◯	◯	Nice

Impression

Please rate your impression of the advisor on the following scales. 1 = left word, 5 = right word.

14. *"Eerie" means strangely unsettling in a way that is hard to explain.* *

Mark only one oval.

	1	2	3	4	5	
Rea:	◯	◯	◯	◯	◯	Eerie

15. *"Freaky"* means very strange or disturbing in an unexpected way. *

Mark only one oval.

	1	2	3	4	5	
Nurr	◯	◯	◯	◯	◯	Freaky

16. *"Supernatural"* means beyond what feels natural or normal. *

Mark only one oval.

	1	2	3	4	5	
Ordi	◯	◯	◯	◯	◯	Supernatural

17. *"Uncanny" means something that feels strangely unsettling, as if almost right but slightly off.* *

Mark only one oval.

	1	2	3	4	5	
Blar	◯	◯	◯	◯	◯	Uncanny

18. *"Hair-raising" means causing a strong feeling of unease or alarm.* *

Mark only one oval.

	1	2	3	4	5	
Une	◯	◯	◯	◯	◯	Hair-raising

19. *"Spine-tingling" means producing a strong feeling of discomfort or dread.* *

Mark only one oval.

	1	2	3	4	5	
Unir	◯	◯	◯	◯	◯	Spine-tingling

20. *"Shocking"* means surprising you in an unpleasant or startling way. *

Mark only one oval.

	1	2	3	4	5	
Bori	◯	◯	◯	◯	◯	Shocking

21. *"Thrilling"* here means a strong, uncomfortable physical sensation in your body. *

Mark only one oval.

	1	2	3	4	5	
Prec	◯	◯	◯	◯	◯	Thrilling

Trust

The following questions use a different scale: 1 to 7, where 1 = Strongly Disagree and 7 = Strongly Agree.

22. This advisor is capable and competent. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

23. This advisor seems knowledgeable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

24. This advisor seems to have the expertise needed to give sound advice on this kind of decision. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

25. This advisor will act in a benevolent (well meaning) manner. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

26. This advisor would act in my best interest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

27. This advisor appears to have good intentions *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

28. This advisor adheres to a set of principles that I find acceptable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

29. This advisor appears to be honest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

30. This advisor seems sincere *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

Response to the Recommendation & Visual Style

31. How much do you agree with the recommendation this advisor gave? *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

32. After watching, which job would you choose? *

Mark only one oval.

- ◯ Job A
- ◯ Job B

33. Please select option 2 regardless of anything else.

Mark only one oval.

- ◯ 1
- ◯ 2
- ◯ 3
- ◯ 4
- ◯ 5

Round 2

Read the scenario below carefully, then answer one quick question before watching the video.

Situation
You have just arrived by train in a city you have never visited. You need to reach your hotel, where colleagues are expecting you in one hour.

ROUTE A
Route A is a direct route on a single line with no transfers. It is straightforward to navigate in an unfamiliar city. If the journey runs on time, you would arrive approximately 10 minutes before your meeting begins.

ROUTE B
Route B involves one transfer at a mid-size station you have not used before. If the connection runs on time, you would arrive approximately 40 minutes before your meeting begins. However, the transfer station is unfamiliar and a missed connection could leave you with very little time.

34. Round 2: Before watching, which route would you choose? *

Mark only one oval.

- ◯ Route A
- ◯ Route B

Video 2

Please watch the video below in full, then continue to the next page.
Make sure your sound is on.

Advisor 2

http://youtube.com/watch?v=3vfPChpASas

The advisor said:
"I'd take Route B. Yes, it has a transfer at a station you have never used. But it gets you there forty minutes early. If something goes wrong, you still have enough time to recover. Route A is direct but leaves only ten minutes. That may not be enough. Take the buffer."

35. I have watched the full video.

Check all that apply.

- ☐ I have watched the video

Round 2 Questionaire

36. Which route did the advisor recommend? *

Mark only one oval.

- ◯ Route A
- ◯ Route B
- ◯ I am not sure

Warmth and Appeal

Below you will see pairs of opposite words. Select the number that best matches your impression of this advisor. A lower number means you lean toward the word on the left. A higher number means you lean toward the word on the right. If you feel neither or somewhere in between, select 3.

37. *

Mark only one oval.

	1	2	3	4	5	
Disli	◯	◯	◯	◯	◯	Like

38. *

Mark only one oval.

	1	2	3	4	5	
Unfr	◯	◯	◯	◯	◯	Friendly

39. *

Mark only one oval.

	1	2	3	4	5	
Unki	◯	◯	◯	◯	◯	Kind

40. *

Mark only one oval.

	1	2	3	4	5	
Unp	◯	◯	◯	◯	◯	Pleasant

41. *

Mark only one oval.

	1	2	3	4	5	
Awfi	◯	◯	◯	◯	◯	Nice

Impression

Please rate your impression of the advisor on the following scales. 1 = left word, 5 = right word.

42. *"Eerie" means strangely unsettling in a way that is hard to explain.* *

Mark only one oval.

	1	2	3	4	5	
Rea:	◯	◯	◯	◯	◯	Eerie

43. *"Freaky"* means very strange or disturbing in an unexpected way. *

Mark only one oval.

	1	2	3	4	5	
Nurr	◯	◯	◯	◯	◯	Freaky

44. *"Supernatural"* means beyond what feels natural or normal. *

Mark only one oval.

	1	2	3	4	5	
Ordi	◯	◯	◯	◯	◯	Supernatural

45. *"Uncanny" means something that feels strangely unsettling, as if almost right but slightly off.* *

Mark only one oval.

	1	2	3	4	5	
Blar	◯	◯	◯	◯	◯	Uncanny

46. *"Hair-raising" means causing a strong feeling of unease or alarm.* *

Mark only one oval.

	1	2	3	4	5	
Une	◯	◯	◯	◯	◯	Hair-raising

47. *"Spine-tingling" means producing a strong feeling of discomfort or dread.* *

Mark only one oval.

	1	2	3	4	5	
Unir	◯	◯	◯	◯	◯	Spine-tingling

48. *"Shocking"* means surprising you in an unpleasant or startling way. *

Mark only one oval.

	1	2	3	4	5	
Bori	◯	◯	◯	◯	◯	Shocking

49. *"Thrilling"* here means a strong, uncomfortable physical sensation in your body. *

Mark only one oval.

	1	2	3	4	5	
Prec	◯	◯	◯	◯	◯	Thrilling

Trust

The following questions use a different scale: 1 to 7, where 1 = Strongly Disagree and 7 = Strongly Agree.

50. This advisor is capable and competent. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

51. This advisor seems knowledgeable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

52. This advisor seems to have the expertise needed to give sound advice on this kind of decision. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

53. This advisor will act in a benevolent (well meaning) manner. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

54. This advisor would act in my best interest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

55. This advisor appears to have good intentions *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

56. This advisor adheres to a set of principles that I find acceptable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

57. This advisor appears to be honest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

58. This advisor seems sincere *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

Response to the Recommendation & Visual Style

59. How much do you agree with the recommendation this advisor gave? *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

60. After watching, which route would you choose? *

Mark only one oval.

- ◯ Route A
- ◯ Route B

Round 3

Read the scenario below carefully, then answer one quick question before watching the video.

Situation

You want to develop a professional skill that would strengthen your CV and improve your career prospects. You have a few weeks available and you are a student with a limited budget.

OPTION A - Paid Online Course
Option A is a structured online course from a well-known learning platform. It follows a clear curriculum and awards a recognised certificate you can add to your CV. It costs €350.

OPTION B - Free Tutorial Series
Option B is a video tutorial series by a practitioner with a strong reputation. It covers the same material at no cost and you set your own pace. However, there is no certificate and no structured curriculum.

61. Round 3: Before watching, which option would you choose?

Mark only one oval.

- ◯ Option A
- ◯ Option B

Video 3

Please watch the video below in full, then continue to the next page.
Make sure your sound is on.

Advisor 3

http://youtube.com/watch?v=oP5tg9IelYQ

The advisor said
"I would take Option A. Yes, 350 euros is a real cost. But the certificate is what matters in here. It is not just about what you learn, it is about what you can show. A recruiter cannot verify a free tutorial series. That is worth it."

62. I have watched the full video.

Check all that apply.

- ☐ I have watched the video

Round 3 Questionaire

63. Which option did the advisor recommend? *

Mark only one oval.

- ◯ Option A
- ◯ Option B
- ◯ I am not sure

Warmth and Appeal

Below you will see pairs of opposite words. Select the number that best matches your impression of this advisor. A lower number means you lean toward the word on the left. A higher number means you lean toward the word on the right. If you feel neither or somewhere in between, select 3.

64. *

Mark only one oval.

	1	2	3	4	5	
Disli	◯	◯	◯	◯	◯	Like

65. *

Mark only one oval.

	1	2	3	4	5	
Unfr	◯	◯	◯	◯	◯	Friendly

66. *

Mark only one oval.

	1	2	3	4	5	
Unki	◯	◯	◯	◯	◯	Kind

67. *

Mark only one oval.

	1	2	3	4	5	
Unp	◯	◯	◯	◯	◯	Pleasant

68. *

Mark only one oval.

	1	2	3	4	5	
Awf	◯	◯	◯	◯	◯	Nice

Impression

Please rate your impression of the advisor on the following scales. 1 = left word, 5 = right word.

69. *"Eerie" means strangely unsettling in a way that is hard to explain.* *

Mark only one oval.

	1	2	3	4	5	
Rea:	◯	◯	◯	◯	◯	Eerie

70. *"Freaky"* means very strange or disturbing in an unexpected way. *

Mark only one oval.

	1	2	3	4	5	
Nurr	◯	◯	◯	◯	◯	Freaky

71. *"Supernatural"* means beyond what feels natural or normal. *

Mark only one oval.

	1	2	3	4	5	
Ordi	◯	◯	◯	◯	◯	Supernatural

72. *"Uncanny" means something that feels strangely unsettling, as if almost right but slightly off.* *

Mark only one oval.

	1	2	3	4	5	
Blan	◯	◯	◯	◯	◯	Uncanny

73. *"Hair-raising" means causing a strong feeling of unease or alarm.* *

Mark only one oval.

	1	2	3	4	5	
Une:	◯	◯	◯	◯	◯	Hair-raising

74. *"Spine-tingling" means producing a strong feeling of discomfort or dread.* *

Mark only one oval.

	1	2	3	4	5	
Unir	◯	◯	◯	◯	◯	Spine-tingling

75. *"Shocking"* means surprising you in an unpleasant or startling way. *

Mark only one oval.

	1	2	3	4	5	
Bori	◯	◯	◯	◯	◯	Shocking

76. *"Thrilling"* here means a strong, uncomfortable physical sensation in your body. *

Mark only one oval.

	1	2	3	4	5	
Prec	◯	◯	◯	◯	◯	Thrilling

Trust

The following questions use a different scale: 1 to 7, where 1 = Strongly Disagree and 7 = Strongly Agree.

77. This advisor is capable and competent. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

78. This advisor seems knowledgeable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

79. This advisor seems to have the expertise needed to give sound advice on this kind of decision. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

80. This advisor will act in a benevolent (well meaning) manner. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

81. This advisor would act in my best interest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

82. This advisor appears to have good intentions *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

83. This advisor adheres to a set of principles that I find acceptable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

84. This advisor appears to be honest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

85. This advisor seems sincere *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

Response to the Recommendation & Visual Style

86. How much do you agree with the recommendation this advisor gave? *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

87. After watching, which option would you choose? *

Mark only one oval.

- ◯ Option A
- ◯ Option B

88. Please select option 4 regardless of anything else.

Mark only one oval.

- ◯ 1
- ◯ 2
- ◯ 3
- ◯ 4
- ◯ 5

Overall Impressions

You have now watched all three advisors.

The following questions ask you to compare them overall. Take a moment to think back on each one before answering.

Advisors in the Videos

89. Imagine each of these advisors could give you advice on any type of decision. * Please rank them from the one you would most want to consult to the one you would least want to consult. Assign each a different rank.

Mark only one oval per row.

	Video 1	Video 2	Video 3
1	◯	◯	◯
2	◯	◯	◯
3	◯	◯	◯

90. Please rank the three advisors from most to least trustworthy overall. *

Mark only one oval per row.

	Video 1	Video 2	Video 3
1	◯	◯	◯
2	◯	◯	◯
3	◯	◯	◯

91. Please rank the three advisors from most to least unsettling overall. *

Mark only one oval per row.

	Video 1	Video 2	Video 3
1	◯	◯	◯
2	◯	◯	◯
3	◯	◯	◯

92. How visually stylised did each advisor appear? *
(1 = Completely realistic, 7 = Very stylised)

Mark only one oval per row.

	1	2	3	4	5	6	7
Video 1	◯	◯	◯	◯	◯	◯	◯
Video 2	◯	◯	◯	◯	◯	◯	◯
Video 3	◯	◯	◯	◯	◯	◯	◯

93. Looking back at all three advisors, what was the most important factor in deciding which one you trusted the most?

Please describe in a few sentences.

94. What do you think this study was investigating?

Thank you for completing this study!

If you have any questions, please contact: nidanur.guenay@uni-konstanz.de

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Google Forms

Parameter Adjustment Study Results

The figures below present the full ranking data from the five-round parameter adjustment study described in Section 4.7. Each bar shows the percentage of participants who assigned a given option to that rank position and options within each round are ordered by Borda score, highest first. The Borda score for each option is the sum of $(N - r + 1)$ across all valid responses, where N is the number of options in that round and r the rank position assigned by a participant. An option placed first by every participant therefore scores N times the number of valid responses. A higher Borda score indicates a more consistently preferred option across the full distribution. The mean rank (MR) is reported alongside each bar, where a lower value indicates higher preference.

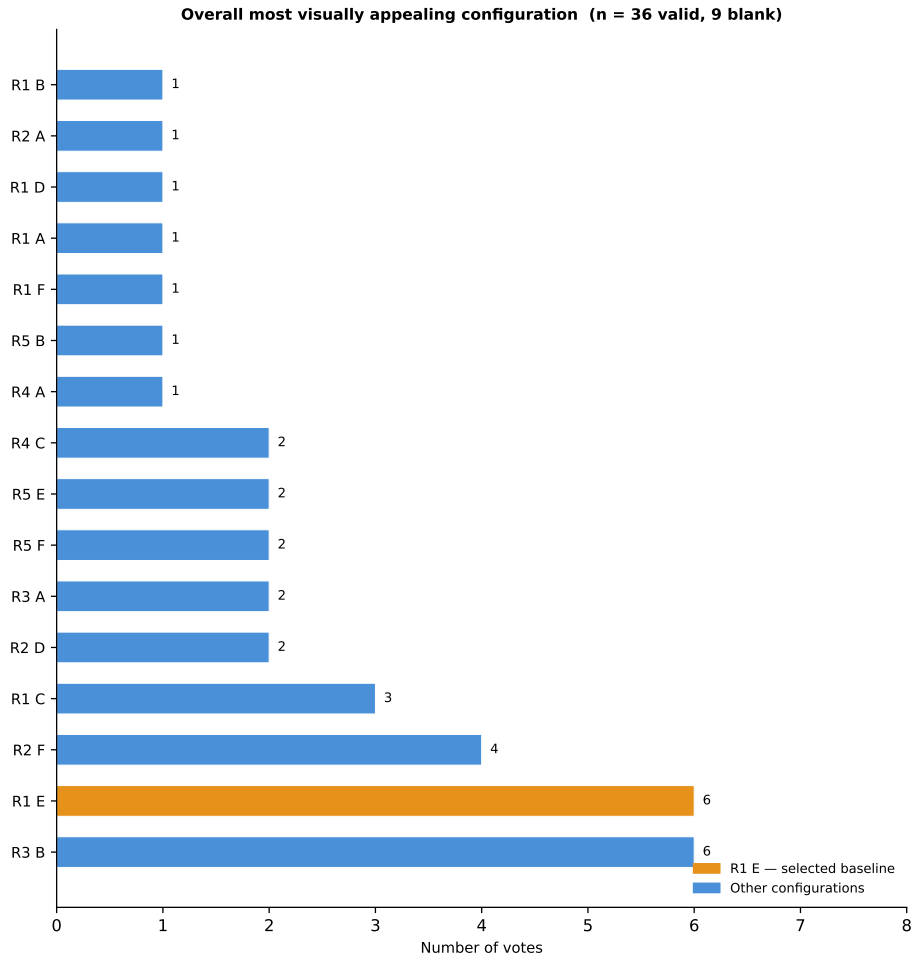

Figure A.1.: Votes for the overall most visually appealing configuration from the final question of the parameter adjustment study (n = 36 valid, 9 blank). The selected study baseline, R1 E, is highlighted in amber.

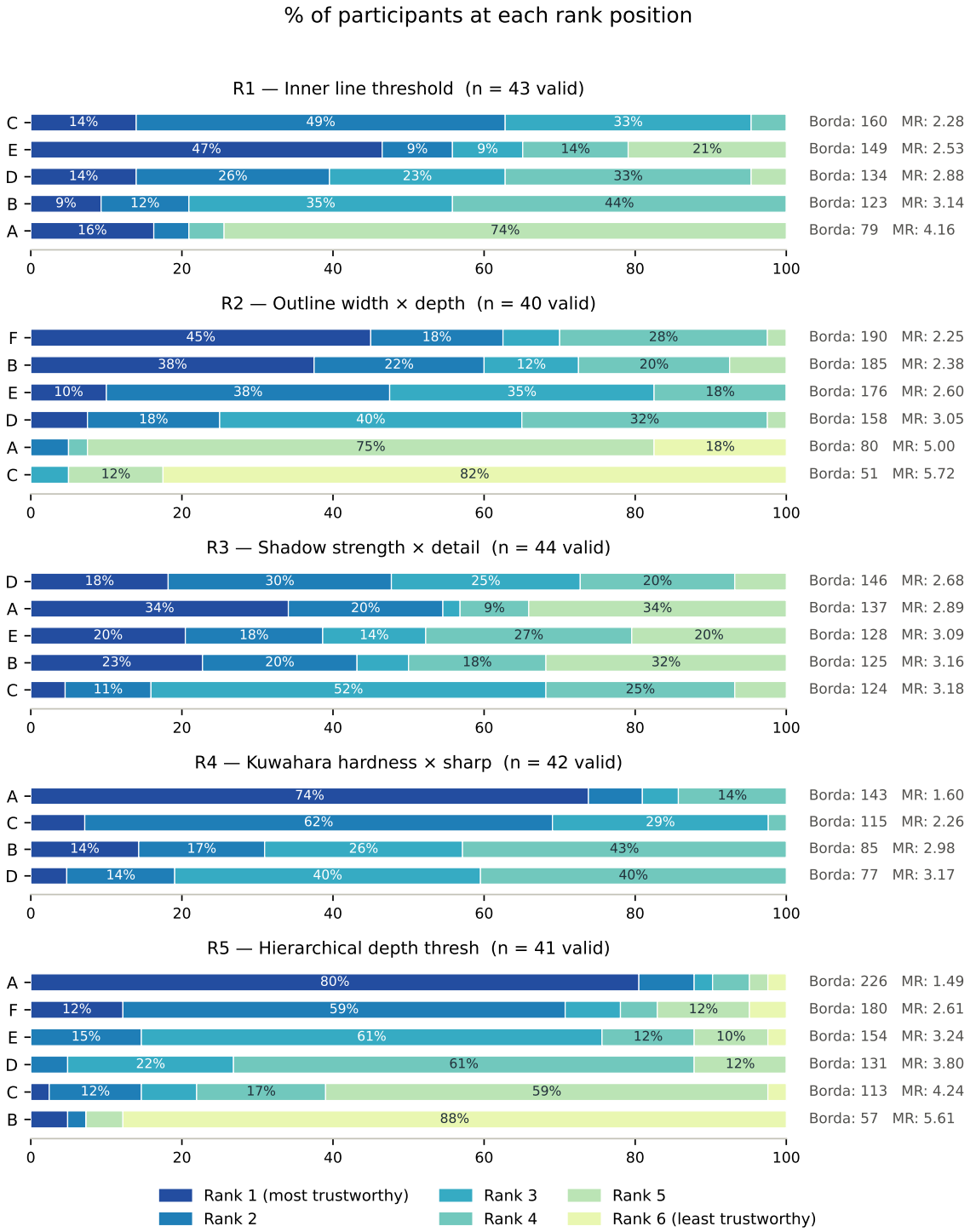

Figure A.2.: Full rank distributions for all five parameter adjustment study rounds. Bars show the percentage of participants placing each option at each rank position, with options within each round sorted by Borda score (highest first). Borda and mean-rank (MR) annotations appear to the right of each bar.